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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 521 — Past Questions & Materials
by Renance

Course: PEE 521
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 500

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Paper 1 of 5 — PEE 521 ADVANCED DEILLING ENGINEERING 2018I2019

Level 500 • Second Semester • PastQuestion

University of Petroleum Resources, Effurun

Department of Petroleum Engineering

PNG 521 – Advance Drilling Engineering

2018/2019 Session Final Examination

Mat. No. _____ Date: _____

Time: 0000 - 0000 hrs (3 hrs)

Basic Instructions

1. Answer Question 1 and Four(4) Others
2. Some relevant charts and graph sheets are provided
3. Where and when required, Show all workings on the charts provided on this question paper
4. This question paper with all charts used in this exam MUST be returned with your answer booklet

2

QUESTIONS 1

a. Define well direction and give with appropriate diagrams, the two notations for deviational well directions in the oil and gas industry 3

- ☐ Deviational Well direction
- ☐ The path of the well from surface location to target location on the horizontal plane relative to the North
- ☐ Direction of tangent plane projection on the HORIZONTAL plane relative to the NORTH reference
- ☐ Direction of the vertical projection of the well bore interval drilled onto a HORIZONTAL Plane measured relative to the NORTH reference
- ☐ A measure of the direction of the wellbore on the HORIZONTAL plane with respect to a NORTH reference
- ☐ Notation for Deviational Well Directions
- ☐ Azimuth System
- Measured Direction Expressed in Clockwise Degrees
- From 0 to 360 with North
- Reference as 0 and 360

$\frac{1}{2}$

- ☐ Quadrant System

Measured Directions Expressed in Degrees From 0 to 90 with North Reference in the Northern Quadrants and South Reference in the Southern Quadrants

1

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2018/2019 Session Final Examination

$\frac{1}{2}$

b. Sketch and label a typical deviated well vertical plan

3

c. A vertical well AKZ21 was being drilled in Agbassa oil field to hit Formation Batza at 15,600 feet from the earth surface. The land rig rotary table elevation is 100 feet. The surface location is N72000 and E61000. The drill string got stuck at 9,000 feet and all efforts to get it freed failed and the drill string was backed-off 8000 ft and cement plug was set over the rest of the drill string to 4,500 ft from rotary table bottom. A side track was thereafter planned to kick-off from top of the cement plug to hit the reservoir at N21000 and E18000 location

i. Design and draw to scale the horizontal plan for this side track well

A. Horizontal section and direction N

$$\square N/S = 72,000 - 21,000 = 51,000 \frac{1}{2}$$

0

$\frac{1}{2}$

$$\square E/W = 61,000 - 18,000 = 43,000$$

7 W

Surface Location

E

$\Delta E / W$

$$A = \tan^{-1} \frac{43,000}{51,000} = 40.10^\circ$$

$\Delta N / S$ 51,000

43,000

$$= \tan^{-1} \frac{43,000}{51,000} = 40.1355^\circ$$

51,000 Target Location

$$= 40.1355^\circ S$$

6

ii. What is the direction of your side track well in quadrant notation?

Direction in Quadrant = S 40.10 W 1

iii. Design and draw the vertical section of the side track well (NOT TO SCALE). Kick-off will be initiated on top of the plug and angle will be built to maximum of 400 at

2

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2018/2019 Session Final Examination

vertical depth of 7,800ft. Thereafter angle will be maintained in a slant section to target.

KOP –

4,500 2

$\Delta N = 51,000$ ft; $\Delta E = 43,000$ ft

ft, I_{max}

$D = \sqrt{\quad}$

$(\Delta N)^2 + (\Delta E)^2 = 51000^2 + 43000^2 = D^2 = \sqrt{4,450,000} = 66,708.32$ ft

VD build

3,300f 9HUW LFDO

6HFW LRQ

6XUIDFH

t,

4,500

3,300 . LFN-RII

7,800

40

0 (QGRI. LFN -2II

15,600 7DUJ HW

9HUW

LFDO' HSW

K 7DUJ HW

66,708

5

4,500

QUESTIONS 2 1

a. With basic definitions ONLY, differentiate between drilling methods and drilling technologies

2

3

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2018/2019 Session Final Examination

i. Drilling Methods consist of the body of techniques and equipment employed to break rock into fragments or core in drilling (constructing or making hole) in the subsurface rock in the oil and gas industry while

ii. Drilling Technology involves the application of equipment, tools and drilling methods in the construction of a well.

b. What is abrasion drilling method?

i. Drilling method in which the rock is worn, gorged, peeled and chipped off by rotating action upon the rock of a shoe or bit armed with some harder material buttons or teeth.

c. Mention and describe the main drilling technology that uses abrasion method of drilling in the Nigerian oil and gas industry

i. The Rotary Drilling Technology

ii. Upon rotation of the bit while drilling fluid is being circulated and weight being put on the bit, the bit teeth on the cones cut the formation into pieces called cuttings. These cuttings are simultaneously swept and carried from the wellbore bottom up the annulus to the surface.

d. Define the core drilling technology, describe the process and state three main applications of core samples in the oil and gas industry

Definition

This is Drilling Technology, involving the application of a rotating hollow centered drill string composed of a special core bit to drill out formation sample.

Core Drilling Technology Description

The process involved a hollow drill, called a core barrel, that is attached to the end of the drillstring and as it drills down, it cuts formation to advance the bore hole while a core of rock forms inside the barrel. On pulling out of the hole, the core sample is obtained.

A side core can also be obtained lowering a coring gun

hollow-point steel bullets mounted along its sides by

4

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short steel cables to the interval of interest and the

bullets are fired individually and the core will be retrieved.

Advantages of this technique are low cost and the ability to sample the formation after it has been drilled.

Application of Core Samples

1. Provision of sample of the formation for the keeps
2. Use of core sample for the determination of formation rock properties 3
3. Use of core sample for the determination of hydrocarbon-in-place
4. Use of core sample for experiments to determine productivity potential
5. Use of core sample for experiments to determine productivity maintenance

QUESTIONS 3 1

a. Define flow regime and aided with appropriate diagrams and sketches, describe the behaviour, characteristic and features of the laminar flow regimes employed in the drilling industry 3

The manner in which drilling fluid behaves or its “character” while flowing in a conduit

It refers to the relation between the different layers to each other in moving fluid

It mostly depends on the relation between the Fluid Velocity & the Critical Velocity

It is the behavior of fluid during flow in conduit.

2

□ In Laminar flow, Fluid layers are parallel to each other in an orderly fashion. 3

□ Flow occurs at low to moderate shear rates when friction between the fluid and the channel walls is at its lowest.

□ It is the flow at the onset of the very low pump rate with low noise, to the rate at which transition to

turbulent begins

5

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2018/2019 Session Final Examination

b. Mention the major area of application of laminar flow in drilling, and give reasons for its application in this area.

n It is a typical flow in the annulus of most wells

n In the annulus, the drilling fluid is required to suspend 3

with it's the rheological properties the cuttings whose density is expected to be higher. This can only be done when the flow is slow, tranquil and laminar

c. A 30 m horizontal pipeline of 7½ cm diameter flows crude oil with a density of 0.93 ton/m³ at 1000 m³/day. The input and discharge pressures are 28 ton/m² and 22.5 ton/m² respectively. If this flow is considered to be laminar, what is the viscosity of the oil

$$Q = 1000 \text{ m}^3/\text{day} = 0.011574 \text{ m}^3/\text{sec}$$

1

$$24 \times 60 \times 60$$

$$D = 7.5/100 = 0.075$$

$$V = 4xQ = 4 \times 0.011574 = 2.62 \text{ m/sec}^2$$

$$\Delta D^2$$

$$\Delta 0.075^2$$

$$\Delta P = 28 - 22.5 = 5.5 \text{ ton/m}^2$$

From

$$= 5.5 \times 0.075^2 =$$

$$0.00012307 = 1.12307 \times 10^{-4} \text{ tsec/m}^2$$

$$32 \times 2.62 \times 30$$

d. Give with reasons the flow regime of the cement slurry during casing and cementing process 1

1. Flow regime is Turbulent

2. Reasons:

a. Cement is pumped at very high pressure which gives rise to turbulent flow 2

b. Cement contains particles of varying sizes that cannot be in layers while flowing

QUESTIONS 4

a. State and define four main types of contracts used in the service industry of the oil and gas industry 1

i. Drilling contracts

This is an agreement between the operator and the service company to drill and complete a well

1

ii. Well services contracts

6

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This is an agreement between the operator and the service company to render certain specialized services within the drilling 1 and completion processes

iii. Seismic acquisition agreements

This is an agreement between government or land owner and seismic company to carry out geophysical and seismic 1 surveys

This is also an agreement between the Seismic company or owner of seismic data and operator for the acquisition of seismic data or license to use seismic data

iv. Master Service Contract

These are “Master Service Agreements” formulated by “big” oil and gas producing companies when they expect an ongoing relationship 1 with a reliable service company. They are generally in their favour. Service companies can also author “Master Service Agreement” for small national oil companies that conduct their own operations

b. List the five(5) main components of drilling service contract

i. The works cope/specifications 2.5

ii. Payment/Fees Terms

iii. Liability or Risk Allocation Clauses

iv. Waiver or Consequential Damages clauses

v. Force Majeure Terms

c. What does “Bury your own Dead” stand for and mention two areas/activities it does NOT apply in the Liability or Risk Allocation Clauses of drilling service contract,

It stands for both operator and contractor separately assuming all risk 2.5

associated with its own equipment and personal, regardless of fault, and agrees to indemnify the other party for claims arising out of injuries suffered to their respective employees and properties.

It does not apply to;

- i. Gross negligence or breach of safety acts 2
- ii. Third parties and their properties

d. What is Seismic Data Acquisition Agreement? Briefly describe the aspects that deal with data acquisition, analysis and interpretation, data ownership and usage 2

i. This is the agreement that gives rights for seismic data acquisition and spells out the process of data sale or license, extent of data ownership or usage.

ii. Acquisition:

Land owner normally gives right to seismic company for 2 seismic investigation(data acquisition) in terms

iii. Data Analysis and Interpretation:

7

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Seismic Agreement are entered into to analyze and process data by whom and with what method and also spelling out the 2

type of data(2-D,3-D, 4-D, etc.)

iv. Data Ownership and Usage:

Seismic Agreement are entered to purchase or use seismic data. The extent of usage is explicitly spelt out and this is 2

quite restrictive with respect to disclosure of seismic data to third parties, the extent of liability of contracted geophysicist to the seismic company, who processes the seismic data

QUESTIONS 5

a. Define burst pressure rating of a casing and mention the 1 pressure in the drilling environment used in its design during casing design and state one reason for using this pressure.

i. Burst is the internal yield pressure or internal 3 pressure resistance of the casing with its couplings

ii. It is the formation pressure - P_f

iii. Because it is based on maximum internal pressure resulting from a kick during drilling of the next

hole section

b. The figure below is the pore pressure and fracture pressure gradient plot of lithology where the Sand Aminutu was discovered with attractive reserves at 14,000 ft

i. On the figure, give a preliminary casing design to drill to the reservoir, sketching on the figure the number of casings and their respective depths

Figure I

4

4500 ft

3

9500 ft

1200 ft

14000 ft

17.5 18

2

8

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c. Provide the detailed design of the 7 inches production casing using ONLY the Burst and Collapse design pressure criteria with the appropriate API Design Factors; and select the long thread coupling (LTC) 7 inches production casing giving its grade and nominal weight

i. Calculate Burst Production casing

Pore pressure at 11,500ft = 10.5 ppg; 2

$$\square P_{burst} = 0.052 \times 10.5 \times 11,500 = 6,272$$

psi

ii. Calculate Collapse pressure

Drilling mud weight at 11,500 ft = 10.75 ppg.

$$\square P_{collapse} = 0.052 \times 10.75 \times 11,500 = 2$$

6,428.5

psi

iii. With $P_{burst} = 6,272$ psi; collapse pressure of 6,428.5

and LTC coupling, from the table

HCL-80 casing is the only that satisfies all design

conditions. 3

Casing Grade ` - HCL -80
 Casing size - 7"
 Casing nominal weight - 26 lb/ft
 Burst rating - 7,200 psi
 Collapse rating - 7,800 psi

QUESTIONS 6

a. Mention and briefly describe the two keys to planning a cost-effective well

i. Avoiding problems:

This is involves 2

i. The mud properties and to good drilling practices (but not by being overly cautious!)

ii. Paying that extra bit to get the most suitable mud system as much as possible.

iii. Avoiding unplanned short cuts that often lead to unnecessary problems

ii. Maximizing progress

i. By optimizing bit/BHA selection, optimum drilling parameters, good forward planning, and good drilling practices. 2

To achieve the above you need

i. Practical Knowledge

ii. Exposure(Experience)

These help drilling engineers to make better and informed decisions themselves rather than subjecting themselves to unduly influenced by the people or environment around them.

9

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b. Define and briefly describe the process of mud logging

i. Mud logging is the process of gathering information on the formation through the observation of the cuttings, fluids and content of the mud coming out of the well during drilling process. 3

ii. Mud loggers take sample of the cuttings and cleaned them up to give to petro physicist who identify the rock ascertain its properties

iii. The mud return fluid is also analyzed to for hydrocarbon content

c. Mention and describe the basic components of the drilling fluid programme in drilling a well

i. Optimum Mud Weight

This is the density of the drilling fluid with which it exert 1 pressure to control the drilling process

ii. Type of Mud

The mud type defines the fluid type and the base fluid

This has application in drilling through specific lithology 1 and the method of drilling (UNB or OVB)

iii. Properties of Mud

The properties of the fluid that enhance the drilling process include

1. Rheological properties such as apparent viscosity, plastic viscosity, yield, etc. These help in cutting 1 suspension and cariability.

2. Flow and pressure. These help in cuttings cariability and provide low regime suitable for the drilling. They also provide fluid hydraulic and impact horse power needed for hole cleaning and ROP increase

d. In planning a directional well, provide a table below with the various sections of the well and state, the appropriate services, tools and 9 technologies that will be applied to drill the sections

Well Section Required Services Technology Tools

Name

1. Vertical to □ Drilling □ Rotary □ Rotary drilling rig

Kick-off Services drilling □ Mud mixing and

Point □ Mud □ Mud treatment equipment

engineering engineering □ Mud logging

tools/equipment

□ Formation □ Mud logging

evaluation

2. Kick-off □ Drilling □ Rotary □ Rotary drilling rig

10

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2018/2019 Session Final Examination

section services drilling □ Mud mixing and

□ Mud □ Mud treatment equipment

engineering engineering □ Deviation tools(Bent
 □ Directional □ Deviation, subs, etc.), mud
 Drilling steering and motors, jetting bits, etc
 sliding
 technology □ MWD tools
 □ Surveying □ Mud logging
 □ Formation □ MWD tools/equipment
 evaluation □ Mud logging
 □ Drilling □ Rotary □ Rotary drilling rig
 3. Slant(Tang services drilling □ Mud mixing and
 ent) □ Mud □ Mud treatment equipment
 section to engineering engineering □ Deviation tools(Bent
 Pay zone □ Directional □ Deviation, subs, etc.), mud
 Drilling steering and motors, jetting bits, etc
 sliding
 technology □ MWD tools
 □ Surveying □ Mud logging
 □ Formation □ MWD tools/equipment
 evaluation □ Mud logging □ LWD tools
 □ LWD

11

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 Department of Petroleum Engineering
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 2018/2019 Session Final Examination

12

Paper 2 of 5 — PEE 521 ADVANCED DRILLING ENGINEERING 2017/2018

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Department of Petroleum Engineering

PNG 521 – Advance Drilling Engineering

2017/2018 Session Final Examination

Mat. No. _____ Date: _____

Basic Instructions

1. Duration: 3 hours
2. Answer Question 1 and Four(4) Others
3. Some relevant charts and graph sheets are provided

QUESTIONS 1

The figure below is the pore pressure and fracture pressure gradient plot of formations where the Sand Taminutu was discovered with attractive reserves at 11500 ft

- a. On the figure, give a preliminary casing design to drill to the 2 reservoir, sketching on the figure the casings and their respective depths.

1

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2017/2018 Session Final Examination

- b. Calculate for the production casing, the Burst design pressure and Collapse design pressure using API Design Factors.

- i. Calculate Burst Production casing

Pore pressure at 11,500ft = 10.5 ppg;

$$\square P_{burst} = 0.052 \times 10.5 \times 11,500 = 6,272$$

psi

- ii. Calculate Collapse pressure

Drilling mud weight at 11,500 ft = 10.75 ppg.

$$\square P_{collapse} = 0.052 \times 10.75 \times 11,500 = 6,428.5$$

psi

- c. Taking into consideration ONLY both the Burst and Collapse pressures, design and select the long thread coupling (LTC) 7 inches production casing giving its grade and nominal weight.

i. With $P_{burst} = 6,272$ psi; collapse pressure of 6,428.5 and LTC coupling, from the table HCL-80 casing is the only that satisfies all design conditions.

Casing Grade - HCL -80

Casing size - 7"

Casing nominal weight - 26 lb/ft

Burst rating - 7,200 psi

Collapse rating - 7,800 psi

d. What will be mud pressure with which you will drill at 6,500 ft with an under-balance of 0.55 ppg?

Mud weight at 6,500 ft = 10.3 ppg. With an under-balance of 0.55 ppg, drilling mud weight will $10.3 - 0.55 = 9.75$ ppg.

$$P_{mud} = 0.052 \times 9.75 \times 6,500 = 3,295.5 \text{ psi}$$

2

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2017/2018 Session Final Examination

QUESTIONS 2

a. With basic definitions ONLY, differentiate between drilling methods and drilling technologies 1

i. Drilling Methods consist of the body of techniques and equipment employed to break rock into fragments or core in drilling (constructing or making hole) in the subsurface rock in the oil and gas industry while

ii. Drilling Technology involves the application of -2 equipment, tools and drilling methods in the construction of a well.

b. What is abrasion drilling method?

i. Drilling method in which the rock is worn, gorged, peeled and chipped off by rotating action upon the rock of a shoe or bit armed with some harder material buttons or teeth.

c. How does the roller-cone bit drill in rotary drilling technology?

i. Upon rotation of the bit while drilling fluid is being circulated and weight being put on the bit, the bit teeth on the cones cut the formation into pieces called

cuttings. These cuttings are simultaneously swept and carried from the wellbore bottom up the annulus to the surface.

d. Briefly describe the percussion drilling technology and give five (5) advantages of drilling with it.

i. This involves the application of other equipment, tools and percussion drilling method to pound the rock into fragments small enough to be readily removed from the hole by the circulating drilling fluid.

ii. ADVANTAGES

i. Improve hole geometry,

ii. Increased ROP and fastest drilling in hard formation

iii. Reduces drillstring stresses,

iv. Generates better and larger cuttings which yields a better representation for geological study

v. Lower drilling cost per foot substantially

vi. Most efficient drilling method in hard formations – generally 2-3 times higher than conventional drilling method and 7.3 times higher in medium-hard granite

vii. Because it involves less contact time with rock – only 1 or 2 percent of total operational drilling

3

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2017/2018 Session Final Examination

time – bit has less abrasion and hence a longer bit life.

viii. Less use of other bottom hole assembly – saving equipment rental cost

ix. Less hole deviation and easier control of deviation problem for straight hole drilling.

e. Define the rotary percussion drilling technology and describe how it drills

i. This is drilling technology that simultaneously implores both abrasion(rotary) and percussion drilling methods to drill the formation.

ii. The rotary percussion uses a mud motor mechanism to offer rotation while a percussion mechanism-commonly

called the hammer - is located directly behind the drill bit reciprocating piston-driven "hammer" to energetically drive a heavy drill bit into the rock.

- iii. The bit face remains in constant contact with the wellbore bottom while the piston behind the bit face impacts the face repeatedly fracturing the formation.
- iv. Drilling fluid (mud, water or air) at up to 180 bar delivery pressure is used to power the impact mechanism of the hammer at a high frequency rate.
- v. As the fluid then leaves the hammer, it loses the pressure and keeps a low flush velocity which is still adequate to bring any cuttings to the surface and clean the bore
- vi. This percussive action increases the instantaneous weight on bit (WOB), even though the mean WOB may be kept at or below typical levels for drilling
- vii. This action results in increased penetration rate of the drill bit, longer drill bit life due to less WOB application, and unplanned borehole deviation may be decreased.

QUESTIONS 3

a. Define drilling service contract

The Drilling Service Contract is an agreement between the 1 operator and the service company to drill and complete a well OR to render certain specialized services within the drilling and completion processes.

b. State and define four main types of contracts used in the service industry of the oil and gas industry

i. Drilling Contracts

4

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2017/2018 Session Final Examination

This is the contract agreement between the operator and the service company to drill and complete a well.

The nature, major parts and operation are dealt with in subsequent sections.

ii. Seismic Acquisition Agreements

These are agreements/licences/contracts between

Seismic companies and government or land owner to carry out geophysical and seismic surveys.

iii. Well services contracts

These are contracts between service companies and operator to carry out specific service(s) in or during the drilling of the well for the operator

iv. Master Service Contract

These are “Master Service Agreements” formulated by “big” oil and gas producing companies or service companies when they expect an ongoing relationship with a reliable service company Or the oil company

c. List and give a brief explanation of the five(5) main components of drilling service contract

i. The works cope/specifications

This clause specifies the

n Wellbore diameter of each section of the well

□ Depth of each section of the well

□ The drilling muds to be used in each section

□ The deviational and directional requirement of each section

□ The equipment, technology and services to be employed in each section.

ii. Payment/Fees Terms

A clause that specifies the rate (cost) for the service and payment methods. It can be per day, per foot, Footage rate, Day work rate, Turnkey rate, Combination and Alliance.

Other fees include Standby rates, Mobilization rates, Commissioning

Rates and Decommissioning rates

iii. Liability or Risk Allocation Clauses

A clause that allocates liability or risk to each of the partners in the drilling contract. Insurance is normally

5

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2017/2018 Session Final Examination

required to cover the cost of these liability and risk

allocations.

iv. Waiver or Consequential Damages clauses

A clause that absolves a party from the consequential damages arising from actions of the other(s) that may affect him or the other(s).

v. Force Majeure Terms

A clause that takes care of an unexpected event that crucially affects somebody's ability to do something and can be used in law as an excuse for not having carried out the terms of an agreement (formal)

d. In a tabular format define footage rate and day work rate payment terms of the drilling contract, give their respective advantages, disadvantages and areas of application

Payment Definition	Advantage	Disadvantage	Areas of Application
Footage Rate	Operator pays for service higher nothing is on total rendered. footage known	number of Increases sometimes about the feet drilled Rate of leads to less area drilling payment to Service safety company. measures.	Put operator It also on his toe decreases and brings quality the best out of him
Day Rate	per Easy Payment Where a lot Work day in determinati could be is known in work and on of work made for the area pay is to be paid days that do made on for. not yield the total Increases any footage. number of operator's Makes days supervision service to ensure company work is done lazy and for the day play pranks		

n

Footage Rate is per Operator Desire for Where very rate foot and pays exactly achieving little or payment for service higher nothing is is on total rendered. footage known number of Increases sometimes about the feet drilled Rate of leads to less area drilling payment to Service safety company. measures. Put operator It also on his toe decreases and brings quality the best out of him

Day Rate is per Easy Payment Where a lot Work day in determinati could be is known in work and on of work made for the area pay is to be paid days that do made on for. not yield the total Increases any footage. number of operator's Makes days supervision service to ensure company work is done lazy and for the day play pranks

6

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Department of Petroleum Engineering

PNG 521 – Advance Drilling Engineering

2017/2018 Session Final Examination

QUESTIONS 4

a. In a sequential format, list the processes involved in well design from Data Acquisition and review to Pre-spudding Meeting. 19

- i. Data Acquisition and Review
- ii. Data Analysis
- iii. Well Design Meeting
- iv. Well Design
- v. Circulate the Design for Comment
- vi. Write the Drilling Program
- vii. Circulate the Program for Comment
- viii. Pre-spud Meeting

b. Give brief description of what wells' hole section summaries in drilling and explain how they are used in well design.

i. Hole Section Summaries are summaries created for each section drilled in previous wells and they contain large quantities on the

i. Interval of formation – (2000 – 2900 ft)

ii. Formation

1. Lithology
2. Compressive strength
3. Competence – loose, caving, hard, etc
4. Stickiness, tight hole, drag, etc on pull
5. Problem associated with drilling through

iii. Drilling parameters

1. WOB
2. RPM
3. GPM
4. Torque
5. Tripping condition

iv. BHA

1. Packed (Locked)
2. Build angle
3. Drop angle
4. Mud motor
- v. Drilling fluid

1. WBM
2. KCL
3. OBM
4. Water, etc
- vi. Comments on
 1. Casing/cement jobs,
 2. Over gauge hole from logs, and
 3. Any problems logging or other worthwhile
 comments

7

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 Department of Petroleum Engineering
 PNG 521 – Advance Drilling Engineering
 2017/2018 Session Final Examination

- ii. Uses in Well Design
 - i. Comparisons can be quickly made in the same formations between several wells.
 - ii. The knowledge of the lithologies and problem areas, with notes on compressive strengths (derived from sonic logs) and other relevant information help in designing with
 1. Preventive measures.
 2. Better bit selection
 3. BHA with high Drillability in the section
 4. Better selection of RPM and GPM
 5. Better mud properties
- c. Design (to scale) the horizontal section of the well with surface location at S15000 and E75000 and the target location at N55000 and E15000. What is the direction (in quadrant notation) of the well and its horizontal displacement?

$$\square E/W = 75,000 - 15,000 = 60,000$$

$$\square N/S = 55,000 + 15,000 = 70,000$$

$$HD = \sqrt{N^2 + E^2} = \sqrt{(70000)^2 + (60000)^2} = 92,195.45 \text{ ft}$$

$$A = \tan^{-1} \frac{60,000}{70,000} = \tan^{-1} 0.857142 = 40.6^\circ = N40.6^\circ W$$

8

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 Department of Petroleum Engineering
 PNG 521 – Advance Drilling Engineering
 2017/2018 Session Final Examination

QUESTIONS 5

b. Give reasons/justifications for the liability or risk allocation clauses in drilling contracts

□ Drilling is a high risky process associated with huge cost.

□ Problem arising from drilling are associated with substantial damage to property and in personal injuries.

□ Problem may also occur at thousands of feet down hole where direct observation is difficult, the nature and cause may be difficult to ascertain and appropriated.

□ This clause endeavour to eliminate uncertainty with regard to responsibility for consequences relating to problems that arise by ensuring that responsibilities for damages are clearly stated.

□ The goal is to ensure that when various scenarios come to pass the partners responsible for damages are well enunciated, and to make sure that there are neither gaps in insurance coverage, nor double coverage.

□ This also avoids protracted litigation

c. Briefly state the purpose of the indemnification clause

The purpose is to exempt (indemnify) parties from penalties and liabilities.

d. Give detailed explanation of what “Knock for Knock” stands for, how it works, the condition under which it does not apply and state reason for its non-application

Both operator and contractor separately assumes all risk associated with its own equipment and personnel, regardless of

9

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Department of Petroleum Engineering

PNG 521 – Advance Drilling Engineering

2017/2018 Session Final Examination

fault, and agrees to indemnify the other party for claims arising out of injuries suffered to their respective employees and properties.

No waiver or indemnity clause will be allowed by law for damages arising from acts of “gross negligence” or breach of safety rules and regulations from any party

e. State and briefly explain three(3) contractor’s indemnities and three(3) operator’s indemnities

Contractors’ Indemnities

□ The infringement of patents or other industrial property rights arising out of or in connection with the Work or Company's use of the Work, except where such infringement necessarily arises from the use of Company Provided Items or Company's instructions

□ The loss or damage suffered by a Third Party in connection with the Work, to the extent that any such loss or damage is caused by the negligence or breach of duty (whether statutory or otherwise) of Contractor Group.

□ The pollution occurring on the premises of Contractor Group or originating from the property and equipment of Contractor Group located above the rotary table, arising from or relating to the performance of the Work.

Operators' Indemnities

□ The loss of or damage to any well or hole

□ The blowout, fire, explosion, cratering or any uncontrolled well condition (including the cost to control a wild well and the removal of debris)

□ The damage to any reservoir, geological formation or underground strata or the loss of oil or gas therefrom

□ The use of radioactive tools in relation to the Work or any contamination resulting therefrom (including retrieval and/or containment and clean up) - This applies regardless of any form of liability, whether strict or by negligence, in whatever form, on the part of Contractor group.

□ The loss or damage suffered by a Third Party in connection with the Work, to the extent that any such loss or damage is not caused by the negligence or breach of duty (whether statutory or otherwise) of Company.

10

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Department of Petroleum Engineering
PNG 521 – Advance Drilling Engineering
2017/2018 Session Final Examination

QUESTIONS 6

A vertical well with N72456.4 and E21995.6 was drilled to top of reservoir sand Bemba at 12500 ft when it blew out. To kill the well,

a relief well was sited at N61986.8 and E18794.7 The relief well was planned to kick-off at 750 ft and build angle to 450 and thereafter maintain inclination to intersect the vertical well at the top of Bemba. We only have a 30 bent sub available at site 19

a. What is the total vertical depth of the relief well?

i. Horizontal Displacement

$$HD = \sqrt{N^2 + E^2} = \sqrt{(72456.4 - 61986.8)^2 + (21995.6 - 18794.7)^2}$$

$$= \sqrt{(10,469.6)^2 + (3200.9)^2} = \sqrt{109,612,524.16 + 10,245,760.81}$$

$$= \sqrt{119,858,284.97} = 10,947.981 \text{ ft}$$

ii. Radius of Curvature $R = 180 \times CL = 180 \times 100 = 1909.9 \text{ ft}$

iii. $\pi \times BUR = 3.142 \times 3$

iv. Build TVD $= R \times \sin I = 1909.9 \times \sin 45 = 1350.5 \text{ ft}$

v. Build Section HD $= R \times (1 - \cos I) = 1909.9(1 - \cos 45) = 559.4 \text{ ft}$

vi. Tangent Section TVD $= THD - BHD = 10947.981 - 559.4 = 10,388.581 \text{ ft}$

vii. $\tan I = \tan 45$

viii. Total TVD $= 750 + 1,350.5 + 10,388.581 = 12,489.081 \text{ ft}$

b. What is the total measured depth of the relief well?

i. Length of Build Section (BUAL)

$$= 45 \times 100 = 45 \times 3.142 \times 1909.9 = 1,500 \text{ ft}$$

$$3180$$

ii. Length of Tangent Section

$$= \text{Tangent Section TVD} = 10,388.581 = 14,691.672 \text{ ft}$$

$\cos I = \cos 45$

iii. Total Measured Depth $= 750 + 1500 + 14,691.672 = 16,941.672 \text{ ft}$

c. Sketch the vertical section of the relief well.

Casing Properties Table

11

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Department of Petroleum Engineering

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2017/2018 Session Final Examination

12

Paper 3 of 5 — ADVANCED DRILLING ENGINEERING PEE 521 2017

Level 500 • Second Semester • PastQuestion

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PNG 521 – Advance Drilling Engineering

2016/2017 Session Final Examination

Mat. No. _____

Date: _____ Duration:

3 hrs

Basic Instructions: $n = 3.32 \log (\frac{\mu}{600} / \frac{\mu}{300})$

a. Answer Question No. 1 and Four(4) others

b. You MUST Return this Question Paper with Your

Answer Scripts

c. Some basic equations are provided below but you are free to use any equation of your choice

[()]

n

$V = (\rho)$

$K 1$

$5.82 \times 10^4 2 - n$

$1.6 3 n + 1 2 - n$

$d 4n NRe =$

$928 \mu d$

1. 2. μ

n

$\mu_{mud} V_{2p} - n 0.0416 IDDP 2. 27 \times 10^{-7} \rho 0 .8 V 1. 8 PV 0. 2 L$

$NRe p = 89 , 100 PDS =$

$Kp 1 d1 . 2$

$3+$

np

3. 4.

Q

$V p = 0. 408$

5. ID2DP 6. $\mu_{Pbit} = (3013 \times \mu_{grad} \times Q2) /$

$(3 \times Dn2) 2$

Moody

7.

$\Delta P f = 1.1486 \times 10^{-9} f .$

L

$D5$

. ρQ^2

8.

$v_n = \sqrt{58.26 \Delta P_{bit}}$

ρ

$[(\)](\)]$

n_n

9.

$PDS =$

$1.6 V_{3n+1}$

d_{4n}

KL

$300 d$

$V = ($

$10.. K_1$

$3.878 \times 10^4 2 - n$

$\rho ($

$2.4 2 n + 1 2 - n$

$dH - d p_{3n}$

Q

$V =$

$11. 12$

$(\)$

2^2

$2.44 d - d p$

$P_{s,opt} = [1 / (n + 1)] P_t h$

$7.7 \times 10^{-5} \rho_0 .8 V_{1.8} P V_{0.2} L$

$[\]]$

n

$12. P = 2$

$(dH - d p)(dH + d p)$

$1.8 P = ($

$2.4 V_{2n+1}$

$dH - d p_{3n}$

$14. KL$

$300 (dH - d p)$

$Q V^2 \rho_{mud}$

$V = \Delta P f = f L$

$15. 2.44 . d_2 16.$

$25.81 De_{ff}$

Fanning

L

$$\Delta P_f = 1.8608 \times 10^{-9} f \cdot 5$$

$$\cdot \rho Q^2$$

$$17.1 \text{ bbl} = 42 \text{ gals } 18. D$$

1

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2016/2017 Session Final Examination

$$\tau_2 \theta 300 \text{ dp } f \mu v$$

$$K = (1 + 0.067 n) =$$

$$\gamma 2n 511 n \text{ dL } 1000(d_2 - d_1)^2$$

19. 20.

$$Q =$$

bbl/day

$$L = \text{ft}$$

$$D = \text{ft}$$

Question 1 $\square = \text{lb/ft}^3$

Figure I below

 $\square P_f$ = illustrates

psi the lithology, pore pressure, formation pressure

margin, kick margin and fracture pressure profiles of formations on top of reservoir ADf-K27 at 14,000 ft. The surface location of the directional well to drill to the reservoir is at N72000 and E61000 and the well will be kicked-off at 1000 ft, angle will be built to maximum of 450 at vertical depth of 3,000ft.

Thereafter angle will be maintained in a slant section to target.

a. If the target location is at N21000 and E18000, graphically design the horizontal plan to drill this well stating its direction in the quadrant format.

N

1. A. Horizontal section and direction

$$\square N = 72,000 - 21,000 = 51,000$$

$$\square E = 61,000 - 18,000 = 43,000$$

Surface Location

7 W E

$$-1 \Delta E / W$$

$$A = \tan 1$$

$$\Delta N / S 40.10$$

$$43,000 51,000 3$$

$$-1$$

$$= \tan 0.843137 43,000$$

$$51,000$$

Target Location

$$= 40.13550$$

S

6

Dire 1

ction

in

1.B Sketch the vertical section of the well(not to scale)

Qua

drankOP -

 $t = 1000$ $\Delta N = 51,000$ ft; $\Delta E = 43,000$ ft

S ft, I max 1

40. = 452

0

 $D = 10(W$ ΔNVD $\sqrt{\sqrt{}}$ $/ S)_{\text{build}} + (\Delta E / W)^2 = 51000^2 + 43000^2 = D = 4,450,000,000 = 66,708.32$ ft

Vertical

3,000f

t, Section

. LFN-RII

(QGRI. LFN -2II 4

2

7DUJ HW

9HUW

LFDO' HSW

K

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1.C In a tabular format and for each lithology section, provide the BHA and Drilling fluid designed programmes you will need to drill this well successfully.

3 6

1.D Design a

Figure

preliminary casing programme for this well

4500 ft 4

9500 ft

1200 ft

14000 ft

17.5 18

1.E, With casing made of steel of 65.4 b/ft³, calculate the Burst design pressure and Collapse design pressure for the production casing using API Design factors
API Collapse design factor = 1.125 and Burst design factor = 1.1

i. Burst design Pressure

3 2

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Pore pressure gradient at 14,000 = 17.5 ppg; □ Pore pressure = $0.052 \times 17.5 \times 14000 = 12,740$ psi

Design Burst pressure = $12,740 \times 1.1 = 14,014$ psi

ii. Collapse design pressure

Drilling mud weight at 14,000 = 18 ppg; □ Mud hydrostatic pressure = $2 \times 0.052 \times 18 \times 14000 = 13,104$ psi

Design collapse pressure = $13,104 \times 1.125 = 14,742$ psi

Question 2

a. Define bottom-hole Assembly (BHA) and state two(2) of its main functions

This is a connection of combination of different drilling tools on top of the drill bit and below the regular drill pipes 1

Functions:

i. To add weigh on the bit

ii. To control drilling 1

iii. To control wellbore direction

b. In a tabular format, give the three(3) main BHA's employed in drilling a well with their respective functions and formations of application

N Type of BHA Function Formation of

o. application

1. Jetting BHA □ Kick-off wells Soft formation

□ Drop wellbore angle

□ Build Wellbore

Angle

2. Mud Motor deviation □ Kick-off wells Hard formations

BHA □ Drop wellbore angle

□ Build Wellbore 9

Angle

□ Maintain wellbore

angle

3. Packed BHA □ Drill vertical well □ Soft formations

□ Maintain wellbore □ Hard
angle in deviated formations
wells slant section

4. Drop BHA □ Drop wellbore angle □ Soft formations
□ Hard
formations

5. Angle Build BHA Build Wellbore angle □ Hard
formations

c. Drilling a hard shale formation section in the well
requires maximum of 100 psi of bit area. The available
bottom hole assembly tools are as follow:

5 pieces of 10 ½" drill collars with internal bore of 3 ½"
and 261.3 lb/ft- 30 ft each. 20 pieces of 5" Heavy weight
drill pipe – 63.78 lb/ft-29 ft each; 9" Jars – 110 lb/ft -29 ft;
10 ¼ " stab – 68.2 lb/ft- 4.5 ft each; 2.5 ft bit sub – 8.5

4

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lb/ft; 2.5 ft -10 ½" x 9" cross-over sub – 25 lb/ft; 10 ½" x 5" cross-
over sub – 8.5 lb/ft; 1½ft 10 ½" bit – 12 lb.

i. Design a packed bottom-hole assembly (BHA) to drill this
section of the well and give its sketch.

5

ii. What is the total length and total weight of your BHA?

BHA Member Weight(lb)

Length (ft)

i. 10 ½" DC – $5 \times 30 \times 261.3 = 39,195$ 150

ii. 10 ½" bit – $1 \times 12 \text{ lb} = 12$ 1½

iii. 9" Jars – $29 \times 110 = 3,190$ 29

iv. 10 ¼ " stab – $5 \times 68.2 \times 4.5 \text{ ft} = 1,534.6$ 22.5

v. bit sub – $2.5 \times 8.5 = 21.25$ 2.5

vi. 10 ½" x 9" cross-over sub = 25 2.5

vii. 10 ½" x 5" cross-over sub = 8.5 2.5

viii. 5" HWDP – $20 \times 29 \times 63.78 = 36,992.4$

580 2

ix. Total = $77,788.75 \text{ lb} = 790.5$

ft

d. While drilling this formation, determine the required weight-on
Bit (WOB) and the tension on the drill string while bit is on

bottom at 14,000 ft.

$$1. \text{ Bit Area} = \frac{\pi D^2}{4} = \frac{3.142 \times 10^2}{4} = 86.59 \text{ in}^2$$

4 4 2

$$2. \text{ Required weight on bit} = 100 \times 86.59 = 8,659 \text{ lb}$$

$$3. \text{ Tension on string} = 77,788.75 - 8,659 = 69,129$$

lb

Question 3

a. In question 1 above, using a derrick safety factor of 1.85 what will be your designed derrick capacity to drill to top of reservoir ADf-K27 at 14,000 ft? 4 ½" drill pipe of 29 ft average length and 25.83 lb/ft with 870 ft 7" drill collars of 240 lb/ft will be used to drill the well. Neglect stabilizers and other BHA members. Steel density is 65.5 ppg

$$\text{Length of Drill pipes} = \text{Well depth} - \text{drill collar length} = 14,000 - 870 = 13,130 \text{ ft}$$

$$\text{Number of drill pipes} = 13130 / 29 = 452.7 = 453$$

$$\text{Weight of 4 ½" Drill pipe} = 25.83 \times 29 \times 453 = 339,328.71 \text{ lb}$$

$$\text{Weight of 870ft DC} = 870 \times 240 = 208,800$$

$$\text{Total weight of drill string} = 548,128.71 \text{ lb}$$

$$\text{Mud weight at 14,000 ft} = 18 \text{ ppg}$$

1

5

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i. Calculate Buoyancy Force

$$\text{BF} = 1 - (\rho_{\text{mud}} / 65.5)$$

1

$$\text{BF} = 1 - (18 / 65.5) = 0.725$$

ii. Calculate Buoyed Weight

$$\text{BW} = \text{BF} \times (\text{in-air weight})$$

1 lb

$$\text{BW} = 0.725 \times 548,128.71 \text{ lb} = 39,739.22$$

iii. Applying a derrick safety factor of 1.85

$$1.85 \times 39,739.22 = 73,517.557 \text{ lb}$$

b. If the consistency index and flow index of the drilling fluid at 9000 ft in well ADf-K27 from question 1 were to be 0.978 and 0.689 respectively, with the BHA in question 2 above, use the SPE Method for Power Law fluid to determine the pressure drop across the bit.

i. Pump pressure = 1500 psi

ii. Annular pressure drop = 150 psi

iii. Flow rate = 1 bbl/min

iv. Uniform bore through drillstring = 3 ½"

Pump rate in gpm = 42 x 1 = 42 gpm

Q 42

V_p = 0.408 2 = 0.408 2

= 1.3988 ≈ 1.4 ft/sec

i.

ID DP 3.5

Now n = 0.689 ≈ 0.7

1 = 2,000

ii.. From Power-law friction factor chart below, at n = 0.7, N_{Rec}

3

0.0027

6

2,000

185,000

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iii.. Calculate N_{Rec}

ρ 1

At 9000ft mud = 13.8 ppg; μ from fig. I

V_p = 2.1 ft/se; K_p = 0.6923; n = 0.281; ID_p = 3.5" 5

[] []

2-n n

μ_{mud} V_p 0.0416 IDDP 13.8 x 1.42 -0.689 0.0416 x 3.5

0.6891

N_{Re p} = 89, 100 = 89, 100 = 185, 181.3

K_p 1 0.978 1

3+ 3+

n_p 0.689

iv.. Determine flow regime

1

□ N_{Rep} □ N_{Rec} and flow is turbulent

v.. Determine friction factor f

From the chart, with n = 0.7 and N_{Rep} = 185,181.3; f =

10.0027

vi.. Calculate

ΔP f

f_p ρ_{mud} V_{2p} 0.0027 x 13.5 x 1.4 2

$$\Delta P_f = L = 9000 = 7.3 \text{ psi}$$

$$25.8 \text{ IDDP} \Delta P = 25.8 \times 3.5$$

vii. Calculate bit

$$\Delta P_{\text{bit}} = P - P_p - P_{\text{pp}} = 1500 - 150 - 7.3 = 1,332.7$$

P_{ann}

c. Determine the appropriate nozzle sizes for the bit to optimum bit and flow performance

$$Q_{\text{Pbit}} = (156 \times W \times Q_2) / (D_{2n} + D_{2n} + D_{2n})^2 \quad D_{2n} = (156 \times W \times Q_2) =$$

$$156 \times 13.8 \times 422 = 942.76$$

$$3 \times 942.76 = 2828.28$$

$$D_n = \sqrt[3]{2828.28} = 14.1 \text{ in} \approx 14 \text{ in}$$

Question 4

7

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2016/2017 Session Final Examination

a. Define and then differentiate between kicks and blow-outs

Kick is any undesirable inflow of formation fluids into the wellbore from the reservoir that normally occurs as a result of a negative pressure differential

1

across the formation

Blow-out is an uncontrolled outburst (sometimes with explosion) of reservoir fluid to the surface

Main difference is that kick is an underground phenomenon while blow-out is a surface phenomenon

b. List the five (5) main causes of kick, explain with appropriate diagrams how each of them causes kick and what can be done to prevent each of them from occurring.

N Kick Diagram How it causes kick Prevention

o. Causes 2

1 Low Drilling Lower mud density Maintain

Fluid Density results in lower PH than higher mud

than that of P_f and P_p in favour of density than

Formation formation that causes formation

Oil or Water

ingress of formation fluid density

fluid into wellbore

2 Abnormal Highly porous Knowledge of

Reservoir formations having full area ahead

Pressure column of saline water and

2 with pressure greater increasing
 than the average mud weight
 formation pressure before drilling
 than the area that formation
 3 Not Drill pipe pulled out of Fill wellbore
 Keeping wellbore during with
 the Hole tripping out have equivalent
 Full on volume and is this mud volume
 Trips 2 equivalent volume is of drillstring
 not filled into the well, members
 it result lower fluid pulled out.
 volume and lower PH
 than P_f and P in
 favour of formation
 that causes ingress of
 formation fluid into
 wellbore
 4 Swabbing Fast and Instantaneous Pull drillstring
 pull of drillstring during at lower rate
 8

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 tripping could result in to wellbore
 vacuum creation at the mud enough
 2 wellbore bottom and time to fill
 this lower PH than P_f wellbore
 and P in favour of bottom
 formation that causes
 ingress of formation
 fluid into wellbore
 5 Lost Highly porous and ϕ Have
 Circulation permeable zone knowledge
 2 unusually “steals” of thieving
 filtrate from mud than zone
 in-hole mud can ahead.
 recuperate. This ϕ Drill with
 results in Lower fluid lower mud
 level in wellbore, lower weight
 PH than P_f and P in ϕ Run
 favour of formation drillstring

that causes ingress of at higher
formation fluid into rate
wellbore □ Use Mud-
Cap Drilling

6

d. Kick Indicators

i. What are kick indicators?

1

Drilling parameters and events that indicates the presence of a kick
downhole

ii. Differentiate between positive and secondary kick indicators

Positive indicators indicate definite presence of a kick while
secondary indicators indicate possible presence of a kick as other 1
drilling parameters and events could have causes these indicators

iii. State and briefly explain how the two(2) main positive
indicators are employed to identify the presence of a kick

i. Increase in Pit Volume $\frac{1}{2}$

$\frac{1}{2}$ off

ii. Increase in Flow Rate or Well Flows with pumps

iii. If any of or all two above occur(s),

1. Drilling is stopped

2. Bit raised from wellbore bottom 2

3. Pumps stopped

4. Watch out for flow

5. If well flows, a kick is in the well

e. Define Pressurized Mud-Cap Drilling(PMD) and explain with
appropriate diagram(s) how it is employed to control drilling in
“thieving” formations

9

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2016/2017 Session Final Examination

Pressurized Mud-Cap Drilling”

involves drilling ahead blind (i.e.,

without returns to the surface) by

pumping different fluid densities

down the drill string and annulus

simultaneously with a full annular 3

fluid column maintained above a

formation that is taking injected

2 from the
fluid and drilled cuttings
drill string.

The annulus fluid

All fluid is lost to the thief zone.

The “mud cap” column is assisted
by surface pressure (made possible
with the use of a stationed Rotating
Control Devices (RCD), maintained
above a formation that is capable of accepting fluid and cuttings.
The mud cap serves as an annular barrier, while the driller uses a lighter,
less damaging and less expensive fluid to drill into the weak zone.

Question 5

a. Define flow regime and give in bullet point format its detailed
application in the determination of frictional pressure drop in the
drillstring 1

This is the manner in which drilling fluid behaves or its “character” while
flowing in drill pipe or conduit.

- Two main flow regimes are employed in the determination of frictional
pressure drop in drill pipe – Laminar and Turbulent flow regimes
- These are determine through fluid velocity and Reynold Number
 - o Fluid velocity and critical velocities are calculated; and if the fluid
velocity is greater than the critical velocity, flow regime is
Turbulent. If not, it is Laminar
 - o The Reynold Number(NR)is also calculated; and if this is greater
than 2100, flow is Turbulent. If not, it is Laminar
- The friction factor(f) is then determined using either Fanning or Moody’s
Charts or equations
 - o For laminar flow $f = 16/NR$ (fanning); $f = 64/NR$ (Moody) 5
 - o For Turbulent Flow, f is determined from the charts mentioned
above with the Reynold Number(NR) and relative roughness(ϵ/D)
- The Rheological Model is then determine using the fluid flow index
 - $n < 1$ a pseudoplastic fluid
 - $n = 1$ a Newtonian fluid
 - $n > 1$ a dilatant fluid

10

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2016/2017 Session Final Examination

- Frictional pressure Drop is calculated with employing the various
equations for the determined flow regime and rheological model

b. A drilling fluid of 11 ppg and 10.8 cp was being used to drill a 9 $\frac{5}{8}$ ".

Flow was initiated with triplex pump at the rate of 85 gpm and 2100 psi pump pressure. A drillstring for the 8700 ft well is made up of 750 ft BHA mainly made up drill collars and stabilizers having average OD of 7 $\frac{1}{2}$ " and ID of 3.5" and 5" drill pipe of the same ID. Average relative roughness of drillstring members is 4.5×10^{-4}

i. Sketch and label the well with the various fluids, drillstring components and physical dimensions

ii.

5

iii. Assuming laminar flow in the annulus, determine the frictional pressure drop in the annulus.

$\rho_m = 11$ ppg, $\mu = 10.8$ cp $D_h = 9\frac{5}{8}$ ". $Q = 85$ gpm, $P_s = 2100$ psi ,
 $H = 8700$ ft $H_{DC} = 750$ ft $H_{DP} = 7950$ ft $OD_{DC} = 7\frac{1}{2}$ ", $OD_{DP} = 5$ ", $ID = 3.5$ ", drillstring relative roughness of 4.5×10^{-4}

DC Annulus ΔP_{ann}

$q = 85$

$v = \frac{q}{1.36} = 0.954$ ft/sec

$12.448 \left(\frac{d_2^2 - d_1^2}{d_2} \right)$

2×2

$2 \times 12.448 \left(\frac{9.625^2 - 7.5^2}{9.625} \right)$

-7.52

2×1

$\mu v = 10.8 \times 0.954$

$\Delta P_{ann} = 2$

$L = 2$

$750 = 1.7113$ psi

$1000 \left(\frac{d_2^2 - d_1^2}{d_2} \right) = 1000 \left(\frac{9.625^2 - 7.5^2}{9.625} \right)$

Drill Pipe Annulus ΔP_{ann}

$q = 85$

$v = \frac{q}{1.36} = 0.513$ ft/sec

$12.448 \left(\frac{d_2^2 - d_1^2}{d_2} \right)$

2×2

$2 \times 12.448 \left(\frac{9.625^2 - 5^2}{9.625} \right)$

11

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$\mu v = 10.8 \times 0.513$

$$\Delta P_{ann} = 2$$

$$L = 7950 = 2.05 \text{ psi}$$

$$1000(d_2 - d_1) = 1000(9.625 - 5)2$$

$$1 \text{ psi}$$

$$\text{Total Annular Pressure drop} = 1.7113 + 2.06 = 3.77$$

iv. Employing Darcy Weisbach equation, determine the pressure drop in the drill pipe

$$q = 85$$

$$v = 2.835 \text{ ft/sec}$$

$$2.448 d^2 = 2.448 \times 3.52$$

$$928 \rho v d = 928 \times 11 \times 2.835 \times 3.5$$

$$N_{Re} = 9,378.6$$

$$\mu = 10.8$$

$$1$$

Since $N_{Re} = 9,378.6 > 2100$, flow is turbulent

$$f = 0.033$$

From Moody Chart, with $N_{Re} = 9,378.6$ and relative roughness $= 4.5 \times 10^{-5}$

$$\rho m = 11 \text{ ppg} = 11/8.33 \times 62.4 = 82.37 \text{ lb/ft}^3; ID = 3.5'' = 3.5/12 =$$

$$0.292 \text{ ft};$$

$$Q = 85 \text{ gpm} = 85/42 \times 1440 = 2,914.286 \text{ bbl/day};$$

$$L = 8700$$

$$\Delta P_f = 1.1486 \times 10^{-9} f D^{5.5} \rho Q^2 = 1.1486 \times 10^{-9} \times 0.033 \times$$

$$82.37 \times 2,914.286^2 = 108,672.87 \text{ psi}$$

$$D = 0.292$$

$$1$$

$$0.033$$

$$2$$

$$9,378.6$$

$$12$$

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PNG 521 – Advance Drilling Engineering

2016/2017 Session Final Examination

Question 6

a. State and define four main types of contracts used in the service industry of the oil and gas industry

i. Drilling contracts

This is an agreement between the operator and the service company to drill and complete a well

ii. Well services contracts

This is an agreement between the operator and the service company to

render certain specialized services within the drilling and completion processes 5

iii. Seismic acquisition agreements

This is an agreement between government or land owner and seismic company to carry out geophysical and seismic surveys

This is also an agreement between the Seismic company or owner of seismic data and operator for the acquisition of seismic data or license to use seismic data

iv. Master Service Contract

These are “Master Service Agreements” formulated by “big” oil and gas producing companies when they expect an ongoing relationship with a reliable service company. They are generally in their favour.

Service companies can also author “Master Service Agreement” for small national oil companies that conduct their own operations

b. In payment clauses of drilling services contract, briefly explain the origin of the “Footage Contract”

Footage drilling contract involves payment by operator per footage drilled by drilling service company.

It started when Col. Blake contracted Mr. Smith to drill for the pool of the oil in Turtlesville PA and payment was for footage drilled by Mr. Smith since nobody knew the depth at which the oil pool will be found

c. Briefly explain the difference between Turnkey and Alliance Contracts

In the Turnkey contract, the drilling contractor is obligated to drill a complete well for a lump sum fixed fee which the operator agrees to pay contractor upon completion of the well. Drilling contractor assumes all cost and responsibilities of the well site. Provides highest risks and rewards.

But in Alliance contract, a lead contractor is employed by operator to subcontract, provides most of the resources of a drilling department, plus areas of specialist expertise and manage all the services needed to drill a well, while the operator remains on site and stays closely involved.

d. What does “Bury your own Dead” stand for and mention two areas/activities it does NOT apply in the Liability or Risk Allocation

Clauses of drilling service contract, 1

It stands for both operator and contractor separately assuming all risk associated with its own equipment and personal, regardless of fault, and agrees to indemnify

13

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PNG 521 – Advance Drilling Engineering

2016/2017 Session Final Examination

the other party for claims arising out of injuries suffered to their respective employees and properties.

It does not apply to;

- i. Gross negligence or breach of safety acts²
- ii. Third parties and their properties

e. List and give a brief explanation of the five(5) main components of drilling service contract

- i. Contract Duration

The length of the contract is commonly determined by the number of wells the operator wants to drill. Could be long or short term

- ii. The Work Scope/Specifications:

This specifies the well each section Wellbore diameter, Depth, drilling muds, deviational and directional requirement, equipment, technology and services

4

- iii. Payment/Fees Terms

The rates to be paid for service rendered by the service contractor and the manner of payment determines the type of service contract

- iv. Liability or Risk Allocation Clauses

This clause assigns responsibility for damages to partners when problems, damages, etc arise. It ensures that responsibilities are well enunciated, and that there are neither gaps in insurance coverage, nor double coverage and eliminates protracted litigation.

- f. With basic definitions ONLY, differentiate between drilling methods and drilling technologies

Drilling Method is a body of techniques and equipment employed to break rock into fragments or core in drilling(constructing or making hole) in the subsurface rock WHILE Drilling Technology is the application of other¹ equipment, tools and drilling methods in the construction of a well.

- g. Briefly describe the various ways drilling methods and technologies are built into one components of the drilling contract you enumerated in 6e above 2

- i. The Work Scope/Specifications:

Generally the drilling method and technology required to drill every section of the well are normally specified in the Work Scope/Specifications. These sections include the kick-off, build, drop sections of a well

14

Paper 4 of 5 — PEE 521 ADVANCED DRILLING ENGINEERING BY CHIEF OGBON

Level 500 • Second Semester • Handout

Federal University of Petroleum Resources Effurun

Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Advance Drilling Engineering

Lesson VII

Drilling Hydraulics

Ogbon Advance Drilling Engineering Slide #2

Lesson VII: Drilling Hydraulics

Introduction to Natural Gas Processing

Sub-Topics

1. Introduction to Drilling Hydraulics

2. Rheological models

□ Bingham Plastic model

□ Power Law Model

□ Herschel-Bulkley (yield-power law [YPL]) Model

3. Applications of Hydraulics in Drilling

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Introduction to Drilling Hydraulics

□ Hydraulics

Introduction to Natural Gas Processing

□ Hydraulics is the study of fluids at rest or in motion, especially with respect to engineering applications.

□ In drilling, hydraulics is centred on the mud system and it serves many purposes and applications including:

□ The Control of sub-surface pressures

□ Providing buoyancy effect to the drill string and casing

□ Minimizing hole erosion due to the mud's washing action during movement

□ Removal of cuttings from the bit and well thereby cleaning the bit and the well.

□ Increasing penetration rate.

□ Application in sizing surface equipment such as pumps,

□ Control of surge pressures created by lowering pipe into the well,

- Minimizing well bore pressure reductions from swabbing when pulling pipe from the well,
- Evaluation of pressure increases in the well bore when circulating the mud,
- Maintaining and controlling the well during kicks.

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Introduction to Drilling Hydraulics

□ Rheology

Introduction to Natural Gas Processing

□ Rheology is the study of how matter flows and change shape.

□ It is used in the understanding and analysis of:

□ Fluid flow velocity profiles

□ Fluid viscosity

□ Marsh funnel viscosity,

□ Apparent viscosity and

□ Plastic viscosity

□ Friction pressure losses

□ Annular borehole cleaning

□ Fluid basic rheological parameters and

□ Hydraulic calculations

Ogbon Advance Drilling Engineering Slide #5

Introduction to Drilling Hydraulics

Introduction to Natural Gas Processing

□ Basic Rheological Parameters in Drilling Hydraulics

□ Shear rate, $\dot{\gamma}$ (sec⁻¹):

This refers to the change in fluid velocity divided by the width of the channel through which the fluid flows. It is most applicable to laminar flow

Shear rate ($\dot{\gamma}$) = V/X

□ Shear stress, τ (lb/100 ft²):

The force per unit area required to move a fluid at a given shear rate.

$F / A = \mu (V / X) = \mu \dot{\gamma}$

The Shear stress is measured in the laboratory with Fann Viscometer

□ Viscosity, μ (centipoises (cP)):

This is the ratio of shear stress to shear rate.

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Introduction to Drilling Hydraulics

Introduction to Natural Gas Processing

□ Basic Rheological Parameters in Drilling Hydraulics

□ Plastic viscosity, PV (cP):

□ PV is the slope of the shear stress/shear rate line above the yield point on the Bingham plastic model.

□ It represents the contribution to total fluid viscosity of a fluid under dynamic flowing conditions.

□ Plastic viscosity is dependent on the size, shape, and number of particles in a moving fluid.

□ A low PV indicates mud capable of drilling rapidly as low viscosity mud is exiting the bit.

□ High PV is caused by a viscous base fluid and excess colloidal solids.

□ Dilution lowers PV.

□ PV is calculated using shear stresses measured at 600 and 300 rpm on the Fann 35 viscometer.

$$PV = \frac{\tau_{600} - \tau_{300}}{\gamma_{600} - \gamma_{300}}$$

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Introduction to Drilling Hydraulics

Introduction to Natural Gas Processing

□ Basic Parameters in Drilling Hydraulics

□ Effective viscosity, μ (cP):

This term takes account of the geometry through which the fluid is flowing and is therefore a more descriptive term of the flowing viscosity.

□ Yield point, YP (lb/100 ft²):

The minimum force required to initiate flow, see Bingham Plastic model.

In practice the yield point is calculated using the above equation

$$YP = \frac{\tau_{300} - PV}{\gamma_{300} - \gamma_{600}} = \frac{(2 \times \tau_{300}) - \tau_{600}}{\gamma_{300} - \gamma_{600}}$$

□ Yield stress (lb/100 ft²):

This is the calculated force required to initiate flow and is obtained when the rheogram (a plot of shear stress vs shear rate) is extrapolated to the y-axis at $\dot{\gamma} = 0$ sec⁻¹.

□ Note that:

The Yield Stress is a time-independent measurement and is usually denoted in the Herschel-Bulkley

(yield-power law [YPL]) model as τ_0 and Bingham

model as YP. It can also be considered a Gel Strength at zero time.)

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Introduction to Drilling Hydraulics

Introduction

- Basic Parameters in Drilling Hydraulics to Natural Gas Processing
- Gel Strength (lb/100 ft²):
- All drilling fluids build a structure when at rest.
- The gel strength is time-dependent measurement of the fluid shear stress when under static conditions.
- Gel strengths are commonly measured after 10 seconds, 10 minutes, and 30 minutes intervals.
- Reynolds Number, NRe:
- This is a dimensionless number which determines whether a flowing fluid is in laminar or turbulent flow.
- A Reynolds number greater than 2,100 marks the onset of turbulent flow in most drilling fluids.
- For laminar flow ($NRe < 2,100$) and for turbulent flow ($NRe > 2,100$).
- Critical Reynolds number, NRec :
- This value corresponds to the Reynolds number at which laminar flow turns to turbulent flow.
- Friction factor (f) :
- This is a dimensionless term used for power law fluids in turbulent flow
- It relates the fluid Reynolds number to a "roughness" factor for the pipe.
- It is used to compute the frictional pressure loss in pipe and annulus.

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Introduction to Drilling Hydraulics

Introduction to Natural Gas Processing

- Basic Parameters in Drilling Hydraulics

Ogbon Advance Drilling Engineering Slide #10

Introduction to Drilling Hydraulics

Introduction to Natural Gas Processing

- Basic Parameters in Drilling Hydraulics

Ogbon Advance Drilling Engineering Slide #11

Introduction to Drilling Hydraulics

Introduction to Natural Gas Processing

- Basic Parameters in Drilling Hydraulics

Ogbon Advance Drilling Engineering Slide #12

Introduction to Drilling Hydraulics

Introduction to Natural Gas Processing

- Fluid Types in Drilling Hydraulics

Drilling fluid are basically either Newtonian or Non-newtonian fluids.

- Newtonian Fluids
- These are fluids characterized by constant viscosity at a given temperature and pressure.
- Examples include:
 - Water
 - Diesel
 - Glycerine
 - Clear brines
- Non-newtonian Fluids
- Viscosity of Non-newtonian fluids changes with temperature and pressure and it depends on the measured shear rates for a given temperature and pressure.
- In drilling, almost all drilling fluids are non-Newtonian because even brines which are used as completion fluids are not truly Newtonian fluids, as a result of the dissolved solids in them which made them practically behave in a non-Newtonian manner.
- Examples include:
 - Most drilling fluids
 - Cement slurries

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Rheological Models

Introduction

- Rheological Models are mathematical to Natural models (equations) applied to Gas Processing describe behaviour of fluids under dynamic conditions across a wide range of shear rates.
- The models characterize the flow behaviour of the two types of fluids used in the drilling industry.
- The models provide practical means of calculating pumping (pressure) requirements for a given fluid.
- The model can be used to calculate frictional pressures, swab and surge pressures, and slip velocities of cuttings in fluids.
- As mentioned previously, most drilling fluids are non-Newtonian; and they are also pseudoplastic, as a result, the hydraulic models apply several approximations to arrive at practical equations used. A pseudoplastic fluid is a fluid that increases viscosity as force is applied. Example is a suspension of cornstarch in water with equal concentration. This cornstarch behaves like water when no force is applied; later it is solidified as force is applied.
- The full mathematical derivations of the models are in the oil and gas literatures.
- The frequently used Rheological Models in the oil & gas industry include:
 - Newtonian.

□ Bingham Plastic.

□ Power Law.

□ Hershel-Bulkley (yield-power law)

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Rheological Models

Introduction to Natural Gas Processing

□ The typical method of illustrating these model is by plotting the sheer stress against the shear rate as shown blow. This plot is referred to as Rheogram.

Rheological Models of Drilling Fluids

□ The above flow curve or rheogram demonstrates that these models do not actually characterize drilling fluids accurately

□ The Hershel-Bulkley Model that will be shown later, is most accurate when characterizing drilling fluids.

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ Terms used in drilling fluid models are shear stress and shear rate.

□ They can be described by considering two plates separated by a specified distance(x) with a fluid within the plates (see a below).

□ If a force is applied to the upper plate while the lower plate is stationary(see b below), a velocity(V) will be reached that will be a function of the applied force(F), the distance between the plates(X), the area of exposure(A), and the fluid viscosity(μ).

a b

□ This force relationship is given as:

$$F / A = \mu (V / X) X$$

F = force applied to the plate;

A = contact area;

V = plate velocity

X = plate spacing

μ = fluid viscosity

□ The quotient of F/A is termed the shear stress (τ), while V/X is shear rate ($\dot{\gamma}$)

$$\tau = \mu \dot{\gamma}$$

□ In drilling operations, the shear stress and shear rate are analogous to pump pressure(P) and flow rate(Q), respectively.

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ In drilling, Rheological models and measurements are normally made with 2-speed rotational viscometer readings to achieve a 2-points oil field and standard approach. A higher quality rheological control approach now used a 6-speed viscometer. □ and □ 300 600

□ The flow curve or rheogram is a plot of the fluid shear stress vs. shear rate from the viscometer

□ The table below shows typical data from viscometer reading using the 6-speed approach at different viscometer rpm

Symbol RPM □ □ □ □ 160

Shear Stress(lb/100 ft²)

Chart Title

Unit rpm (-) S-1 lb/100 ft² Pa 140

120

600 140 1022 148.4 71.05

100

300 98 511 103.9 49.75 80

200 78 340 82.7 39.6 60

40

100 54 170 57.2 27.39

20

6 16 10.2 17 8.14 0

3 13 15.1 13.8 6.61 -200 0 200 400 600 800 1000 1200

Shear Rate(s⁻¹)

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

160

Shear Stress(lb/100 ft²)

Chart Title

140

120 Newtonian Fluid

100

80

60

40 Non-newtonian

20

0

-200 0 200 400 600 800 1000 1200

Shear Rate(s⁻¹)

□ Generally, the shear rate is indicative of the shearing intensity in the conduit or the change in velocity between the fluid layers across the flow path

□ The rate of change between the share stress and shear rate is commonly referred to as the viscosity

□ For the Newtonian Model

□ The viscosity is constant for all shear rates and the shear stress is directly proportional to the shear rate with the constant of proportionality being the viscosity(□)

□ □ □ □ □ □ shear stress; □ □ absolute viscosity ; □ □ shear rate

□ For non-newtonian fluid, shear stress is not linearly related to shear rate and its viscosity is NOT constant with shear rate

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Summary of Laminar Flow Equations for Pipes and Annuli

Ogbon Advance Drilling Engineering Slide #19

Summary of Frictional Pressure Loss Equation

Ogbon Advance Drilling Engineering Slide #20

Summary of Frictional Pressure Loss Equation

Ogbon Advance Drilling Engineering Slide #21

Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ For the Bingham Plastic Model

□ □ □ $y = \tau_p + \mu_p \dot{\gamma}$

where

□ □ Shear stress in lb/100 ft²

□ τ_p Yield point in lb/100 ft²

□ μ_p Fluid plastic viscosity in cp

□ □ Shear rate in sec⁻¹

□ □ $\mu_p = \frac{\tau - \tau_p}{\dot{\gamma}}$ if $\tau > \tau_p$

□ 0 if $\tau < \tau_p$

□ □ $\mu_p = \frac{\tau - \tau_p}{\dot{\gamma}}$ if $\tau < -\tau_p$

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ For the Bingham Plastic Model

□ Employing the 2-point viscometer reading approach, API guidelines require □ τ_p and □ μ_p be calculated using the following equations:

□ $\mu_p = \frac{\tau_1 - \tau_2}{\dot{\gamma}_1 - \dot{\gamma}_2}$

$$\tau_y = \tau_{300} - \tau_p = (2 \times \tau_{300}) - \tau_{600}$$

□ Since the Bingham Plastic Model assumes true plastic behaviour, the fluid fitting flow index n has to be 1 which unfortunately does always occur and predicted yield stress becomes inaccurate to the tune 40 - 90%

□ It is more accurate in Oil Base Mud than in Water base mud

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Rheological Models

□ Development of Drilling Fluids Rheological Models

□ For the Power Law Model

The Power Law model is a standard mathematical expression used to describe a non-linear curve. The equation for drilling fluids is ;

□ K is a measure of the thickness of the fluid and it is very closely related to the viscosity for low shear rates.

□ K is more of a measure of the amount of solids in the fluid at larger shear rates.

$$K = \tau_{300} .$$

$$511n$$

□ n , the power-law exponent or the flow behavior index is a measure of the degree of deviation from Newtonian behavior.

$n < 1$ a pseudoplastic fluid

$n = 1$ a Newtonian fluid and $K = \tau$

$n > 1$ a dilatant fluid

$$n = 3.32 \log (\tau_{600} / \tau_{300})$$

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ For the Power Law Model

$$\log \tau_2 / \tau_1 = n$$

$$n = 3.32 \log 600$$

$$\log \tau_2 / \tau_1 = \tau_{300}$$

$$\tau_2 = 300$$

$$K =$$

$$\tau_2^{1/n} 511 n \text{ where}$$

□ τ Calculated shear stress in lb/100 ft²

□ τ_2 Shear stress at higher shear rate

□ τ_1 Shear stress at lower shear rate

K Fluid's consistency index in cp or lb/100 ft² sec

n Fluid's flow index

□ $\dot{\gamma}$ Shear rate in sec⁻¹

□ τ_2 Shear stress at higher shear rate

□ 1 □ Shear stress at higher shear rate

□ This model describes fluid rheology of polymer-based drilling fluids(viscosified clear brines) and some other fluids viscosified with biopolymers that do not exhibit yield stress.

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ For the Power Law Model

Example-3: Use the following data to compute PV, YP, n and K.

□600 = 64, □300 = 35,

Solution:

$$PV = \square 600 - \square 300 = 64 - 35 = 29 \text{ cp}$$

$$YP = \square 300 - PV = 35 - 29 = 6 \text{ lb/100ft}^2$$

$$n = 3.32 \log (\square 600 / \square 300)$$

$$n = 3.32 \log (64 / 35) = 0.87$$

$$K = \square 300 / 511n$$

$$K = 35 / 5110.87 = 0.154$$

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ For the Power Law Model

□ It allows for more plastic or pseudo – fluid behavior.

□ Power law slip velocity is generally less than Bingham one.

□ Calculating slip velocity by Bingham provide adequate hole cleaning

□ It is more accurate in water base mud and we shall use it to mostly calculate pressure losses in water base muds.

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ Computation of Frictional Losses in the Drilling Conduit

□ The main frictional losses to be overcome by the circulating pressure(PP) include

□ Drillstring frictional pressure losses (PDP)

□ Annular frictional pressure losses(PANN)

□ Bit pressure drop(Pbit)

And

$$PP = PDP + PANN + Pbit$$

□ The determination of these frictional pressure drop will be handled differently for each rheological model

□ Bingham Plastic Model Frictional Pressures

□ This model is used primarily to compute friction pressure associated with laminar flow and its restriction is based on its inability to accurately describe shear stresses associated with high shear rates.

□ Since both the laminar and turbulent flows are frequently used in the drilling industry their calculations are presented below.

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ Bingham Plastic Model Frictional Pressures

□ Velocity in the flow in the drill pipe

$$V = Q / (2.448 d^2)$$

where;

V = fluid velocity, ft/sec;

Q = flow rate, gal/min;

d = pipe diameter, inch

The critical velocity (Vc) for laminar and turbulence determination is computed;

$$V_c = [1.08 PV + 1.08 \sqrt{(PV)^2 + 12.34d^2 YP}] / \sqrt{d}$$

□ = mud weight, ppg

Friction pressures for laminar flow can be calculated as follows:

$$P_f = [(PV L V) / 1500 d^2] + [(YP L) / 225 d]$$

L = section length, ft

Friction pressures for Turbulent flow is calculated as;

$$P_f = (\sqrt[0.75]{V^{1.75} PV^{0.25} L}) / 1800 d^{1.25}$$

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ Bingham Plastic Model Frictional Pressures

In the annulus, same series of operations is performed but slightly different equations to account for the hole geometry,

$$V = Q / [2.448 (d_h^2 - d_p^2)]$$

d_h = casing or hole ID, inch ;

d_p = pipe or drill collar OD, inch

$$V_c = [1.08 + 1.08 \sqrt{(PV)^2 + 9.26(d_h - d_p)^2 YP}] / \sqrt{(d_h - d_p)}$$

For laminar flow;

$$PfA = [(PV L V) / 1000 (dh - dp)^2] + [(YP L) / 200(dh - dp)]$$

For turbulent flow;

$$PfA = (0.75 V^{1.75} PV^{0.25} L) / 1396 (dh - dp)^{1.25}$$

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ Bingham Plastic Model Frictional Pressures

Example-4:

Calculate friction pressures for flow rates of 100 and 200 gpm.using the Bingham model.

Pipe ID = 3.5 inch; MW = 12.9 ppg; PV = 29 cp;YP = 6 lb/100ft² ; Length = 10000 ft

Solution:

$$1. V_a = Q / (2.448 d^2)$$

$$V_{a100} = 100 / [2.448 (3.5)^2] = 3.33 \text{ ft/sec (at 100 gal/min)}$$

$$V_{a200} = 200 / [2.448 (3.5)^2] = 6.66 \text{ ft/sec (at 200 gal/min)}$$

$$2. V_c = [1.08 PV + 1.08 (PV)^2 + 12.34d^2 YP r] / \pi d$$

$$V_c = [1.08 (29) + 1.08 (29)^2 + 12.34(3.5)^2 (6) (12.9)] / (12.9) (3.5) = 3.37 \text{ ft/sec}$$

For the flow rate of 100 gal/min, the actual velocity (V_a) is slightly less than the critical velocity (V_c) of 3.37 ft/sec. Use the laminar flow equation.

(Note that the difference between V_a and V_c is small. Therefore, it might be advisable in some cases to consider calculating pressure losses for laminar and turbulent flow and use the larger value.

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Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ Bingham Plastic Model Frictional Pressures

Example-4:

Calculate friction pressures for flow rates of 100 and 200 gpm.using the Bingham model.

Pipe ID = 3.5 inch; MW = 12.9 ppg; PV = 29 cp;YP = 6 lb/100ft² ; Length = 10000 ft

Solution:

$$Pf = [(PV L V) / 1500 d^2] + [(YP L) / 225 d]$$

$$Pf = [(29)(10000) (3.33) / 1500 (3.5)^2] + [(6) (10000) / 225 (3.5)] = 128.6 \text{ psi}$$

At a flow rate of 200 gal/min, the actual velocity of 6.66 ft/sec is significantly greater than the critical velocity of 3.37 ft/sec. Therefore, use the turbulent flow equation;

$$Pf = (0.75 V^{1.75} PV^{0.25} L) / 1800 d^{1.25}$$

$$Pf = (12.9)^{0.75} (6.66)^{1.75} (29)^{0.25} 10000 / 1800 (3.5)^{1.25} = 505.7 \text{ psi}$$

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Rheological Models

Introduction to Natural Gas Processing

- Development of Drilling Fluids Rheological Models
- Bingham Plastic Model Frictional Pressures
- The laminar and turbulence equations can be used to illustrate the basic difference between these two flow systems.
- In the laminar equations, a value for the yield point (YP) is a significant part of the pressure loss, particularly when it is observed that the PV value is divided by a squared diameter.
- The turbulent flow equations do not contain a YP term, so the yield point is one of the forces creating the inter-particle attractions, causing the mud to move in laminae.
- When the shear force exceeds the yield stress, turbulence begins and the yield point is not a factor thereafter.

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Rheological Models

Introduction to Natural Gas Processing

- Development of Drilling Fluids Rheological Models
- Power Law Model Frictional Pressures
- Power Law calculations follow the same sequence as the Bingham model.
- Actual and critical velocities are compared to determine the flow regime before calculating the pressure loss.
- If V_a and V_c differ significantly, choose the appropriate flow equation.
- When $V_a \approx V_c$ make both pressure loss computations and choose the larger.
- The equations presented here are those of Moore et al.
- Calculating friction pressures in the drill string using the Power law equations for laminar and turbulent flow are accomplished respectively:

$$PDS = [(1.6V / D) ((3n + 1) / 4n)]^n (KL / 300d) \quad \square \text{ laminar-}$$

$$PDS = [2.27 (10^{-7}) \square 0.8 V^{1.8} PV^{0.2} L] / d^{1.2} \quad \square \text{ turbulent}$$

- Since both the laminar and turbulent flows are frequently used in the drilling industry their calculations are presented below.

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Rheological Models

Introduction to Natural Gas Processing

- Development of Drilling Fluids Rheological Models
 - Power Law Model Frictional Pressures
 - For Drill Pipe Flow
- For computation simplicity, $NR = 3000$ is assumed for turbulence criteria.

Basic assumptions for friction factor correlations result in the critical velocity equation:

$$V_c = [5.82 (104) K / \mu]^{1/2-n} [(1.6/d) ((3n+1) / 4n)]^{n/2-n}$$

$$PDS = [(1.6V / D) ((3n + 1) / 4n)]^n (KL / 300d) \mu \text{ laminar-}$$

$$PDS = [2.27 (10^{-7}) \mu^{0.8} V^{1.8} PV^{0.2} L] / d^{1.2} \mu \text{ turbulent}$$

□ For Annular Flow

$$PA = [(2.4V / (dH - dp)) ((2n + 1) / 3n)]^n [(KL / 300 (dH - dp))] \mu \text{ laminar-}$$

$$PDS = [7.7 (10^{-5}) \mu^{0.8} V^{1.8} PV^{0.2} L] / [(dH - dp)^2 (dH + dp)^{1.8}] \mu \text{ turbulent-}$$

$$V_c = [3.878 (104) K / \mu]^{1/2-n} [(2.4/ (dH - dp)) ((2n+1) / 3n)]^{n/2-n}$$

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Rheological Models

Introduction

□ Development of Drilling Fluids to Natural Gas Processing

Rheological Models

□ For the Power Law Model

Example-5:

Calculate friction pressures using the Power law model and flow rate of 125 gal/min using products of Example 3

Pipe ID = 3.5 inch; MW = 12.9 ppg; PV = 29 cp; YP = 6 lb/100ft² ; Length = 10000 ft

Compute the friction pressures for the system.

If $V_a \leq V_c$, compute the pressure drop for laminar and turbulent flow and choose the larger value.

$n = 0.87$; $K = 0.154$; Pipe ID = 3.5 inch; MW = 12.9 ppg;

Length = 10000 ft; PV = 29 cp;

Solution:

1. Determine the actual velocity at 125 gal/min;

Recall $V = Q/2.448 D^2$

$$V_a = [125 / 2.448 (3.5)^2] = 4.168 \text{ ft/sec (250 ft/min)}$$

2. Compute critical velocity;

$$V_c = [5.82 (104) K / \mu]^{1/2-n} [(1.6/d) ((3n+1) / 4n)]^{n/2-n}$$

$$V_c = [5.82 (104) 0.154 / 12.9]^{1/2-0.87} [(1.6/3.5) ((3(0.87)+1) / 4(0.87))]^{0.87/2-0.87}$$

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$V_c = 183 \text{ ft/min}$ Advance Drilling Engineering Slide #36

Rheological Models

Introduction to Natural Gas Processing

□ Development of Drilling Fluids Rheological Models

□ For the Power Law Model

Example-5:

Solution:

For purpose of illustration in this example, assume $V_a \approx V_c$ (250 ft/min $\approx$ 183 ft/min)

4. Calculate laminar flow pressure losses;

$$PDS = [(1.6V / D) ((3n + 1) / 4n)]^n (KL / 300d)$$

$$PDS = [(1.6 \times 250 / 3.5) ((3(0.87) + 1) / 4(0.87))]^{0.87} ((0.154) (10000) / 300(3.5)) \\ = 95.4 \text{ psi}$$

5. Calculate turbulent flow pressure losses;

$$PDS = [2.27 (10^{-7}) \approx 0.8 V^{1.8} PV^{0.2} L] / d^{1.2}$$

$$PDS = [2.27 (10^{-7}) 12.90.8 250^{1.8} 290.2 10000] / 3.5^{1.2} = 158.6 \text{ psi}$$

Since 158.6 psi > 93.4 psi, assume the pressure loss is the greater value and the flow is turbulent

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Rheological Models

Introduction to Natural Gas Processing

- Development of Drilling Fluids Rheological Models
- Summary of Velocity Equations in Pipe and Annulus

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Rheological Models

Introduction to Natural Gas Processing

- Development of Drilling Fluids Rheological Models
- For the Herschel-Bulkley(Yield Power Law -YPL) Model
- Since most drilling fluids exhibit yield stress, the Herschel-Bulkley (yield-power law [YPL]) model has been considered to be most accurate model in describing the rheological behavior of drilling muds
- The following equation is employed by the YPL model to describe fluid behaviour:

$$\tau = \tau_o + K \dot{\gamma}^n$$

where

τ Shear stress in lb/100 ft²

τ_o Yield stress (ie. Shear stress at zero shear rate) in lb/100 ft²

K Fluid's consistency index in cP or lb/100 ft² secⁿ

n Fluid's flow index

η_p Fluid plastic viscosity in cp

$\dot{\gamma}$ Shear rate in sec⁻¹

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Rheological Models

Introduction to Natural Gas Processing

- Development of Drilling Fluids Rheological Models
- For the Herschel-Bulkley(Yield Power Law -YPL) Model

- K and n values in the YPL model are calculated differently than their counterparts in the power law model.
- The YPL model reduces to the Bingham model when $n = 1$ and it reduces to the power law model when $\tau_o = 0$.
- An obvious advantage the YPL model has over the power law model is that, from a set of data input, only one value for n and K are calculated.
- Generally, a minimum of three shear-stress/shear-rate and measurements are needed to arrive at a solution and a computer algorithm is required to obtain solutions.
- The Model accuracy improved with additional data input.

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulics is applied to

- Determine the total hydraulic frictional losses at maximum designed flow rate along the circulating path during drilling and this is used to determined the required pump developed pressure and size.
- Determine the pressure drop through bit nozzles which in turn is used to determine
 - Optimal bit nozzle sizes
 - Optimal hydraulic horsepower
 - Optimal hydraulic impact force
 - Optimal hole cleaning drilling conditions
- Optimum Equivalent Circulation Density(ECD) required in the annulus to stay inside the pressure window.
- Determination of the Buoyancy effect of drilling fluid on drill string and casing during drilling operations
- Make well control calculations

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

□ Hydraulic Frictional Pressure Drop

- When fluid flows through a conduit, frictional forces develop resulting in fluid energy dissipating.
- These frictional forces are referred to as pressure drops, and are normally denoted as a pressure per unit length.
- The longer a pipe, annulus or conduit, the greater the frictional pressure drop.
- The factors affecting the magnitude of pressure drop include:
 - The pipe length

- Flow rate (flow regime type laminar or turbulent)
- Fluid rheological properties
- Pipe eccentricity
- Pipe/annulus geometry
- Pipe roughness, etc. Length
- During drilling, the drilling fluid is driven in and out of the well by the pump pressure.
- In the process the developed pump pressure is dissipated in overcoming the frictional losses along the flow path and the pressure drop at the bit.

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

- Hydraulic Frictional Pressure Drop
- The total hydraulic frictional losses at maximum designed flow rate occur at the followings;
- Surface Connections
- Drill String
- Annulus
- The drill Bit

□ Total System Pressure Loss is given as

$$\Delta P_s = c Q^n$$

Q = flow rate (gal/min)

n = Flow exponent a variable power (1.86)

c = a constant that depends on mud flow properties, hole geometry and pipe geometry

- The above figure give typical pressure drop profile along the drill string
- Proper hydraulics design requires that the greater portion of the frictional pressure drop occurs at the Drill bit since this is used to do work

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

- Hydraulic Frictional Pressure Drop
- Surface Connections Pressure Losses
- Pipe connection from the pump to the stand pipe, the stand pipe, the rotary hose, the swivel and the kelly i.e. from pump to drill pipe
- Generally, pressure drop along the pump-stand pipe is small and therefore considered negligible.
- The basic equation for its determination is
- $\Delta P_{sc} = E \cdot 0.8 Q^{1.8} PV^{0.2} \text{ psi}$

Where

γ_m = mud weight (lbm/gal)

Q = volume rate (gpm)

E = a constant depending on type of surface equipment used

PV = plastic viscosity (cP)

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

☐ Hydraulic Frictional Pressure Drop

☐ Surface Connections

☐ In practice, there are four types of surface equipment; each type is characterised by the dimensions of standpipe, kelly, rotary hose and swivel with each having its own pressure losses

☐ Table 1 and below summarise the four types of surface equipment and their respective values of the constant E

☐ Refers to the pressure required to overcome the friction between the drilling fluid and annulus

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

☐ Drill String Frictional Pressure Drop

☐ Drill String and Annular Pressure Losses

☐ Frictional pressure losses are determined as already enumerated in this course

☐ Drill Bit Pressure Losses

☐ As the rate of volume pumped (Q) varies, the pressure losses at the bit change as follows:

$$\Delta P_{bit} = (156 \times W \times Q^2) / (D_{2n}^5 + D_{2n}^4 + D_{2n}^3) \text{ -psi-}$$

D_{2n} = nozzle size number, 1/32 inch

Q = flow rate, gal/min and W = mud weight, ppg

☐ The jet velocity in the nozzles, V_n , can be calculated from

$$V_n = (418.3 \times Q) / (D_{2n}^5 + D_{2n}^4 + D_{2n}^3) \text{ ft/sec}$$

☐ Therefore the bit pressure drop can also be shown as;

$$\Delta P_{bit} = \Delta V_n^2 / 58.26 \text{ - psi}$$

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

☐ Two main sets of equations using the Power Law and Bingham Models in

determining drilling fluids hydraulic parameters are used by the American Petroleum Institute (API) and the Society of Petroleum Engineers (SPE) – Applied Drilling Engineering textbook.

□ Both sets of equations are valid for drilling fluid behavior in laminar and turbulent flow regimes but differ only in approach.

□ Calculations by the two sets of equation are treated below for Power Law Model.

□ American Petroleum Institute Method for Power Law Fluids

□ Pipe Fluid Velocity Where;

Q V_p = internal drill pipe average mud velocity – ft/sec

V_p □ $0.408 \sqrt[2]{Q}$ V_a = annular velocity average mud – ft/sec

ID DP Q = pump flow rate (out put) – gpm

IDH = internal diameter of wellbore – inches

□ Annular Velocity $ODDP$ = outside diameter of tubular – inches

Q $IDDP$ = inside diameter of tubular – inches

V_a □ 0.408

IDH^2 □ $IDDP^2$

2

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ American Petroleum Institute Method for Power Law Fluids

□ Plastic Viscosity, Yield Point, n and K

For high shear rate applicable in inside drill pipe calculations, n and K are determined from the Fann rates of 600 and 300 rpm, ie □600 and □ 300

□□ 600 □

n_p □ $3.32 \times \log$ □ □ and

□□ 300 □

$510 \times$ □ 300

K_p □

$511n_p$

For low shear rate applicable in annular calculations, n and K are determined from the Fann rates of 100 and 3 rpm, ie □100 and □ 3

□□100 □

n_a □ $0.657 \times \log$ □ □ and

□ □3 □

$510 \times$ □ 3

K_a □ Where;

K_a □ consistency index in the annulus

5.11n p

KP □ consistency index in the drillpipe

n □ flow index in the annulus

n □ flow index in the drillpipe

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ American Petroleum Institute Method for Power Law Fluids

□ Wellbore Interval Hydraulics

Deff = IDDP

Daeff = IDH - IDDP

n □1 n

□ 96 V p □ □ 3n p □ 1 □

□ep □ 100 K p □□ □□ □□

□ 4n □

□ IDDP □ □ p □

n□ □ 1 n□

□ 144 Va □ □ 2n pa □ 1 □

□ea □ 100 K a □□ □□ □□ □□

□ IDH □ IDDP □ □ 3 na □

Deff V □ mud

N Re □ 928

□e Where;

Deff = internal drill pipe effective diameter- inches

Daeff = annular effective diameter - inches

□ep = Inside drill pipe effective viscosity - cp

□es = annular effective viscosity - cp

V = either annular velocity(Va) or

drill pipe internal velocity(Vp) - ft/sec

IDDP = inside diameter of tubular - inches

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ American Petroleum Institute Method for Power Law Fluids

□ Wellbore Interval Hydraulics

□ Friction Factor (f)

If the Reynolds number is greater than 2100 the flow is turbulent and the friction factor is:

a log n □ 3.93

f ; where; a

$N_{Re} b$

50

$1.75 \log n$

b

7

If the Reynolds number is less than 2100 the flow is laminar and the friction factor is:

16

f Where

$N_{Re} D$ effective diameter of the hole in inches eff

f friction factor

ΔP Interval Pressure Loss D mud density in lb/gal mud

V either V_a for inside annulus or V_p for inside

V^2 mud drillpipe

$\Delta P_f = f L$

25.81 Deff

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

ΔP SPE Method for Power Law Fluids

Power Law Inside Drill Pipe for each Hydraulic Interval

Q Fluid Velocity Inside Drill Pipe

Q

$V_p = 0.408 \frac{Q}{ID_{DP}}$

ID_{DP}

Where;

V_p = internal drill pipe average mud velocity – ft/sec

V_a = annular velocity average mud – ft/sec

Q = pump flow rate (out put) – gpm

IDH = internal diameter of wellbore – inches

$ODDP$ = outside diameter of tubular – inches

$IDDP$ = inside diameter of tubular – inches

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

ΔP SPE Method for Power Law

Fluids

Power Law Inside Drill Pipe for
each Hydraulic Interval

□ Determine Flow Regime –
Laminar or Turbulent

1. Determine N_{Rec} (critical
Reynolds number) from the
Power Law friction factor
figure using the lowest N_{Re}
that intersect the straight line
for a given value of n or $N_{Rec} =$
2100

2. The friction factor graph
below shows friction factors
versus Reynolds numbers for
power law fluids having
different values of n .

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ SPE Method for Power Law Fluids

Power Law Inside Drill Pipe for each Hydraulic Interval

□ Determine Flow Regime – Laminar or Turbulent

3. Calculate N_{Rep} for pipe

n

□ □ Where

$2 \square n$

□ mud V_p

□ 0.0416 IDDP □ IDDP = inside diameter of the drillpipe or DC – in

$N_{Rep} \square 89,100$ K = consistency index in the drillpipe,

$K_p \square 3 \square 1$

□ ρ_{mud} = the mud density in lb/gal mud - ppg

□ n_p □ n_p = flow index n inside the drillpipe

4. If $N_{Rep} < N_{Rec}$, the flow is laminar.

If $N_{Rep} \square N_{Rec}$ the flow is turbulent

5. Determine friction factor from figure

□ Turbulent Flow Pressure Drop Where

Frictional pressure drop inside IDDP = inside diameter of the drillpipe or DC – in

drill pipe or drill collar f_p = friction factor inside the drillpipe p

$f_p \square \rho_{mud} V_p^2 L$ = length of the drillpipe in feet

ρ_{mud} = the mud density in lb/gal mud - ppg

$\rho_{\text{Pf}} \times L \times V_p$ = average mud velocity inside the drillpipe in ft/sec

25.8 IDDP

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

ρ_{SPE} Method for Power Law Fluids

Power Law Inside Drill Pipe for each Hydraulic Interval

ρ_{Laminar} Flow Pressure Drop

Frictional pressure drop inside drill pipe or drill collar

n_p

$\rho_1 \times$

$\rho_3 \times$

$K_p \times V_p \times \rho$

$n_p \times$

$\rho_0.0416 \times$

$\rho \times$

$\rho_{\text{Pf}} \times \rho \times L$

$\rho_1 \times n \times$

144,000 ID DP p

Where

IDDP = inside diameter of the drillpipe or DC - in

K_p = consistency index in the drillpipe,

n_p = flow index n inside the drillpipe

V_p = average mud velocity inside the drillpipe

in ft/sec

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

ρ_{SPE} Method for Power Law Fluids

Power Law in the Annulus for each Hydraulic Interval

$\rho_{\text{Determine}}$ Average Fluid Velocity in the Annulus

Where;

V_p = internal drill pipe average mud velocity - ft/sec

$Q \times V_a$ = annular velocity average mud - ft/sec

$\rho_0.408$

V_a

$\rho_{\text{IDH2}} \times \rho_{\text{ODDP}}$

2

□ Q = pump flow rate (out put) – gpm

IDH = internal diameter of wellbore – inches

ODDP = outside diameter of tubular – inches

IDDP = inside diameter of tubular – inches

□ Determine Fluid Flow Regime - Turbulent or Laminar

1. Determine NRec from the Power Law friction factor figure using the lowest Nre that intersect the straight line for a given value of n or NRec = 2100

2. Calculate NRea na

□ □

2 □ na □

□ mud Va 0.0208 □ IDH □ ODDP □ □

N Re a □ 109,100 □ □

Ka □ 3□

1 □

□ □ na □ □

3. If NRea < NRec, the flow is laminar.

If NRea □ NRec the flow is turbulent.

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ SPE Method for Power Law Fluids

Power Law in the Annulus for each Hydraulic Interval

□ For Turbulent Flow Frictional Pressure Drop in the Annulus

f a □ mud Va²

□ Pfa □ Where

21.1 IDH □ ODDP

L

□ ODDP = outside diameter of the drillpipe or DC – in

IDH = inside diameter of the wellbore– in

fa = friction factor inside the annulus

L = length of the drillpipe in feet

□ mud = the mud density in lb/gal mud - ppg

Va = average mud velocity in annulus in ft/sec

□ For Laminar Flow Frictional Pressure Drop in the Annulus

na □ 2 □ 1 / na □

n

a

K Va □ □

$\Delta P_{fa} \approx 0.0208 \Delta L$

144,000 $\Delta IDH \approx ODDP \approx$

$\Delta 1 \approx na \approx$

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Δ Drill Bit Nozzle Sizes, Velocity and Pressure Drop

Δ Drill Bit Pressure Losses

Δ As the rate of volume pumped (Q) varies, the pressure losses at the bit change as follows:

$\Delta P_{bit} = (156 \times W \times Q^2) / (D_{2n} + D_{2n} + D_{2n})^2$ -psi-

D_{2n} = nozzle size number, 1/32 inch

Q = flow rate, gal/min and W = mud weight, ppg

Δ The jet velocity in the nozzles, V_n , can be calculated from

$V_n = (418.3 \times Q) / (D_{2n} + D_{2n} + D_{2n})$ ft/sec

Δ Therefore it can be shown as;

$\Delta P_{bit} = \Delta V_{2n} / 58.26$

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Δ Drill Bit Nozzle Sizes, Velocity and Pressure Drop

Δ The determine of the bit nozzle sizes, velocity and pressure drop are essential parameters for

Δ Optimal hydraulic horsepower

Δ Optimal hydraulic impact force

Δ Optimal hole cleaning drilling conditions

Δ The above are basic ingredients for drilling optimization

Δ Nozzle area

Δ Nozzle velocity

Q

Δ Bit pressure drop

ΔP_{bit}

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Δ Drill Bit Nozzle Sizes, Velocity and Pressure Drop

Δ Bit hydraulic horsepower

$\Delta P_{bit} \times Q$

□ Bit hydraulic horsepower per unit bit area

□ Percent pressure drop at bit

□ Pbit

□ Jet impact force

x QGPM x

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ Equivalent Mud Weight(EMW)

□ Sometimes, drilling operations often involve several fluid densities in the well and several pressures resulting from fluid circulation especially during kick control operations.

□ The practice of bringing these complex pressures and fluid density arrangements on a common basis is addressed by the concept of "equivalent mud weight"

□ The Equivalent Mud Weight(EMW) is the density of a fluid that would provide the same pressure at the bottom of a stratified fluid static system with no surface pressure. $h = 3500 \text{ ft}$

1

□1 = 8.81 ppg

Example

Suppose a 17,000-ft well has three mud weights as shown in the figure. Calculate the EMW at wellbore bottom

n $h_1 = 8500 \text{ ft}$

EMW □ □ h

i □1

i i □1 = 9.23 ppg

EMW = $0.052(8.81 \times 3500 + 9.23 \times 8500 + 10.15 \times 5000)$

= $0.052(30,835 + 78,455 + 50,750)$

= 8,322.08 psi $h_1 = 5000 \text{ ft}$

□1 = 10.15 ppg

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ Equivalent Circulation Density in Annulus(ECD)

□ The ECD usually considers the hydrostatic pressures and the friction pressure resulting from fluid movement.

□ Generally, the sum of the frictional pressure drop for each annular interval

regardless of inclination is given as;

n

$\Delta P_{fa} = \Delta P$

$i = 1$

f_i

The Annular Equivalent Circulation Density(ECD) for a Vertical Well is given as

Where

ΔP_{fa}

ΔP_{fa} = pressure drop in the annulus - psi

$L =$ the length of the interval - ft

ECD =

L = Total well depth

$L_i \times 0.052$

L

L_i = length of interval - ft

L_{vi} = vertical length of the interval - ft

ρ_{mud} = density of the mud in lb/gal

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

Equivalent Circulation Density in Annulus(ECD)

Take Home Exercise

A 12 1/4" 17,000-ft well has three mud weight columns in the annulus and one mud weight column in the drill string with a uniform 4 1/2" ID as

shown in the figure. Calculate the ECD if the pump rate is 150 gal/min and the friction factors in the columns are as shown.

$h_1 = 3500$ ft

$\rho_1 = 8.81$ ppg

$f = 0.00681$

$h_1 = 8500$ ft

$\rho_1 = 9.23$ ppg

$f = 0.00651$

$h_1 = 5000$ ft

$\rho_1 = 10.15$ ppg

$f = 0.00639$

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ Annular Velocity

□ This is the minimum annular velocity required clean the hole of its cuttings.

□ It is important to avoid solids build-up and increasing the hydrostatic head, which might cause mud losses to the formation.

$$V_{ann} = (24.5 \times Q) / (D_2^2 - D_1^2)$$

where;

D_2 = ID of hole - inch

D_1 = OD of pipe - inch

V_{ann} = Annular Velocity - ft/sec

□ Density and Velocity

□ Drilled cuttings typically have a density of about 21 lb/gal.

□ Since the fluid density is less than 21 lb/gal the cuttings will tend to settle, or 'slip' relative to the drilling mud.

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ Slip Velocity:-

□ This is the rate at which cuttings fall back toward bottom.

□ Newtonian Fluids:

Terminal velocity of a small spherical particle settling (slipping) through a Newtonian fluid under Laminar flow conditions is given by STOKES' LAW:

□ Particle Slip Velocity:

$$v_s = \frac{138 (\rho_s - \rho_f) d_s^2}{\mu}$$

where; v_s = slip velocity, ft/s

ρ_s = density of solid particle, lbm/gal

ρ_f = density of fluid, lbm/gal

d_s = diameter of particle, in

μ = fluid viscosity, cp

□ Stokes' Law gives acceptable accuracy for a particle Reynolds number

$$< 0.1 \quad Re = \frac{\rho_f v_s d_s}{\mu}$$

□ For $N_{Re} > 0.1$ an empirical friction factor may be used.

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

Surge Pressure

□ When a string of pipe is being lowered into the wellbore, drilling fluid is being displaced and forced out of the wellbore.

□ The pressure required to force the displaced fluid out of the wellbore is called the surge pressure.

□ It is a positive value produced by pipe movement down ward.

□ An excessively high surge pressure can result in breakdown of a formation

□ Swap pressure

□ Swab pressure is the temporary reduction in the bottomhole When pipe is being withdrawn a similar reduction in pressure is experienced.

□ This is called a swab pressure, and may be high enough to suck fluids into the wellbore, resulting in a kick.

□ It is a negative value produced by pipe movement up ward.

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ Buoyancy Effect

□ This is the amount of support by way of weight reduction provided by the drilling fluid on the total weight of the drill string or casing in the drilling fluid

□ When the drill string or casing is lowered into the well, the drilling fluid will support, or buoy, some of the pipe weight. This effect is termed buoyancy, or buoyant forces.

□ The buoyed weight of the drill string or casing (i.e its weight in the drilling fluid) will therefore be less than its weight in-air.

□ Buoyant forces are a function of the volume and weight of the displaced fluid. Heavier mud has greater buoyant forces than low-density mud.

$$BW = BF \times (\text{in-air weight})$$

$$BF = 1 - (\rho_{\text{mud}} / 65.5)$$

$$BW = \text{buoyed weight}$$

$$BF = \text{buoyancy factor}$$

$$\rho_{\text{mud}} = \text{mud density, lb/gal and}$$

$$\rho_{\text{steel}} = 65.5 \text{ lb/gal}$$

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ Buoyancy Effect

Example

Assuming casing string is run into a well that contains 11.7 ppg mud and the casing is filled with mud as it is run. The casing weight and section length are as indicated in the table below. If the engineer uses a derrick safety factor of 2, will the 1,000,000 lb derrick capacity be satisfactory?

Casing weight, lb/ft Section length, ft

47.0 4500

53.0 5500

47.0 3000

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Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

Hydraulic Parameter Calculations

□ Buoyancy Effect

Solution:

1. Calculate the casing string weight, in air

$$4500 \text{ ft} \times 47.0 \text{ lb/ft} = 211500 \text{ lb}$$

$$5500 \text{ ft} \times 53.0 \text{ lb/ft} = 291500 \text{ lb}$$

$$3000 \text{ ft} \times 47.0 \text{ lb/ft} = 141000 \text{ lb}$$

Total casing string weight in air is = 644000 lb.

2. Calculate Buoyancy Force

$$BF = 1 - (\rho_{\text{mud}} / 65.5)$$

$$BF = 1 - (11.7 / 65.5) = 0.82$$

3. Calculate Buoyed Weight

$$BW = BF \times (\text{in-air weight})$$

$$BW = 0.82 \times 644000 \text{ lb} = 528964 \text{ lb}$$

4. Applying a derrick safety factor of 2.

$$2 \times 528964 = 1,057,928 > 1,000,000 \text{ lb}$$

Therefore, the derrick load will exceed the design criteria if a factor of 2 is used. The actual design factor is $1000000 / 528964 = 1.89$

Ogbon Advance Drilling Engineering Slide #68

Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

□ Hydraulic Frictional Pressure Drop

□ Field method of optimizing bit hydraulics

□ Prior to POH current bit for next bit change, run the pump at four or five speeds and record the resulting stand pipe pressure

- From current nozzle size and stand pipe pressure and mud weight determine pressure loss across the bit
- Subtract Pbit from stand pipe pressure to get Pc Plot a graph of Pc against Q on log-log paper and determine the slope to get n
- For the next bit run calculate Pbit using BHHP or IF equations
- Select nozzle size to for the value of Pbit calculated
- For particular rig and field the index n will not vary widely if the same parameters are used
- $P_s = c Q^n$
- Q = flow rate (gal/min)
- n = a variable power (1.86)
- Ogbon Advance Drilling Engineering c = a constant Slide #69

Federal University of Petroleum Resources Effurun

Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Advance Drilling Engineering

Lesson

Drilling Optimization

Ogbon Advance Drilling Engineering Slide #2

Drilling Hydraulics Optimization

Introduction to Natural Gas Processing

- The optimization of drilling hydraulics is achieved when the drilling hydraulics program
 - Maximizes the cleaning of bottom hole
 - Effectively evacuates drilled cuttings as they are generated
 - Results in higher penetration rate
- Drilling parameters necessary to achieve optimum drilling hydraulics include;
 1. Mud pump Output Volume (and Volumetric Efficiency)
 2. Mud pump Output in terms of Hydraulic Power
 3. Pressure losses in:
 - Surface connections
 - Drill pipe (inside and outside)
 - Drill collars (inside and outside)
 - Bit Nozzles
 4. Proper nozzle sizes selection which guarantees bottom hole cleaning efficiency
 5. Velocity of mud passing through the bit nozzles (Nozzle velocity, V_n)

6. Velocity of mud rising in the annulus (Annular Velocity, Vann)

□ Proper bottom-hole cleaning eliminates excessive regrinding of drilled solids, and results in improved penetration rates

Ogbon Advance Drilling Engineering Slide #3

Drilling Hydraulics Optimization

Introduction to Natural Gas Processing

□ Mud Pump Output Volume Rate(Q):

The output volume flow rate, Q, depends on:

□ pump line size

□ pump stroke length

□ pump speed (strokes/min.)

□ pump volumetric efficiency

□ The Double Acting Duplex Pump

$$Q = 0.0068 \times L \times (2D^2 - d^2) \times \text{SPM} \times (\text{Vol. Eff.} / 100) \quad \square \text{ gal/min}$$

□ For Single Acting Triplex Pump

$$Q = 0.0102 \times L \times D^2 \times \text{SPM} \times (\text{Vol. Eff.} / 100) \quad \square \text{ gal/min}$$

where

L = stroke length, inch

D = inside diameter of liner, inch

d = rod diameter, inch

SPM = stroke per minute

Vol. Eff. = $\geq 95\%$ - triplex pump; $\geq 90\%$ - duplex pump

Ogbon Advance Drilling Engineering Slide #4

Drilling Hydraulics Optimization

Introduction to Natural Gas Processing

□ Mud Pump Output Volume Rate(Q):

□ Pump Output Hydraulic Power

□ This hydraulic power is generally assumed to equal 85 % of mechanical or electrical input power.

HHP = $P_t \times Q$ where;

1714 P_t = circulating pressure, psi

Q = pump output volume, gal/min

$P_t = \square P_s + \square P_{bit}$

□ P_s = pressure losses in the system, psi

HHP_{bit} = $\square P_{bit} \times Q$ □ P_{bit} = pressure losses over the bit nozzles -psi

1714

HHPs = $\square P_s \times Q$

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Total HHP = HHPs + HHP_{bit}

$$HHP_{bit} = Q (\rho P_t - \rho P_s)$$

1714

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Drilling Hydraulics Optimization

Introduction to Natural Gas Processing

□ Mud Pump Output Volume Rate(Q):

The output volume flow rate, Q, depends on:

□ pump line size

□ pump stroke length

□ pump speed (strokes/min.)

□ pump volumetric efficiency

□ The Double Acting Duplex Pump

$$Q = 0.0068 \times L \times (2D^2 - d^2) \times SPM \times (\text{Vol. Eff.} / 100) \text{ gal/min}$$

□ For Single Acting Triplex Pump

$$Q = 0.0102 \times L \times D^2 \times SPM \times (\text{Vol. Eff.} / 100) \text{ gal/min}$$

where

L = stroke length, inch

D = inside diameter of liner, inch

d = rod diameter, inch

SPM = stroke per minute

Vol. Eff. = $\geq 95\%$ - triplex pump; $\geq 90\%$ - duplex pump

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Drilling Hydraulics Optimization

Introduction to Natural Gas Processing

□ Mud Pump Output Volume Rate(Q)

□ Pressure Drop Across Bit Nozzles

□ As stated already in drilling hydraulics chapter, the rate of volume pumped (Q) varies, the pressure losses at the bit change as follows:

$$\Delta P_{bit} = (156 \times W \times Q^2) / (D_n^2 + D_n^2 + D_n^2) \text{ -psi-}$$

D_n = nozzle size number, 1/32 inch

Q = flow rate, gal/min and W = mud weight, ppg

□ The jet velocity in the nozzles or maximum nozzle V_n, can be given as

$$V_n = (418.3 \times Q) / (D_n^2 + D_n^2 + D_n^2) \text{ ft/sec}$$

□ Therefore it can be shown as;

$$\Delta P_{bit} = \Delta V_n^2 / 58.26$$

 ΔP v_n □ C_b

$$58.26 \Delta P \text{ OR } 8.074 \times 10^{-4} \Delta P$$

v_n = bit

C_d - Nozzle discharge co-efficient (0.98 – 0.95)

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Drilling Hydraulics Optimization

Introduction to Natural Gas Processing

□ Mud Pump Output Volume Rate(Q)

□ Annular Velocity

The minimum annular velocity required cleaning the hole. It is important to avoid solids build-up and increasing the hydrostatic head, which might cause mud losses to the formation.

$$V_{ann} = (24.5 \times Q) / (D_2^2 - D_1^2)$$

where'

V_{ann} = Annular velocity in ft/sec

D_2 = ID of hole - in

D_1 = OD of pipe - in

$$V_{ann} = (2.448 \times Q) / (D_2^2 - D_1^2)$$

where'

V_{ann} = Annular velocity in ft/min

D_2 = ID of hole - in

D_1 = OD of pipe - in

Ogbon Advance Drilling Engineering Slide #8

Hydraulics Optimization

Introduction to Natural Gas Processing

□ Optimization

Two basic principle approaches are employed to achieve efficient removal of cuttings below the bit and to achieve the best penetration rate:

1. To maximize hydraulic power expended at the bit; assuming that cuttings removal depends on the fluid energy dissipated.
2. To maximize the hydraulic impact force; assuming that formation is best removed when the mud hits the bottom of the hole with the greatest force.

□ It is also important to be able to know the proportion of the circulating pressure of system pressure loss in the event of a kick, because this value is needed to determine the circulating pressures at various killing speeds.

Recall that

System Pressure Loss, $\Delta P_s = c Q^n$

□ Generally, if the value of "n" is known, the proportion of the circulating pressure which will be lost in the drill string under optimum hydraulics conditions ($P_{s,opt}$) can be determined.

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Hydraulics Optimization

Introduction to Natural Gas Processing

□ Hydraulic Optimization

□ The equation for the respective approaches mentioned above are different

□ For the maximum Hydraulic Power at the Bit approach, the equation is given as

$$P_{s,opt} = [1 / (n + 1)] P_t$$

and for the maximum Impact Force approach it is

$$P_{s,opt} = [2 / (2 + 1)] P_t$$

□ If the value of “c” is known the corresponding optimum circulation rate(Q_{opt}) can be determined.

□ With the optimum circulation rate(Q_{opt}) and pressure ($P_{s,opt}$) known, bit nozzle sizes can therefore be selected to obtain the correct ΔP_{bit} ($\Delta P_{bit} = P_t - P_{s,opt}$)

□ Theoretically Derived Values of “n” and “c”

The values of “n” and “c” can be theoretically calculated if two pressures are obtained at two different pump rates by using the following equation

$$n = \log (P_{s1} / P_{s2})$$

$$\log (Q_1 / Q_2)$$

$$c = P_{s1} / Q_1^n$$

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Hydraulics Optimization

Introduction to Natural Gas Processing

□ Hydraulic Optimization

□ Note that small inaccuracies in pressure readings and pump stroke counts can result in considerable errors in “n” and “c”.

□ Graphical Derivation of Values of “n” and “c”

Now;

$$P_s = c Q^n$$

Taking log of both side of the equation gives

$$\log P_s = \log c + n \log Q$$

□ From the above equation, plotting the values of “ P_s ” and “ Q ” on log-log paper gives a straight line curve whose slope is “n” and “c” can be calculated from

$$c = P_s / Q^n$$

□ The above equation is good for any combination of P_s and Q 's.

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Hydraulics Optimization

Introduction to Natural Gas Processing

□ Hydraulic Optimization

Example 6

Before starting a trip out at 9460 ft, to change the bit (12 ¼ with 3 x 14 nozzles), the following readings were taken:

Pump Rate (spm) Circulating pressure (psi)

160 3480

135 2640

110 1810

88 1230

Mud pump types □ two single acting triplex pumps, with 6" liner, 10" stroke, 97.5% volumetric efficiency. The mud gradient in use is: 0.676 psi/ft.

1. Calculate P_s for each circulation rate (Q).
2. Plot $P_s - Q$ on log - log paper and determine n and calculate c .
3. For max. Hydraulic horsepower at the bit, calculate $P_{s \text{ opt}}$ and Q_{opt} . assuming the max. pump pressure to be used is 3250 psi.
4. Determine the size of the nozzles for the next bit, to obtain optimum hydraulics.
5. Calculate bit and total hydraulic horse power for the next bit run.
6. Calculate annular velocity around 5" drill pipe.
7. Calculate nozzle velocity.

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Hydraulics Optimization

Introduction to Natural Gas Processing

□ Hydraulic Optimization

Example 6

Solution:

1. Calculate P_s

- a. Calculate Q

For a triplex pump

$$Q = 0.0102 \times L \times D2 \times \text{SPM} \times \text{Vol. Eff.}$$

$$Q_1 = 0.0102 \times 10 \times 62 \times 160 \times 0.975 = 573 \text{ gal/min}$$

$$Q_2 = 0.0102 \times 10 \times 62 \times 135 \times 0.975 = 483 \text{ gal/min}$$

$$Q_3 = 0.0102 \times 10 \times 62 \times 110 \times 0.975 = 394 \text{ gal/min}$$

$$Q_4 = 0.0102 \times 10 \times 62 \times 88 \times 0.975 = 315 \text{ gal/min}$$

- b. Calculate ΔP_{bit}

To obtain P_s it is necessary to calculate P_{bit} for each circulating rate, from

$$\Delta P_{\text{bit}} = (3013 \times \Delta \text{grad} \times Q^2) / (D_n^2 + D_n^2 + D_n^2)^2 - \text{psi}$$

Δgrad □ mud weight gradient in psi/ft

D_n □ nozzle size number, 1/32 inch

$$\Delta P_{\text{bit},1} = (3013 \times 0.676 \times 573^2) / (142 + 142 + 142)^2 = 1928 \text{ psi}$$

$$\Delta P_{\text{bit},2} = (3013 \times 0.676 \times 483^2) / (142 + 142 + 142)^2 = 1552 \text{ psi}$$

$$\Delta P_{bit,3} = (3013 \times 0.676 \times 3942) / (142 + 142 + 142) \times 2 = 899 \text{ psi}$$

$$\Delta P_{bit,4} = (3013 \times 0.676 \times 3152) / (142 + 142 + 142) \times 2 = 647 \text{ psi}$$

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Hydraulics Optimization

Introduction to Natural Gas Processing

☐ Hydraulic Optimization

Example 6

Solution:

c. Calculate ΔP_s

And because, $\Delta P_s = P_t - P_b$, it follows that,

$$\Delta P_{s1} = 3480 - 1928 = 1552 \text{ psi}$$

$$\Delta P_{s2} = 2640 - 1370 = 1270 \text{ psi}$$

$$\Delta P_{s3} = 1810 - 911 = 899 \text{ psi}$$

$$\Delta P_{s4} = 1230 - 583 = 647 \text{ psi}$$

2. Plot P_s against Q on log-log paper.

Recall

$P_s = c Q^n$ and

$$\log P_s = \log c + n \log Q$$

a. Measured slope of straight line curve; $n = 1.50$

b. Calculate “c” using $P_s = c Q^n$ and $Q_3 = 394 \text{ gal/min}$ and $P_{s3} = 899 \text{ psi}$

$$C = P_{s3} / Q_n = 899 / 394^{1.5} = 0.115$$

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Hydraulics Optimization

Introduction to Natural Gas Processing

☐ Hydraulic Optimization

3. For maximum hydraulic pressure at the bit.

$$P_{s,opt} = [1 / (n + 1)] P_t$$

$$P_{s,opt} = [1 / (1.5 + 1)] P_t = 0.4 P_t$$

As maximum pump pressure is given as 3250 psi,

$$P_{s,opt} = 0.4 \times 3250 = 1300 \text{ psi}$$

$$Q_{opt} = (P_s / c)^{1/n} = (1300 / 0.115)^{1/1.5} = 504 \text{ gal/min}$$

4. Calculate ΔP_{bit}

$$\Delta P_{bit} = P_t - P_s = 3250 - 1300 = 1950 \text{ psi}$$

5. Calculate Optimum Nozzle Size

$$\Delta P_{bit} = (3013 \times \mu_{grad} \times Q^2) / (3 \times D_n^2)^2$$

$$(3 \times D_n^2)^2 = 3013 \times 0.676 \times 504^2 / 1950$$

$$(3 \times D_n^2)^2 = 514$$

$$D_n^2 = 171$$

$$D_n = 13$$

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Hydraulics Optimization

Introduction to Natural Gas Processing

□ Hydraulic Optimization

6. Calculate Horsepower;

a. Total Hydraulic Horsepower

$$HHPT = (P \times Q) / 1714$$

$$HHPT = (3250 \times 504) / 1714 = 955 \text{ hp}$$

b. Total Hydraulic Horsepower

$$HHP_{Bit} = (\Delta P_{Bit} \times Q) / 1714$$

$$= (1950 \times 504) / 1714 = 573 \text{ hp}$$

7. Calculate the annular velocity around the 5" drill pipe

$$V_{ann} = (24.5 \times Q) / (D_2^2 - d_2^2)$$

$$V_{ann} = (24.5 \times 504) / (12.252^2 - 5^2) = 99 \text{ ft/sec}$$

8. Calculate Nozzle velocity

$$V_n = (418.3 \times Q) / (3 \times D_n^2)$$

$$V_n = (418.3 \times 504) / (3 \times 13^2) = 416 \text{ ft/sec}$$

Ogbon Advance Drilling Engineering Slide #16

Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

□ Drill Bit Nozzle Sizes, Velocity and Pressure Drop

□ The determine of the bit nozzle sizes, velocity and pressure drop are essential parameters for

□ Optimal hydraulic horsepower

□ Optimal hydraulic impact force

□ Optimal hole cleaning drilling conditions

□ The above are basic ingredients for drilling optimization

□ Nozzle area Where

D is the mud density in lb/gal mud

Press is the pump pressure in psig Pump

PO is the pump output in gal/min GPM

Jet is the nozzle diameter in 32nds of an inch

□ Nozzle velocity

A is the area of the bit Bit

A is the total nozzle area in in

Q V is the nozzle velocity in ft/sec

□ P_{bit} is the bit pressure drop in psi

□ Bit pressure drop

□ P_{bit} = good for newtonian & Non-newtonian

Ogbon Advance Drilling Engineering Slide #17

Applications of Hydraulics in Drilling

Introduction to Natural Gas Processing

□ Drill Bit Nozzle Sizes, Velocity and Pressure Drop

□ Bit hydraulic horsepower

□ $P_{bit} \times Q$

□ Bit hydraulic horsepower per unit bit area

□ Percent pressure drop at bit Where

D is the mud density in lb/gal mud

□ P_{bit} Press is the pump pressure in psig Pump

PO is the pump output in gal/min GPM

Jet is the nozzle diameter in 32nds of an inch i

A is the area of the bit Bit

A is the total nozzle area in in N

□ Jet impact force 2

$\times Q_{GPM} \times V$ is the nozzle velocity in ft/sec N

□ P_{bit} is the bit pressure drop in p

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Drilling Hydraulics

Introduction to Natural Gas Processing

•Newtonian Fluids

•The model used initially to describe drilling mud was the Newtonian model,

• $\tau = \mu \frac{dv}{dy}$

•It stated that pump pressure (shear stress) would increase proportionally to

•shear rate. If a constant of proportionality is applied to represent fluid

•viscosity,

• $\tau = \mu \frac{dv}{dy}$

•Unfortunately, drilling mud usually cannot be described by a single

•viscosity term. They require two or more points for an accurate representation

•of behavior. As a result, the Newtonian model generally is not used in hydraulics

•plans.

•Bingham Plastic

•The Bingham model was developed to describe more effectively drilling mud

•presently in use. Bingham theorized that some amount of stress would be

•required to overcome the mud's gel structure before it would initiate movement

• $\tau = \mu \frac{dv}{dy} + \tau_y$

• τ_y = yield stress,

• μ = fluid viscosity

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Drilling Hydraulics

Introduction to Natural Gas Processing

- In practical terms, the equation states that a certain pressure would be applied to the mud to initiate movement. Flowing mud pressures would be a function of the initial yield pressure and the fluid viscosity.
- Shear rates are normally taken at 300 and 600 rpm rates on the viscometer.
- The fluid viscosity (μ_p) and the yield stress (τ_y) are calculated as follows:
- $\mu_p = q_{600} - q_{300}$
- q_{600}, q_{300} = readings at 600 and 300 rpm, respectively.
- $\tau_y = q_{300} - \mu_p$
- The fluid viscosity is termed plastic viscosity (PV) due to the plastic nature of the fluid and is measured in centi-poise (cp). The size, shape, and concentration of particles affect the plastic viscosity in the mud system. As mud solids increase, the plastic viscosity increases. The plastic viscosity is a mud property that is not affected by most chemical thinners and can be controlled only by altering the state or number of solids.
- The yield stresses τ_y , is given the name of yield point and is measured in lb/100 ft². It is a function of the inter-particle attraction of the solids in the mud. Chemical thinners, dispersants, and viscosifiers control the yield point.

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Drilling Hydraulics

Introduction to Natural Gas Processing

- Power Law
- The Power Law model is a standard mathematical expression used to describe a non-linear curve. The equation for drilling fluids is ;
- $\tau = K (g)n$
- K = consistency index; n = flow behavior index
- The flow behavior index is descriptive of the degree to which the fluid is non-Newtonian.
- $n = 3.32 \log (q_{600} / q_{300})$
- $K = q_{300} / 511n$
- Example-3: Use the following data to compute PV, YP, n and K .
- $q_{600} = 64, q_{300} = 35,$
- Solution:
- $PV = q_{600} - q_{300} = 64 - 35 = 29 \text{ cp}$
- $YP = q_{300} - PV = 35 - 29 = 6 \text{ lb/100ft}^2$
- $n = 3.32 \log (q_{600} / q_{300})$
- $n = 3.32 \log (64 / 35) = 0.87$
- $K = q_{300} / 511n$
- $K = 35 / 511 \cdot 0.87 = 0.154$

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Drilling Hydraulics

Introduction to Natural Gas Processing

- Friction Pressure Determinations
- Pumping a drilling fluid requires overcoming frictional drag forces from
- fluid layers and solids particles. The pump pressure (Pp) can be described as the
- summation of the frictional forces in the circulation system:
- $P_p = P_{DS} + P_B + P_A$
- Pp = pump pressure, psi;
- PDS = drill string friction pressure, psi;
- PB = bit pressure drop, psi;
- PA = annulus pressure, psi
- The pressure drop in the bit results from fluid acceleration and not solely
- friction forces. As a result, it will be discussed in a separate section. Equations
- to determine friction pressures vary according to the flow regimes, such as
- laminar and turbulent. In addition, Bingham Plastic and Power Law models differ
- in form. Since these models are frequently used in drilling applications, they will
- be presented in the following sections. Newtonian-based equations will not be
- presented.
- Bingham Plastic Friction Pressures
- The Bingham Plastic model is used primarily to compute friction pressure
- associated

Ogbon with laminar flow. This restriction

Advance Drillingis based on its inability to

Engineering Slide #22

Average Fluid Velocity

Pipe Flow Annular Flow

q

$v \propto v$

$2 \cdot 448 d^2$

$2 \cdot 448 d^2 \propto d^1$

2^2

WHERE

v = average velocity, ft/s

q = flow rate, gal/min

d = internal diameter of pipe, in.

d_2 = internal diameter of outer pipe or borehole, in.

d_1 = external diameter of inner pipe, in.

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Velocity in drill collars

q (g a l/m in)

$v^2 \propto 2^2$

$2 \cdot 448 d^2$ (in)

400

$v_2 \approx 2$

$\approx 26.14 \text{ ft/sec}$

$2.448 * (2.5)$

Velocity in mud pits, $v_1 \approx 0$

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$p_2 \approx p_1 \approx 0.052 \approx (D_2 \approx D_1)$

$\approx 8.074 * 10^{-4}$

$\approx (v \approx v) \approx P_p \approx P_f$

2

2

2

1

$p_2 \approx 0 \approx 0.052 * 12 (10,000 - 0)$

$- 8.074 * 10^{-4} * 12 (26.142 \approx 0.2) \approx 3,000 \approx 1,400$

$\approx 0 \approx 6,240 \approx 6.6 \approx 3,000 \approx 1,400$

Pressure at bottom of drill collars = 7,833 psig

NOTE: KE in collars ≈ 0

May be ignored in many cases

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Drilling Hydraulics

Introduction to Natural Gas Processing

$\approx$ Bit Hydraulics

- The term critical velocity is used to define the single velocity at which the
- flow regime changes from laminar to turbulent. This variable is the most
- important since all other members are considered constant in a typical equation.
- Since no single Reynolds number defines the transitional zone, it follows that a
- range of critical velocities may be necessary to determine the flow regime.
- In practical applications, a critical velocity (V_c) and an actual velocity (V_a)
- are calculated. If $V_a < V_c$ the flow is laminar. If $V_c < V_a$ the flow is turbulent. If
- $V_a @ V_c$ calculations are made with both flow regimes and the larger pressure
- losses are used.

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$p_2 \approx p_1 \approx 0.052 \approx (D_2 \approx D_1)$

$\approx 8.074 * 10^{-4}$

$\approx (v \approx v) \approx P_p \approx P_f$

2

2

2

1

$$p_2 - p_1 = 0.052 \times 12 (10,000 - 0) - 8.074 \times 10^{-4} \times 12 (26.142 - 0.2) = 3,000 - 1,400 = 1,600 \text{ psi}$$

Pressure at bottom of drill collars = 7,833 psig

NOTE: KE in collars = 0

May be ignored in many cases

Ogbon Advance Drilling Engineering Slide #27

$$p_2 - p_1 = 0.052 \times (D_2^2 - D_1^2)$$

$$= 8.074 \times 10^{-4}$$

$$\times (v_1^2 - v_2^2) + P_p - P_f$$

$$2$$

$$2 \times 1$$

$$2$$

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Fluid Flow Through Nozzle

Assume: $D_2 = D_1$, $v_1 = 0$

$$P_p = 0, v_2 = v_n$$

$$P_f = 0$$

$$= 4$$

$$p_2 - p_1 = 8.074 \times 10^{-4} \times v^2$$

$$n$$

$$= p$$

$$\text{and } v_n = 4$$

$$8.074 \times 10^{-4}$$

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$$P_f = 0, P_{if}$$

Equation may be written as

$$= p$$

$$v_n = C_d \times 4$$

$$C_d = 0.95$$

$$8.074 \times 10^{-4}$$

This accounts for all the losses in the nozzle.

$$1,000$$

$$\text{Example: } v_n = 0.95 \times 4$$

$$= 305 \text{ ft/sec}$$

$$8.074 \times 10^{-4} \times 12$$

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For multiple nozzles in //

V_n is the same for each nozzle even if
the d_n varies!

This follows since ρp is the same across each
nozzle.

$$\rho p q$$

$$v_n \propto d_n^2 \text{ \& } v_n \propto$$

$$8.074 \times 10^{-4} \times 3.117 A t$$

$$8.311 \times 10^{-5} \times q^2$$

$$\Delta p_{bit} \propto 2^2$$

$$C A d t$$

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Hydraulic Horsepower

HHP of pump putting out 400 gpm at 3,000 psi = ?

Power $\propto$ rate of doing work

$$\propto F \cdot s/t$$

$$\propto \rho p \cdot A \propto$$

$$q$$

$$A$$

$$PH \propto q \cdot p$$

In field units:

$$q \cdot p = 400 \times 3,000$$

$$\text{HHP} \propto 700 \text{ hp}$$

$$1714$$

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Hydraulic Impact Force

What is the HHP Developed by bit?

Consider:

$$C D \propto 0.95$$

$$q \propto 400 \text{ gal/min}$$

$$\propto 12 \text{ lb/gal}$$

$$\Delta p_n \propto 1,169 \text{ psi}$$

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Impact = rate of change of momentum

$$\propto m v \propto m \propto q v_n$$

$$F_j \propto \rho q v \propto$$

$$\propto t \propto t \propto 3.2 \times 10^6$$

$$C D \propto 0.95$$

$$q \propto 400 \text{ gal/min}$$

$F_j = 0.01823 c d q^{0.5} p$

$q = 12 \text{ lb/gal}$

$\Delta p_n = 1,169 \text{ psi}$

$F_j = 0.01823 * 0.95 * 40012 * 1,169^{0.5} = 820 \text{ lb f}$

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Optimum Bit Hydraulics

- Under what conditions do we get the best hydraulic cleaning at the bit?
 - Maximum hydraulic horsepower?
 - Maximum impact force?

Both these items increase when the circulation rate increases.

However, when the circulation rate increases, so does the frictional pressure drop.

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Jet Bit Nozzle Size Selection

- Nozzle Size Selection for Optimum Bit Hydraulics:

- Max. Nozzle Velocity
- Max. Bit Hydraulic Horsepower
- Max. Jet Impact Force

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Jet Bit Nozzle Size Selection

□ Proper bottom-hole cleaning

Will eliminate excessive regrinding of drilled solids, and

Will result in improved penetration rates

□ Bottom-hole cleaning efficiency

□ Is achieved through proper selection of bit nozzle sizes

Ogbon Advance Drilling Engineering Slide #38

Jet Bit Nozzle Size Selection

- Optimization -

Through nozzle size selection, optimization may be based on maximizing one of the following:

Bit Nozzle Velocity

Bit Hydraulic Horsepower

Jet impact force

- There is no general agreement on which of these three parameters should be maximized.

Ogbon Advance Drilling Engineering Slide #39

Maximum Nozzle Velocity

ΔP_b

From ΔP_c

v nEq. (4.31)

$8.074 \times 10^{-4} \Delta P_b$

d

v n i.e.

ΔP_b

so the bit pressure drop should be maximized in order to obtain the maximum nozzle velocity

Ogbon Advance Drilling Engineering Slide #40

Maximum Nozzle Velocity

This (maximization) will be achieved when the surface pressure is maximized and the frictional pressure loss everywhere is minimized, i.e., when the flow rate is minimized.

ΔP_b is maximized when 1& 2 above are satisfied, at the minimum circulation rate and the maximum allowable surface pressure.

Ogbon Advance Drilling Engineering Slide #41

Maximum Bit Hydraulic Horsepower

The hydraulic horsepower at the bit is ($\Delta P_b q$) is maximized. maximized when

$\Delta P_p + \Delta P_d = \Delta P_b$

$\Delta P_b = \Delta P_p + \Delta P_d$

where ΔP_d may be called the parasitic pressure loss in the system (friction).

Ogbon Advance Drilling Engineering Slide #42

Maximum Bit Hydraulic Horsepower

The parasitic pressure loss in the system,

$\Delta P_d = \Delta P_s + \Delta P_{dc} + \Delta P_{pa} \approx 1.75$

if the flow is turbulent.

In general, $\Delta P_d \propto q^m$

where $0 \leq m \leq 2$

Ogbon Advance Drilling Engineering Slide #43

Maximum Bit Hydraulic Horsepower

$\rho_b \text{ bit} \leq \rho_p \text{ pump} \leq \rho_d \text{ pd} \leq c_q \text{ m}$

$m \leq 1$

$\rho_b \text{ bit} \leq \rho_p \text{ pump} \leq c_q$

$\rho_{PHbit} \leq$

1714 1714

d_{PHbit}

$\rho \leq 0$ when $\rho_p \text{ pump} \leq c (m \leq 1) \leq q \leq 0m$

dq

Ogbon Advance Drilling Engineering Slide #44

Maximum Bit Hydraulic Horsepower

$\rho_p \text{ pump} \leq c (m \leq 1) \leq q \text{ m}$

≤ 0

i.e., when $\rho_p \text{ pump} \leq (m \leq 1) \leq p \text{ d}$

$\leq 1 \leq$

i.e., when $\rho_p \text{ d} \leq \rho_p \text{ pump}$

$\rho_m \leq 1 \leq$

ρ_{PHbit} is maximum when

$\leq 1 \leq$

$\rho_{pd} \leq \rho_p \text{ pump}$

$\rho_m \leq 1 \leq$

Ogbon Advance Drilling Engineering Slide #45

Maximum Jet Impact Force

The jet impact force is given by Eq. 4.37:

$F_j \leq 0.01823 \text{ c d q} \leq \rho_b \text{ bit}$

$\leq 0.01823 \text{ c d q} \leq (\rho_p \text{ pump} \leq \rho_p \text{ d})$

Ogbon Advance Drilling Engineering Slide #46

Maximum Jet Impact Force

$F_j \leq 0.01823 \text{ c d q} \leq (\rho_p \text{ pump} \leq \rho_p \text{ d})$

But parasitic pressure drop,

$\rho_p \text{ d} \leq c_q \text{ m}$

$\rho F_j \leq 0.01823 \text{ c d} \leq \rho_p \text{ q} \leq c_d \text{ q}$

2 m^2

Ogbon Advance Drilling Engineering Slide #47

Maximum Jet Impact Force

Upon differentiating, setting the first derivative to zero, and solving the resulting quadratic equation, it may be seen that the impact force is maximized when,

2

$\frac{1}{2} \rho v^2$

$m \frac{v^2}{2}$

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Federal University of Petroleum Resources Effurun

Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Advance Drilling Engineering

Lesson III

Well Planning - 2

Ogbon Advance Drilling Engineering Slide #2

Directional Well Planning

Directional Well Planning

Ogbon Advance Drilling Engineering Slide #3

Directional Well Planning

□ Basic Features of Well Planning

□ Well Nudging

□ Slots Allocation

□ Major Sections of Deviated Wells

□ Deviated Well Trajectories

□ Well Horizontal and Vertical Sections Design

□ Borehole Calculations

Ogbon Advance Drilling Engineering Slide #4

Planning Well Trajectory

□ Factors Guiding Well Trajectory Planning

□ Rig Location

□ Target Location

□ Geological Features

□ Number of Targets to Be Penetrated

□ Targets Relative Lateral Displacement

From One Another

- Existing Well Paths in the Field
- Required Angle of Penetration Into the Target
- Drilling Practicability – Torque & Drag Profiles

□ Available Technology

□ Technology of Tools and Equipment

Ogbon □ Economics Advance Drilling Engineering Slide #5

Planning Well Trajectory

□ Factors Guiding Well Trajectory Planning

Target

□ Geologist Specified

□ Acceptable Radius of Tolerance
(Specified per Company Policy)

Tolerance Radius

□ Target Choice of Multiple Pay
Zones Should Ensure Drillability of
Well.

□ Slot Allocation

□ Number of Targets for the well

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Planning Well Trajectory

Lead Angle

□ Additional angle to wellbore direction to the “left” during kick-off to
offset anticipated roller cone “right hand walk” during rotary section
drilling.

□ As much as 200 may be required in extreme walk scenarios

□ No longer required due to use of steerable systems and PDC bits

□ A lot of wells can now be kicked-off without “Lead Angle”

Geological Structures

Formation Competence/Casing Programme

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Planning Well Trajectory

Slots Allocation to Targets

General Guiding Factors

□ As much as possible, wells with targets in the same general directions,

such as North East or so, should be allocated slots in the North East quadrant of the platform excepts when other field projects such as water injection program demands otherwise

- Planned water injection wells should be allocated slots to guarantee easy manifolding

- Wells with smallest horizontal displacements targets should be allocated innermost slots. (These well are normally associated with deeper Kick-Off points)

- Well with larger horizontal displacement targets should be allocated outer slots (these wells are normally associated with shallow Kick-Off points)

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Planning Well Trajectory

Slots Allocation to Targets

Directional Wells on Stationary Platform

Layout of Wells Directions of Targets

Layout of Targets on the Geological Structure

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Planning Well Trajectory

Major Sections of the Deviated Well

- Vertical Section

- Kick-off Point and Build Section

- Tangent Section

- Drop-off Section

- Horizontal Section

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Planning Well Trajectory

- Kick-off Point Section

Kick-off Point General Guiding Factors

- Shallow Kick-off Profile

(Associated with Higher Build-up Rate)

- Deep Kick-off Profile

- Horizontal Profile

- BUR □ 1 – 2.5 dogleg

- Build-up and Drop-off Section

- Units of BUR and DOR

0/30 ft.

0/100 ft.

0/10 m

0/30m

□ Tangent Section

□ Maintaining angle in Confined to an Imaginary

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Planning Well Trajectory

Directional Well Trajectory (Patterns)

Shallow K/O Maintaining Deep Kick-off

course to Target Shallow Maintaining Course S-Shape type

K/O Maintaining course to target

to Target

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Planning Well Trajectory

□ Horizontal Patterns

A directional well drilled with □ Extended Reach

inclination built progressively from 00

to 900 □ (Horizontally) □ Medium Radius

□ Long Radius With Horizontal

Section

□ Inverted Long Radius

S-Shape

□ S-Shape

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Planning Well Trajectory

□ Multibranch/Multilateral Type

□ Definition:

Directional Well With a Single Trunk Connecting Two or More Wells of Any Trajectory

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Planning Well Trajectory

Other Well Trajectory Planning Features

□ Formation

□ Harder Formation Provide Required Stability Integrity

□ Soft Formation Results in Caving In, Washout, Etc

□ Existing Wells

□ To Prevent Collision

□ To Prevent Magnetic Interference

□ To Ensure Accurate Magnetic Survey Spacing

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Planning Well Trajectory

Other Well Trajectory Planning Features

□ Azimuth or Hole Direction

□ Strictly Directional Wells Require ONE Azimuth

□ Complex Wells

□ Require Changing (Multiple) Azimuths

□ Well Penetrating Multiple Targets Get Laterally Displaced Resulting:

□ Incorrect Inclination Value Obtained as HD is Measured as Straight Line

□ Produce One Vertical Section for Each Azimuthal Change

□ Actual HD Measure by DART Computer Programme

□ Horizontal Wells

□ Multilateral Wells

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Planning Well Trajectory

Example of Multiple Targets Vertically & Laterally Displaced Requiring Complex Design

Six Vertical Wells

Single Well "A"

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Planning the Directional Well

Two main sections of a directional well are planned;

□ The Vertical Section

□ Horizontal Section

□ Vertical Section

□ Well Course Projection onto a Specific Vertical Plane – the Well Plan Vertical Plane

□ A Plot of a Graph of Vertical Co-ordinates versus Projection of the Horizontal Displacement onto that Plane

K

Ao □

Proposed Direction Vertical Plane - ABJK D

Actual Well Direction Vertical Plane - ABCD

J

Actual Well Path - AC B □

Projected Well Vertical Path on

Proposed Vertical Plane (Vertical Section) - AJ C

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Planning the Directional Well

□ Vertical Section

□ REQUIRED INFORMATION

□ Surface Location Depth

□ Zero on Land

□ Rotary Table Elevation (RTE) Height from Sea Bed Floor on Offshore

□ Target Location Depth

□ Total Target Displacement (D)

□ Kick-off Point(KOP) Depth KOP

□ Build-up Rate

□ Drop-off Rate

D

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Planning the Directional Well

□ Horizontal Section

□ Required Information

□ Surface Co-ordinates

HS

□ Target Co-ordinates

□ Target TVD

□ Target Area Limit

□ Geological Obstacles (if any)

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Planning the Directional Well

□ Methods

□ Graphical Method

□ Required Data:

□ Given Target Location Co-ordinates Target

□ Given Surface Location Co-ordinates

□ N/S

□ STEP I

A

□ Choose Convenient Scale

Surface Location

□ STEP II

□ Determine □ N/S and □ E/W
(Origin)

□ E/W

□ STEP III

□ Take Surface Location

□ Plot □ N/S on Y-axis

□ Plot □ E/W on X-axis

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Planning the Directional Well

□ STEP IV

□ Draw Line ST from Surface to Target Location

□ ST is Displacement

□ STEP V

□ Measure Angle GST (□ GST)

□ □ GST is the Azimuth or Direction of the Well

□ Always Measure from Surface Location to Target

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Planning the Directional Well

□ Calculation Method

□STEP I

□Determine N/S and E/W as in Step I Above

□STEP II

□Calculate Displacement (D)

$$D = \sqrt{N^2 + S^2} + \sqrt{E^2 + W^2}$$

□STEP III

□Calculate Azimuth or Direction (A)

□E / W

$$A = \tan^{-1}$$

□N / S

□STEP IV

□Draw the Plan as in Graphical Method Above

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Planning the Directional Well

□as

Ogbon Advance Drilling Engineering Slide #24

Planning the Directional Well

□as

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Planning the Directional Well

□ The Overview Vertical Section of a Directional Well

Kick-Off

Point

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Planning the Directional Well

□Graphical Method

□STEP I

□Select Appropriate Scale

□STEP II

□Calculate True Vertical Depth

$$\text{TVDT} = \text{BRT}(\text{BDF}) + \text{TVDGIVEN}$$

□STEP II

□Plot BRT(BDF), TVD, D, KOP & Target on Graph

□STEP III

□Construct Build Arc at KOP Elevation

180

$$R = \text{CI} ; \text{ where BUR} = 0 / \text{CI}$$

$$\text{BUR}$$

e.g. BUR = 50/100 ft; CI = 100 ft

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Planning the Directional Well

□ Graphical Method

□ STEP IV

□ Construct Tangent Line From Build

Arc to Target

□ Measure Tangent (Slant Section)

Line Length (SL)

□ STEP V

□ Measure Maximum Inclination (IMAX)

□ STEP VI

□ Calculate Length of Build Arc (BUAL)

$I_{MAX} \propto R \propto I_{MAX}$

Build Arc Length (BUAL) □ □

BUR 180

□ STEP VII

□ Calculate Total Measure Depth (MT)

□ $MT = KOP + BUAL + SL$

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Planning the Directional Well

□ Step III has to be Repeated for Drop □ Always Design to Avoid Obstacles

Section of S-Type

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Planning the Directional Well

Horizontal Projection (Section)

□ Need to Avoid Other Wells in the

Same Multi-Well Platform

Necessitates Kicking-off wells in different Directions other than the Straight Line Horizontal Projection.

□ This Generates 3-Dimensional Turn

Horizontal projection as Indicated on Figure Below.

Example of 3-Dimensional Turn on the Horizontal Plan

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Planning the Directional Well

Worked Example

□ Question

A vertical well with N72456.4 and E21995.6 was drilled to top of reservoir sand Bemba at 11250 ft when it blew out. To kill the well, a relief well was sited at N61986.8 and E18794.7 The relief well was planned to kick-off at 600 ft and build angle to 500 and thereafter maintain inclination to intersect the vertical well at the top of Bemba. We only have a 30 bent sub available at site

□ Requirements

- What is the horizontal displacement of the relief well?
- What will be the azimuth of the relief well?
- What is the total vertical depth of the relief well?
- What is the total measured depth of the relief well?
- Construct both the horizontal and vertical section of the relief well.
- Assuming the Bemba has pore pressure of 9000 psi, what will be the mud weight with which Bemba must be intersected to kill the blowing well?

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Planning the Directional Well

□ Answer

Worked Example

1. Relief Well Horizontal Displacement

$$HD = \sqrt{N^2 + E^2} = \sqrt{(72456.4 - 61986.8)^2 + (21995.6 - 18794.7)^2}$$

$$= 10469.62 \approx 3200.92 \approx 10947.98 \approx 10948 \text{ ft}$$

2. Relief Well Azimuth

$$A = \tan^{-1} \Delta E. = \tan^{-1} 3200.9 = 170$$

$$\Delta N 10469.6$$

3. Vertical Depth of Relief Well

$$i. \text{ Radius of Curvature } R = 180 \times CL = 180 \times 100 = 1909.6 \text{ ft}$$

$$\pi \times BUR 3.142 \times 3$$

$$ii. \text{ Build TVD} = R \times \sin I = 1909.9 \times \sin 50 = 1463.1 \text{ ft}$$

$$iii. \text{ Build Section HD} = R \times (1 - \cos I) = 1909.9(1 - \cos 50) = 682.2 \text{ ft}$$

$$iv. \text{ Tangent Section TVD} = THD - BHD = 10948 - 682.2 = 8614 \text{ ft}$$

$$\tan I \tan 50$$

$$v. \text{ Total TVD} = 600 + 1463.1 + 8614 = 10677.1 \text{ ft}$$

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Planning the Directional Well

Worked Example

□ Answer

4. Relief Well Measured Depth I MAX □ R I MAX

$$i. \text{ Length of Build Section(BUAL) Build Arc Length(BUAL)} \approx \pi$$

BUR 180

$$= 50 \times 100 = 50 \times 3.142 \times 1909.9 = 1666.7 \text{ ft}$$

3 180

ii. Length of Tangent Section

$$= \text{Tangent Section TVD} = 8614 = 13401 \text{ ft}$$

$\cos I \cos 50$

$$\text{iii. Total Measured Depth} = 600 + 1666.7 + 13401 = 15668 \text{ ft}$$

□ What is the Horizontal Displacement of the Tangent Section ?

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Planning the Directional Well

Take Home Exercise

□ Question

An oil discovery has been made 11980 ft under the FUPRE Administrative block. The Admin, Block coordinates are N88000 and W32000. As a drilling engineer of the FUPRE Consult you have been saddled with the responsibility of designing a well to be located at the Ugbomro Cooperatives Society Farm land with S45000 and W66000 coordinates. Kick-off depth, maximum inclination and available bend-sub are respectively 1000ft, 480 and 2.50.

□ Requirements

- What is the horizontal displacement of the relief well?
- What will be the azimuth of the relief well?
- What is the total vertical depth of the relief well?
- What is the total measured depth of the relief well?
- Construct both the horizontal and vertical section of the relief well.
- If the pore fluid in this area has average specific gravity of 1.21 and company policy is to drill with overbalance of 200 psi, what will be your drilling fluid mud weight?

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Federal University of Petroleum Resources Effurun

Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Advance Drilling Engineering

Lesson IV

Fluid Flow in Drill Pipes

Ogbon Advance Drilling Engineering Slide #2

Lesson IV: Fluid Flow in Pipes

Introduction to Natural Gas Processing

Sub-Topics

- Introduction
- Flow Regime in Pipes
- Friction Factor of Flow in Pipe
- Phases of Fluid Flow
- Phase Behaviour
- Applications of Two-Phase Flow
- Challenges of Two-Phase Flow
- Flow in Pipes

Ogbon Advance Drilling Engineering Slide #3

Introduction to Flow in Pipes

Introduction to Natural Gas Processing

- The study of fluid flow in pipes is a necessity to the understanding and execution of the drilling process.
- First of all, the well is a conduit so also are the casings and drill pipes through which drilling fluid flow during the drilling process.
- Reservoir fluid is also lifted to the surface from the reservoir through a metal conduit called tubing.
- To successfully drill a well, it sometimes requires the application of different types of drilling fluids such as water, water-base mud, oil-base mud, oil, air, foam, synthetic drilling fluids, etc. These exhibit varied rheological properties that impart differently on the flow through the well.
- Drilling also generates cuttings, formation and reservoir fluid mixing up with the mud resulting in multi-phase that impart on the property of the mud and thus the flow.

Ogbon Advance Drilling Engineering Slide #4

Introduction to Flow in Pipes

Introduction to Natural Gas Processing

- Proper understanding of the geometry, flow patterns, flow regimes, rheological properties, flow behavior of fluid flow in pipe is therefore imperative for successful drilling of oil and gas well.
- Above all, fluid flow enable the drilling engineer to determine the total frictional and pressure drop across the entire flow path which enables the determination of pump capacity.

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Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

The manner in which drilling fluid behaves or its “character” while flowing in drill pipe is commonly referred to as Flow Regime.

It refers to the relation between the different layers to each other in moving fluid and it mostly depends on the relation between the Fluid Velocity & the Critical Velocity

It is the behavior of fluid during flow in conduit. There are three main and basic types of flow regimes:

□ Types of Flow Regime

There are three main types of flow regime

Ogbon Advance Drilling Engineering Slide #6

Flow Regime in Pipes

Introduction to Natural Gas Processing

□ Flow Regime

□ Laminar Flow

□ This is the flow behavior that exists at the onset of the very low pump rate with low noise.

□ Fluid layers In laminar flow are parallel to each other in an orderly fashion.

□ Flow occurs at low to moderate shear rates when friction between the fluid and the channel walls is at its lowest.

□ It is a typical flow in the annulus of most wells.

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Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

□ Laminar Flow

□ Characteristics of Laminar Flow

□ It is characterized by the movement of individual layers of fluid stream through the pipe and annulus with low friction pressures and minimum hole erosion.

□ Every fluid molecule follows a straight path that is parallel to the boundaries of the tube

□ The center layers moves faster than the layers near the wellbore or pipe resulting in parabolic velocity profile at the tip of the flow y-axis

□ Flow pattern is linear (no radial flow)

Laminar Flow Velocity Profile in pipe is parabolic indicating that flow velocity at the walls is slower than in the center
x-axis

Ogbon Advance Drilling Engineering Slide #8

Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

□ Laminar Flow

□ Characteristics of Laminar Flow

□ Velocity has only one single component - horizontal

□ Velocity at wall is ZERO and Maximum at the center

Velocity distribution in laminar motion of fluid in horizontal circular pipe

□ Produces minimal hole erosion

□ Mud properties are mainly responsible for the flow behavior and strongly affect pressure losses

□ It is the preferred flow type for annulus (in vertical wells)

□ Laminar flow is sometimes referred to as sheet flow, or layered flow because its flow profile is in variations in the layers velocities

Ogbon Advance Drilling Engineering Slide #9

Flow Regime in Pipes

Introduction to Natural Gas Processing

□ Flow Regime

□ Laminar Flow

□ The effect of Laminar Flow Variations in Velocity Layers

□ Variations are controlled by the shear resistant capabilities of the mud.

□ A high yield point for the mud tends to make the layers move at more uniform rates.

□ Cuttings removal is often discussed as being more difficult with laminar because the cuttings appear to move outward from the higher-velocity layers to the more acquiescent area with low velocity and resulting in less ability to remove cuttings.

□ A common procedure for minimizing the problem is to increase the yield point, which decreases layer velocity variations.

□ Problem can also be resolved by sweeping the cuttings off the annulus by pumping a 10-20-bbl high-viscosity plug.

□ Generally, flow in a very smooth tube or around a smooth, streamlined object will be laminar at low velocity

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Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

□ Laminar Flow

□ Hagen - Poiseuille derived the laminar flow pressure losses and velocity distributions based on momentum and mass conservation to obtain the flow and pressure gradient equations in straight and circular pipes below;

$$\Delta P \propto r^4 \propto Q$$

$$Q \propto$$

$$8 \frac{\Delta P}{L} = \frac{32 \mu Q}{r^4}$$

r = pipe radius

u = flow velocity

The pressure gradient is L = pipe length

ΔP = pressure drop

$\Delta P \propto Q$ $\propto U$ Q = flow rate

$$\propto \frac{1}{r^4}$$

U = average flow velocity

The pressure drop is

□ The average velocity is given from the above equation as

Ogbon Advance Drilling Engineering Slide #11

Flow Regime in Pipes

Introduction to Natural Gas Processing

□ Flow Regime

Example

A straight stretch of horizontal pipe having a diameter of 5 cm is used in the laboratory to measure the viscosity of crude oil (density = 0.93 ton/m³). During a test run a pressure difference of 1.75 ton/m² is obtained from two pressure gages, which are located 6 m apart on the pipe. Oil is allowed to discharge into a weighing tank, and a total of 550 kg(0.55ton) of oil is collected for a duration of 3 min. Determine the viscosity of the oil.

SOLUTION: The discharge of oil flow in the pipe is

$$Q = \frac{\text{Vol}}{\text{Time}} = \frac{M}{\text{Time}} = \frac{0.550}{0.003395} \text{ m}^3/\text{sec}$$

$$\text{Time} = 0.93 \times 3 \times 60$$

$$V = 4Q = 4 \times 0.003395 = 1.729 \text{ m/sec}$$

$$r^2 = 0.052$$

From

$$\mu = \frac{1.75 \times 0.052}{32 \times 1.729 \times 6} = 1.32 \times 10^{-5} \text{ tsec/m}^2$$

$$32 \times 1.729 \times 6$$

Ogbon Advance Drilling Engineering Slide #12

Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

□ Turbulent Flow

□ With high flow rate, the flow streamlines are not steady and straight; and with the particles being pushed forward by current eddies, the flow seems to be laminar in nature and structure and it becomes turbulent.

□ Turbulence occurs when the increased velocity between the layers create higher shear rates and strengths that exceed the ability of the fluid to remain in laminar flow.

□ Turbulent flow layered structure is characterized by disorderly, chaotic, erratic and continuous swirling into the walls with the velocity at the walls being significantly higher than the wall layer in laminar flow.

Chaotic continuous swirling flow behaviour of turbulent flow in pic and diagram

Ogbon Advance Drilling Engineering Slide #13

Flow Regime in Pipes

Introduction to Natural Gas Processing

□ Flow Regime

□ Turbulent Flow

□ In turbulent flow the force leading to stability – viscosity – is small relative to that leading to instability which is inertia

□ The friction between the fluid and the channel walls is highest with turbulent flow and it is typically inside the drill pipe and drill collars.

□ Turbulent flow occurs in the drill pipe and around the drill collars but increases hole erosion in the annulus.

□ Generally, at high velocity, even flow in smooth pipes or around smooth object will experience turbulence

Ogbon Advance Drilling Engineering Slide #14

Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

□ Turbulent Flow

□ Characteristics of Turbulent Flow

□ Flow pattern is random (flow in all directions)

□ Instantaneous and erratic streamlines direction change

□ Fluid behavior is chaotic, erratic and irregular but can be determined.

□ Turbulent flow results in higher pressure losses (takes more energy)

□ Provides excellent hole cleaning but can result in hole erosion too.

- Mud properties have little effect on pressure losses
- It is the usual flow type inside the drill pipe and collars
- Thin laminar boundary layer at the wall
- Every fluid molecule follows very complex path that lead to a mixing of flow components

Ogbon Advance Drilling Engineering Slide #15

Flow Regime in Pipes

Introduction to Natural Gas Processing

- Flow Regime
- Turbulent Flow
- Turbulence Criteria

Several common methods can be used to establish turbulence criteria including the

- The Reynolds number.
- Intersection of the flow rate vs. pressure loss calculations for laminar and turbulent flow.

The most common approach employed in the oil & gas industry to determine turbulence and other flow regimes is the Reynold's Number(NRe)

The viscosity of the fluid considered is dynamic viscosity.

Density can vary with respect to pressure, because it is assumed that the fluid is only weakly compressible

We will assume that the fluid is Newtonian, meaning that the viscosity is constant with respect to shear rate. This is true, or very nearly so, for a wide range of drilling fluids in drilling.

Ogbon Advance Drilling Engineering Slide #16

Flow Regime in Pipes

Introduction to Natural Gas Processing

- Flow Regime
- Turbulent Flow
- Turbulence Criteria
- The Reynold's Number(NRe)

□ The Reynolds number is a dimensionless parameter that is useful in characterizing the degree of turbulence in the flow regime and is needed to determine the Moody friction factor.

- It reveals whether a particular flow is laminar, transitional or turbulent
- The Reynolds number approach is used almost exclusively in the oil and gas industry
- Mathematically the equation for the Reynold's number – NRe- in SI units

is Where;

ρ UD inertia D = pipe ID (m)

N Re μ

μ Viscous forces U = flow velocity (m/s)

μ = Liquid viscosity (pascal seconds)

ρ = liquid density (kg/m³)

and in field units

Where;

928 ρ Ud D = pipe ID - in

N Re μ U = avg. flow velocity - ft/s

μ ρ = liquid viscosity - cp.

ρ = liquid density - lbm/gal

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Flow Regime in Pipes

Introduction to Natural Gas Processing

μ Flow Regime

μ Turbulent Flow

μ Turbulence Criteria

μ The Reynold's Number(NRe)

μ Flow Reynolds' number is also commonly expressed in terms of the fluid's Kinematic Viscosity (ν ; lower case Greek letter nu),

ν μ m/s or ft/s

μ Substituting in the Reynolds' number equation, it becomes

UD

R μ

μ Reynold's Number for Gas Flow in field units

Reynold's number for gas is normally expressed as

2 0 ,100 Q g S Where;

N Re μ D = pipe ID - in

D μ S = specific gravity of gas at std. cond. relative to air

μ = liquid viscosity - cp.

Qg = gas flow rate - MMscf/day

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Flow Regime in Pipes

Introduction to Natural Gas Processing

μ Flow Regime

μ Turbulent Flow

μ Turbulence Criteria

μ The Reynold's Number(NRe)

μ For Transition Flow NRe 2000 - 3000

μ Flow is unstable

□ It can be laminar, but small obstructions and surface roughness can make it turbulent

□ It may oscillate randomly between being laminar and turbulent.

$NRe \leq 2000$

$NRe \ 2000 - 3000$

$NRe \geq 3000$

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Flow Regime in Pipes

Introduction to Natural Gas Processing

□ Examples

1. Given two pipes, one with a diameter of 10 cm and the other with a diameter of 1 m, at what velocities will the flows of water in each pipe become turbulent? Water viscosity of water is 1.005×10^{-3} pascal seconds and density is 998.2 kg/m^3 at 20°C .

Solution:

Since flow is turbulent,

$\mu = 1.005 \times 10^{-3}$

$NRe = \frac{\rho U D}{\mu} = 1.007 \times 10^6 \text{ m}^2/\text{s}$

$\rho = 998.2$

Solve for

$D = 0.1 \text{ m}$ diameter pipe $D = 1.0 \text{ m}$ diameter pipe

$2000 \leq 1.007 \times 10^6$ $2000 \leq 1.007 \times 10^6$

$U = U$

0.1 m

$U = 0.02 \text{ m/s} = 2 \text{ cm/s}$ $U = 0.002 \text{ m/s} = 2 \text{ mm/s}$

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Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

□ Transition Flow

□ This is the “Grey” area between Laminar and Turbulent flows.

□ It is the dynamic behaviour of fluid flow when it changes from laminar to turbulent or changes from turbulent to laminar.

□ Every fluid molecule follows wavy but parallel path that is not parallel to the boundaries of the tube

Transition from Laminar to Turbulent

Wavy structure of the Transition flow streamline

$(2000 \leq NRe \leq 3000)$

However, it is often difficult to estimate the flow rate at which laminar flow gives rise to turbulence flow because at intermediate velocities, flow may

oscillate back and forth indefinitely between laminar and turbulent.

□ Above all, turbulence may occur in various stages depending on the flow rate and its geometry, pipe diameter and roughness, etc.

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Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

□ Transition Flow

□ The Critical Velocity of fluid, is the single velocity at which the flow changes from laminar to turbulent and vice versa.

□ This variable is the most important for the determination of the transition regime.

□ Since no single Reynolds number defines the transitional zone, it follows that a range of critical velocities may be necessary to determine the flow regime.

□ In practical applications, a critical velocity (V_c) and an actual velocity (V_{act}) are calculated.

□ If $V_{act} < V_c$ □ flow is laminar.

□ If $V_{act} > V_c$ □ flow is turbulent.

□ If $V_{act} \approx V_c$

□ Generally, to determine pressure losses in transition zone, calculations are made with both flow regimes and the larger pressure losses are used.

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Flow Regime in Pipes

□ Flow Regime

Introduction to Natural Gas Processing

□ Transition Flow

□ Early transitional state of flow ($2000 < N_{Re} < 2300$) for which the flow becomes wavy; but this flow is still laminar

□ Nevertheless for values of N_{Re} between about 2000 and 3000, flow is generally unstable;

□ It can be laminar

□ Can become turbulent due to obstructions and surface roughness

□ It may oscillate randomly between being laminar and turbulent.

□ Furthermore, transitional flow can continue up to 6,000 N_{Re} depending on:

□ The direction of flow

□ Either increasing or decreasing flow

□ The rate of change of the flow rate

□ Transition friction factor for both smooth and rough pipe is given by the equation below where f is determined from the Fanning Friction Chart

$$1/D \leq D/\epsilon$$

$$\log \left(4.0 + \frac{10.71}{\text{Re}^{0.787}} \right) \leq \log \left(4.67 + \frac{1}{\text{Re}^{0.25}} \right)$$

$$f = \frac{16}{\text{Re}}$$

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Flow Regime in Pipes

Introduction to Natural Gas Processing

□ Special Notes of Flow Regimes

□ It should be noted however, that a clear definition of the drilling fluid behavior in the drill pipe during the drilling process is practically impossible because while the drilling fluid flow is predominantly laminar, the flow near the pipe walls during rotation is quite turbulent

□ With low pump rates at the onset of flow, flow is characterized with silence until a critical velocity is reached with pump rate increase.

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Flow Regime in Pipes

Introduction to Natural Gas Processing

□ Special Notes of Flow Regimes

□ Beyond the critical velocity as a result of pump rate increase with increasing pump pressure, flow becomes noisy and turbulent flow starts.(see fig. below)

Reynold's number

Exceeds 2000

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Friction Factor of Flow in Pipe

Introduction to Natural Gas Processing

□ Friction Factor

□ Another factor that affects the rate of flow in pipe is the roughness(ϵ) of the pipe internal surface.

□ The roughness imposes friction or restriction to the flow and manifests in the form of frictional pressure drop

□ To determine the frictional factor (coefficient) f , a dimensionless group (ϵ/D) is needed in addition to the Reynold's number

□ With ϵ/D and NRe , f is determined from either Fanning Chart or Moody Chart

□ Laminar Flow

□ For the laminar flow with $\text{NRe} \leq 2000$

16

$f = \frac{16}{\text{Re}}$ Fanning Chart

Re

64

f □ □ Moody Chart

Re

□ Note that Moody friction factor is 4 times the Fanning friction factor

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Friction Factor of Flow in Pipe

Introduction to Natural Gas Processing

□ Friction Factor

□ Turbulent Flow

□ Turbulent flow losses and velocity distributions require experimental results.

□ Experiments give the relationship between the friction factor and the Reynolds number.

□ For the turbulent and transition flow, the relationship between f and NRe can be determined from the Fanning Chart as follows:

□ For Smooth pipe with NRe □ 3000

1

f

□ □

□ $4.0 * \log Re * f \approx 0.4$ OR $f \approx 0.079 Re^{-0.25}$ Paul Blasius approx.□ For Rough pipe with $[(D/\epsilon)/(Re\sqrt{f}) < 0.01]$

1 D

□ $4.0 * \log \left[2.28 \left(\frac{D}{\epsilon} \right)^{1.1486} \times 10^{-9} \right]$

f .

L

.□Q 2

f □ f D

D 5

L

□ Transition Flow NRe 2000 - 3000 □ $1.8608 \times 10^{-9} f \cdot \frac{L}{D^5}$

D5

1 D □ D/ϵ □□ $4.0 * \log \left[2.28 \left(\frac{D}{\epsilon} \right)^{1.1486} \times 10^{-9} \right]$

f □ □ Re f □ □

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Friction Factor of Flow in Pipe

Fanning Friction Factor Chart

1 D

1 D □ D/ϵ □ □ $4.0 * \log \left[2.28 \left(\frac{D}{\epsilon} \right)^{1.1486} \times 10^{-9} \right]$

f

$$\frac{1}{f} = 4.0 \cdot \log \left(\frac{2.28}{Re} + \frac{4.0}{Re^{0.25}} \right)$$

$$\frac{1}{f} = \frac{1}{Re} \left(2.28 + \frac{4.0}{Re^{0.25}} \right)$$

$$f = \frac{16}{Re}$$

$$f = 0.079 Re^{-0.25}$$

$$f = 16/Re$$

$$f = 0.079 Re^{-0.25}$$

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Friction Factor of Flow in Pipe

Fanning Friction Factor Chart

Fanning Friction Factor Chart

$$\frac{1}{f} = \frac{D}{\epsilon} + \frac{1}{D} \left(2.28 + \frac{4.0}{Re^{0.25}} \right)$$

$$\frac{1}{f} = \frac{D}{\epsilon} + \frac{1}{D} \left(2.28 + \frac{4.0}{Re^{0.25}} \right)$$

$$\frac{1}{f} = \frac{D}{\epsilon} + \frac{1}{D} \left(2.28 + \frac{4.0}{Re^{0.25}} \right)$$

$$\frac{1}{f} = \frac{D}{\epsilon} + \frac{1}{D} \left(2.28 + \frac{4.0}{Re^{0.25}} \right)$$

$$f = \frac{16}{Re}$$

f

$$f = 16/Re$$

$$f = 0.079 Re^{-0.25}$$

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Friction Factor of Flow in Pipe

Moody Friction Factor Chart

Darcy-Weisbach friction factor

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Friction Factor of Flow in Pipe

Moody Friction Factor Chart

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Friction Factor of Flow in Pipe

ϵ Must be

D dimensionless! Pipe Roughness

pipe material pipe roughness ϵ (mm)

glass, drawn brass, copper 0.0015

commercial steel or wrought iron 0.045

asphalted cast iron 0.12

galvanized iron 0.15

cast iron 0.26

concrete 0.18-0.6

rivet steel 0.9-9.0

corrugated metal 45

PVC 0.12

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Friction Factor of Flow in

Introduction to Natural Gas Processing

□ Friction Factor

□ Turbulent Flow

□ Generally, the flow rate equation for both the Moody and Fanning friction factor application are as follows

L

$$\Delta P_f = 1.1486 \times 10^{-9} f D^{.5}$$

.Q² where

D fD = Moody friction factor

L f = Fanning friction factor

$$\Delta P_f = 1.8608 \times 10^{-9} f^{.5} .Q^2$$

D Q = bbl/day

L = ft

D = ft

□ = lb/ft³

□Pf = psi

Example

Determine the frictional pressure drop through 4½" ID wrought iron drill pipe flowing 0.5 bbl/min 1.3 specific gravity mud in a well of 5200 ft. Fluid viscosity is 11.2 cp

_ Where;

928 ρ u d D = pipe ID - in

N Re □ ū = avg. flow velocity - ft/s

μ □ = liquid viscosity - cp.

□ = liquid density - lbm/gal

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Friction Factor of Flow in

Introduction to Natural Gas Processing

□ Friction Factor

□ Turbulent Flow

Example

Determine the frictional pressure drop through 4½" ID wrought iron drill pipe flowing 0.5 bbl/min 1.3 specific gravity mud in a well of 5800 ft. Fluid viscosity is 11.2 cp

Where;

D = pipe ID - in

U = 4Q = 0.5 x 5.615 x 4 = 0.423 ft/sec ū = avg. flow velocity - ft/s

□ D2 60 x 3.142 x 0.3752 □ = liquid viscosity - cp.

$\rho = \text{liquid density} - \text{lbm/gal}$

$\rho = 1.3 \times 8.33 = 10.829 \text{ ppg}$

$928 \rho u d 10.828 \times 0.423 \times 4.5$

$N_{Re} = 928 \times 1,707.78$

$\mu 11.2$

Flow is laminar

and

From Moody friction chart $f_D = 64 = 64 = 0.0375$

$N_{re} 1,707.78$

9^2

$Q = \text{bbl/day}$

$L 1.1486 \times 10^0 0.0375 \times 5,800 \times 1.3 \times 62.4 \times 720$

$\Delta P_f = 1.1486 \times 10^9 f_D . 5$

$\Delta Q^2 = 1,416.68 \text{ psi } L = \text{ft}$

$D 0.3755 D = \text{ft}$

$\rho = \text{lb/ft}^3$

$\Delta P_f = \text{psi}$

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Phases of Fluid Flow

Introduction to Natural Gas Processing

A phase is simply one of the state of matter and can be either a gas , a liquid or a solid.

Multiphase flow is a simultaneous flow of several phases .

Two-phase flow is the simplest case of multiphase flow .

Two-phase flow involving a mixture of gas or vapor and liquid is very common in various natural occurrences, and in industrial and scientific applications, and these involve a sequence of different two-phase flow configuration or flow patterns .

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Phases of Fluid Flow

Introduction to Natural Gas Processing

Multi-Phase Flow

Different fields of multi phase flow

<http://www.potto.org/fluidMech/phase.php>

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Phases of Fluid Flow

Two Phase Flow Introduction to Natural Gas Processing

The laws governing two phase flow are identical to those for single phase flow, but the equations are more complex and/or more numerous

than those of single phase flow.

□ The description of the two-phase flow is complicated due to the existence of interface between the phases depending on a large number of variables such as:

- Fluid quality.
- Phase physical properties .
- Fluid flow patterns .
- Pipe geometry .
- Orientation of flow - Vertical, inclined or horizontal
- General classifications divide two-phase flow into four groups depending on the mixtures of phases in the flow.
- Gas-liquid,
- Gas-solid,
- Liquid-solid
- Immiscible liquid-liquid mixtures - Emulsion
- Three-phase flows (e.g. gas-oil-water Three-phase flows (e.g. gas-oil-water

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Phases of Fluid Flow

Introduction to Natural Gas Processing

- Two-Phase Flow
- Types of Components-Two-Phase Flow
- Two-Phase Two-Components Flow

When the two components are different in phase, e.g.

- Gas-Liquid Mixture flow
- Smoke: Gas-Sooth mixture flow
- Cloud, Fog, Snow, Dust storm, etc.
- Two-Phase Single-Components Flow

When the two components are same in phase, eg.

- Liquid-Vapour Mixture flow
- Single-Phase Two-Components Flow

When the two components are the same phase but immiscible

- Liquid-liquid mixtures - Emulsion
- The last case is technically not a two-phase mixture, it is rather a single phase two-component flow, but for all practical purposes it can be considered as a two-phase mixture.

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Flow Behaviour

Introduction to Natural Gas Processing

There are several ways of determining the behavior of fluid flow in Pipes including

□ The flow Pattern and Two Phase Flow Regime Mapping

□ Flow Patterns/Regimes

□ Flow Patterns or Flow Regimes are the internal phase distributions and different interfacial structures during fluid flow

□ The flow patterns have their respective different characteristics and associated pressure drops.

□ Flow Patterns/Regimes in Horizontal and Vertical Flow

Flow patterns/regimes for horizontal flow is complicated by gravitational forces which act on the liquid phase causing it to be displaced towards the bottom of the channel.

The various patterns/regimes include;

□ Bubble flow .

□ Slug flow .

□ Plug flow .

□ Stratified flow (layered, separated) .

□ Wavy flow (ripple flow, cresting) .

□ Semi-annular flow .

□ Annular flow (ringed) .

□ Spray flow (mist, froth, dispersed) .

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Flow Behaviour

Introduction

□ Flow Patterns/Regimes in Horizontal Flow to Natural Gas Processing

□ Segregated flow

□ Stratified

□ Annular

□ Wavy

□ Intermittent

□ Slug flow

□ Plug flow

□ Bubble flow

□ Distributive flow

□ Bubble/mist flow

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Flow Behaviour

□ Vertical Flow in Pipes Introduction to Natural Gas Processing

□ Bubble flow.

□ Slug flow.

□ Churn flow.

□ Annular flow

□ Ripple Flow.

□ Mist Flow .

Slug Bubble Separated Annular

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Flow Behaviour

□ Vertical Flow in Pipes

□ Bubble flow

□ Continuous liquid phase with dispersed bubbles of gas

□ Slug flow

□ Large gas bubbles

□ Slugs of liquid (with small bubbles) in between

□ Churn flow

□ Bubbles start to coalesce

□ Up and down motion of liquid

□ Annular flow

□ Gas becomes the continuous phase

□ Droplets in the gas phase

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Flow Behaviour

□ Solid-Liquid Flow

□ Flow is mainly controlled by fluid velocity

Types of flow patterns in solid-liquid flow.

<http://www.potto.org/fluidMech/phase.php>

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Flow Behaviour

Introduction to Natural Gas Processing

□ Two Phase Flow Regime Mapping

□ Two Phase Flow Regime Mapping

□ Mapping of flow patterns that occur in pipe flow has always been a popular means of describing the behaviors of flow at different conditions.

□ The superficial velocity of the gas and liquid are usually put on the two different axes, and supply an efficient method of comparing and contrasting the effects of different flow conditions .

Horizontal Flow Vertical Flow

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Two Phase Flow Applications

Introduction to Natural Gas Processing

The practical importance in many common engineering and industrial applications are in:

- Oil industry where two phase flow occurs in pipelines carrying oil and natural gas.
- Separation of reservoir fluid
- Steam generators and condensers, steam turbines (Power Plants).
- Refrigeration .
- Coal fired furnaces .
- Fluidized bed reactors .
- Liquid sprays .

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Challenges of Two-Phase Flow

Introduction to Natural Gas Processing

- Challenges of Two-phase Flow
- Severe liquid Slugging
- Hydrates
- Scale
- Wax/Paraffin deposit
- Emulsion
- Corrosion – Erosion (CO₂”, H₂S,Sand)
- Generally, in drilling fluids dynamics, the Darcy–Weisbach equation is mostly employed to determine the flow in the well.

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Flow in Pipes

- The Darcy-Weisbach Equation Introduction to Natural Gas Processing
- Generally, in drilling fluids dynamics, the Darcy–Weisbach equation is mostly employed to determine the flow in the well.
- The equation, named after Henry Darcy and Julius Weisbach, relates the frictional pressure loss(head) in the pipe length to the average velocity of the fluid flow.
- The other factor in the equation is the dimensionless Darcy friction factor also known as Darcy–Weisbach friction factor or Moody friction factor.

$L V^2$

$\Delta P_f \propto f D \cdot \rho$

D^2

where the pressure loss due to friction Δp (Pa) is a function of:

- the ratio of the length to diameter of the pipe, L/D ;
- the density of the fluid, ρ (kg/m³);
- the mean velocity of the flow, V (m/s), equal to the volumetric flow rate per unit

cross-sectional wetted area(m/s);

- Darcy Friction Factor(f_D); a (dimensionless) coefficient of laminar or turbulent flow

□ Using Fanning friction factor the above equation becomes

L

$$\Delta P_f = f \cdot L \cdot \frac{V^2}{D}$$

D

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Flow in Pipes

□ The Darcy-Weisbach Equation

Introduction to Natural Gas Processing

□ From the above equation, flow rate can be derived by substituting Q/A for V

2

$$\Delta P_f = f \cdot L \cdot \frac{Q^2}{A^3}$$

□ □ In Field Units

$$L = A$$

$$\Delta P_f = f \cdot D \cdot \frac{Q^2}{A^3}$$

$$D^2 Q = \text{bbl/day}$$

$$L = \text{ft}$$

In SI Units $D = \text{ft}$

$$\Delta P = \text{lb/ft}^3$$

$$\Delta P_f = \text{psi}$$

2

$$\Delta P_f = D^2 \cdot \frac{Q^2}{A^3}$$

$$\Delta P_f = \frac{D^2 \cdot Q^2}{A^3}$$

□ Substituting $\frac{4}{\pi} \cdot \frac{16}{A}$ for A

L

$$L \cdot \frac{16 Q^2}{\pi^2 A^3} = \Delta P_f \cdot \frac{1.1486 \times 10^{-9} f \cdot D \cdot L}{\pi^2 A^3}$$

$$\Delta P_f = f \cdot D \cdot \frac{Q^2}{A^3} \cdot \frac{16}{\pi^2} \cdot \frac{1}{D^5} \cdot \frac{D^5}{A^3}$$

$$D \cdot \frac{D^2 \cdot D}{19.74}$$

L

$$L \cdot \frac{1.8608 \times 10^{-9} f \cdot D \cdot L}{\pi^2 A^3} = \Delta P_f \cdot \frac{1.1486 \times 10^{-9} f \cdot D \cdot L}{\pi^2 A^3}$$

$$\Delta P_f = f \cdot D \cdot \frac{Q^2}{A^3} \cdot \frac{16}{\pi^2} \cdot \frac{1}{D^5} \cdot \frac{D^5}{A^3}$$

$$D \cdot D$$

$$\Delta P_f = \frac{16 Q^2}{\pi^2 A^3}$$

$$\Delta P_f = \frac{16 Q^2}{\pi^2 A^3}$$

$$\Delta P_f = \frac{16 Q^2}{\pi^2 A^3}$$

$$\Delta P_f = \frac{16 Q^2}{\pi^2 A^3} \cdot \frac{1.1486 \times 10^{-9} f \cdot D \cdot L}{\pi^2 A^3}$$

$$Q = 2.47 \cdot \frac{1}{0.6173} \cdot \frac{f \cdot D \cdot L}{\pi^2 A^3}$$

$$\Delta P_f = \frac{16 Q^2}{\pi^2 A^3}$$

□ Note that this friction factor is four times that of the Fanning so attention must be paid to which "friction factor" chart or equation to be employed

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Take Home Exercise

Introduction to Natural Gas Processing

Starting with the units Darcy-Weisbach frictional pressure and flow equations,
derive same equations in field units

$$Q = \text{bbl/day}$$

$$L = \text{ft}$$

$$D = \text{in}$$

$$\rho = \text{lb/gal}$$

$$\Delta P_f = \text{psi}$$

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Flow in Pipes

Introduction to Natural Gas Processing

□ The Darcy-Weisbach Equation

□ In using friction factor charts, inspect if the formula is included

□ If not, then observe the value of the friction factor for laminar flow at a
Reynolds number of 1000.

□ If the value of the friction factor is 0.064, then the Darcy friction factor is plotted
in the Moody diagram and If the value of the friction factor is 0.016, then the
Fanning friction factor is plotted in the Moody diagram.

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Flow in Pipes

Introduction to Natural Gas Processing

□ Pressure Loss Calculation in Pipe and Annulus

□ The prediction of friction pressure losses is very important in many oil field
operations, including drilling, completion, fracturing, acidizing, workover and
production.

□ Many theoretical and experimental studies have dealt with the flow of fluids
through pipes and annuli for friction pressure loss calculations.

□ Most of these studies have concentrated upon the fluids rheological models,
pipe roughness, and geometrical parameters.

□ However, the important effect of tool joint in conjunction with the drillpipe in
estimating the friction pressure loss in annuli has yet not been investigated.

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Fluid Flow in Pipes

□ Average Fluid Velocity

Pipe Flow Annular Flow

$$q$$

$$v$$

$$2.448 \frac{q}{d_2^2 - d_1^2}$$

$$2.448 \frac{q}{d_2^2 - d_1^2} \text{ ft/s}$$

Where;

v = average velocity, ft/s

q = flow rate, gal/min d_2 = 7" d_1 = 5"

d = internal diameter of pipe, in.

d_2 = internal diameter of outer pipe or borehole, in.

d_1 = external diameter of inner pipe, in.

μ In field units - psi/ft,

μ cp., ft/sec, in

dp_f μv

$$dL \frac{1000}{(d_2^2 - d_1^2)^2}$$

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Flow in Pipes

Introduction

Example of Annular Flow Pressure Drop to Natural Gas Processing

□ Compute the frictional pressure loss for a 7" x 5" annulus, 10,000 ft long, using the slot flow representation in the annulus. The flow rate is 80 gal/min. The viscosity is 15 cp. Assume the flow pattern is laminar

□ The average velocity in the annulus

--

$$q = 80$$

$$v = \frac{q}{1.362} \text{ ft/s}$$

$$2.448 \frac{q}{d_2^2 - d_1^2}$$

$$2.2$$

$$2.448 \frac{q}{(7^2 - 5^2)}$$

$$dp_f = \mu v$$

$$1000 \frac{dp_f}{d_2^2 - d_1^2}$$

$$2$$

$$dL$$

$$dp_f = (15) (1.362) (10,000)$$

$$\Delta p_f = D$$

$$dL \frac{1000}{(7^2 - 5^2)^2}$$

$$\Delta p_f = 51 \text{ psi} \approx 51.0750$$

□ A somewhat more accurate answer, using an exact equation for a circular annulus, results in a value of 50.9792 psi.

□ Difference = 0.0958 psi i.e., within 0.2%

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Federal University of Petroleum Resources Effurun

Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Advance Drilling Engineering

Lesson III

Well Planning 1

Ogbon Advance Drilling Engineering Slide #2

Lesson III: Well Planning

Introduction to Natural Gas Processing

Sub-Topics

- Planning Process Overview
- Transportation/General Logistics Programme
- Drilling Rig Selection
- Service Contracts
- Drilling Programme
- Directional Well Plan
- Casings Programme
- Bit Selection Programme
- Drilling Fluid Programme
- Formation Evaluation Programme
- Well Control Programme
- Completion Programme

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Well Planning

• Introduction

- Generally, people planning and supervising wells should have significant theoretical and practical knowledge; and wellsite exposure with relevant experience
- Without the practical knowledge and experience, - the feel for drilling that comes from years on the rig – drilling personnel are unlikely to become first class drilling people
- These people tend to use their PCs as a senior partner to make decisions for them rather than as a tool to help them make better and informed decisions themselves.
- People without practical knowledge will be unduly influenced by the people around them.
- The success or failure of a well, from a drilling viewpoint, is heavily dependent on the quality of well planning prior to spudding the well.

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Well Planning

• Introduction

□ The well planning efforts should be directed at the two keys to drilling a cost-effective well:

□ Avoiding problems:

This is involves

□ The mud properties and good drilling practices (but not by being overly cautious!)

□ Paying that extra bit to get the most suitable mud system as much as possible for that section of the well.

□ Avoiding unplanned short cuts that often lead to unnecessary problems

□ Maximizing progress

□ By optimizing bit/BHA selection, optimum drilling parameters, good forward planning, and good drilling practices.

□ The success of drilling any well is determined first by;

□ The effort devoted to producing the best possible well plan,

□ The competence of the supervision of the drilling process - bearing in mind the above two keys.

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Well Planning

□ Well Planning Involves:

□ Land lease acquisitions

□ Several studies

□ Evaluation of programs for the prospect.

□ Well Planning pre-requisites:

□ Economics

□ Cost estimates

□ Control

□ Objectives

□ Cost Effective Drilling

□ Fast and Efficient Drilling

□ Generation and Gathering of Maximum Information

□ Ensuring Maximum Safety Operation and Standard

□ Making Drilling Environmentally Friendly

Ogbon Advance Drilling Engineering Slide #6

Well Planning

Activities Needed to Deliver Oil and Gas Well

•Exploratory drilling

Exploration Cellar Installation

- Reserve Estimation?

Well Construction

Conductor/Stove

Economic Analysis Pre-Well Planning Pipe

☐Drilling Programmes Drilling of Intervals

☐Selection of Drilling Rig/Equipment

☐Surface Casing

☐Transportation/General Logistics

☐Intermediate Hole/Casing & Cementing

Well Program Alliance ☐Production hole Formation

Well Planning

☐ Evaluation(Cores, Logs)

Contract Day rate ☐Production Casing Cemented

Turnkey

Well Completion

- Site Survey/Clearance

(Onshore) Installation of Surface Facilities

- Satellite Navigation

(Offshore)

Ogbon Advance Drilling Engineering Slide #7

Well Planning Process Overview

Data Acquisition and Review

Data Analysis

Well proposal

☐ Prepare hole section summaries

☐ Offset data

☐ Question/follow-up

☐ Area experience

☐ Area reference data

Well Design Well Design Meeting

Final status of the well, including: ☐ Assemble team:

☐ Mud, Hardware, casing, wellhead, Xmas ☐ Discuss all aspects of the design tree, sand control, completion design ☐ Agree on who does what

☐ Directional requirements

☐ Document major decisions made

Write the Drilling Program

☐ Methods by which the well design

Circulate the Design for Comment will be safely and efficiently

Distribute for comment, including: implemented

☐ Requesting department ☐ Show the assumptions and

- Drilling department decisions made while writing the
- Other qualified reviewers program (technical justification)

Pre-spud Meeting

- Onshore briefing Circulate the Program for Comment
- Distribute the program Distribute for comment, including:
- Offshore briefing □ Requesting department
- Distribute the program to each □ Drilling department person in supervisory position □ Other qualified reviewers (drillers, geologists, T/P, mud □ Approving body(ies) loggers, etc.) Advance Drilling Engineering

Ogbon Slide #8

Well Planning

- Data Acquisition and Review
- The quality of the well planning in turn is heavily dependent on the quality and completeness of the data used in planning.
- For drilling engineer to be successful, he must be a natural detective, snooping around for every snippet of useful data to analyse.
- The Sources of Data and Required Analysis and Review on them are given below:

Well proposal checklist.

This is a formal request from the exploration or production department containing a package of information for drilling that will define what the well should achieve and where it should be.

The well proposal checklist should contain the following elements as relevant to the particular well:

1. Well objectives (exploration, appraisal, development, or workover)
2. Envisaged timescale (earliest/latest spud date desired)

Ogbon Advance Drilling Engineering Slide #9

Well Planning

- Data Acquisition and Review

Well proposal checklist.

3. Essential Data for Well Design:

- Geological/seismic and lithology/petrophysical correlations, including all possible faults that may be encountered
- Well directional information and targets
- Pore pressure and fracture gradients vs. depth plot
- Shallow gas information (e.g., from shallow seismic surveys and offset)
- Required zonal isolation of reservoirs
- Likely temperature profile with depth

☐ Preparation work required in advance of running the completion, including permanent packers, gravel packs, completion fluid specifications, etc.

☐ Perforation intervals; perforation type (if known)

☐ Completion or logging sump required below the bottom perforation depth

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Well Planning

☐ Data Acquisition and Review

Well proposal checklist.

☐ Completion profile fully defining all elements of the completion hardware with depths; tubing, packers, subsurface safety valves (SSSVs), nipples, electric submersible pumps (ESPs), etc.

☐ Completion pressure testing requirements

☐ Future stimulation work and well interventions, including wireline/coiled tubing work, workovers, or recompletions (e.g., on another zone once primary zone is depleted), including fluids pumped, pressures used during stimulations, possible gas lift, etc.

☐ Temperatures and pressures anticipated during the production life of the well

☐ Likely reservoir fluid composition; any H₂S or CO₂ possible?

☐ Options envisaged for future Xmas tree and completion status on handover from drilling (e.g., plugged and depressured, killed, valve configuration, etc.)

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Well Planning

☐ Data Acquisition and Review

Well proposal checklist.

4. Specialist functions should specify:

☐ Wireline logging program

☐ Coring program

☐ Geological surveying/mud logging requirements

☐ Other evaluation requirements (e.g., palaeontologist services, etc.)

☐ Production test requirements

☐ Final desired status of well; handover to production; suspended or plugged and abandoned

5. Approval signature of the head of the sponsoring department

☐ This is to ensure accountability. It may also be necessary for the department head to give you an account code to write the time against.

☐ Finally review the proposal and ensure that all necessary elements are present as per the above checklist.

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Well Planning

□ Data Acquisition and Review

Sources of Offset Data

- Wild Cat Wells
- Company Sources
- Final well reports
- Daily drilling reports
- Coring program
- Personnel that worked on the wells
- Sources outside the company
- Service companies for Bit, Mud, Formation logs, Directional data, etc.
- Retired or personnel no longer with the company
- Maps of regional geology, structures, local water sources, etc.
- Worldwide, governments and regulatory bodies where well information are stored.

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Well Planning

□ Data Acquisition and Review

Sources of Offset Data

- Hole Section Summaries
- These are summaries created for each section drilled in previous wells
- They contain large quantities of offset data in a way that is meaningful “at a glance.”
- Information from each offset well is shown side by side in columns, which are divided in rows by formation.
- Comparisons can be quickly made in the same formations between several wells.
- The left-hand side describes the formation and shows the lithologies and problem areas, with notes on compressive strengths (derived from sonic logs) and other relevant information.

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Well Planning

□ Data Acquisition and Review

Sources of Offset Data

- Hole Section Summaries
- For each well there are three adjacent columns.
- The left column shows depths of formation tops and thicknesses.
- The center column shows a representation of each bit run, including BHAs used, bit grades, etc.
- An arrow shows each bit run and information regarding the run is

written in.

- The right-hand column shows data on parameters, rate of penetration (ROP), mud properties, and short notes where needed.

- Below each well section is a box to include comments on casing/cement jobs, over gauge hole from logs, and any problems logging or other worthwhile comments

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Well Planning

- Data Acquisition and Review

Example of Hole Sections Summary

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Well Planning

- Data Acquisition and Review

Sources of Offset Data

- Field Operational Notes

- These are daily drilling notes of the drilling engineer consisting of all available data relating to each formation which are summarized for future reference when planning and drilling

- Because drilling programs rarely give sufficient information to the drilling supervisor about the formations he/she is expected to drill through writing and updating field operational manual.

- These serve as major source of information for the well planning

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Well Planning

- Data Analysis Plot of Drilling Activities and Days

- Plot drilling mud records of depth vs. day

- Cost per foot plots

- Analysis of Plots can be prepared that detect lithological changes

- Pore pressure plots

- ROP vs. Depth plots

- Sizing and Trajectory of the Hole

- Changes in plot trend show anomalies that need to be taken care of

Ogbon Advance Drilling Engineering Slide #18

Well Planning

□ Data Analysis

Plot of Drilling Rate in each
Depth Interval

□ Write the Drilling Program

□ Methods by which the well design will be safely and efficiently implemented

□ Show the assumptions and decisions made while writing the program
(technical justification)

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Well Planning

□ 3. Circulate the Design for Comment

□ Distribute for comment, including:

□ Requesting department

□ Drilling department

□ Other qualified reviewers

□ Pre-spud Meeting

□ Onshore briefing

□ Distribute the program

□ Offshore briefing

□ Distribute the program to each person in supervisory position for
Comment

□ Distribute for comment, including:

□ Requesting department

□ Drilling department

□ Other qualified reviewers

□ Approval once finalized

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Well Planning

□ Well Design Meeting

□ Summarize and evaluate all relevant offset information gotten from Data Acquisition
and Analysis

□ Sources of offset data

□ Hole section summaries

□ Field operational notes

□ Identify all potential hazards (surface and subsurface) and potential drilling problem
areas and determine how they can be addressed

□ Identify completion design or drill stem test (DST) requirements,

□ Choose casing points that allow kick tolerances to be maintained, casing properties
(outside diameter, weight, grade, connections, etc.) for each casing string, taking into
account the directional plan; minimize potential downhole hazards, and minimize
potential drilling problems.

- Specify the cementing requirements.
- Define the wellhead and Xmas tree requirements.
- Identify long lead time items and obtain approval to place orders in good time.
- Check that the supporting infrastructure (roads, airfields, support bases, etc.) is in place and fit for purpose; flag up if infrastructure needs upgrading.
- Define the earliest possible start date (logistics, permits/approvals, rig availability, weather, infrastructure, etc.).

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Well Planning

- Design Well and Circulate Well Design for Comment
- The well design defines the desired final status of the well.
- The design, therefore, defines casing sizes, grades, weights, connections, and setting position (relative to depths or formation tops).
- Cement tops and particular requirements will be noted. It will define whether the well is to be completed, tested, suspended, or abandoned.
- Define and specify the requirement for permanent packers, perforation intervals, downhole sand control measures such as gravel packs, completion fluid and required completion design, wellhead, and Xmas tree will be specified.
- Surface location and directional requirements are also part of the well design.
- Check all items on the proposal and ensure that the well is designed to meet them; obtain dispensations or amendments if necessary from the sponsoring department.
- Estimate times and costs to prepare an authorization for expenditure (AFE) and time/depth curve. (Note: a more accurate estimate can be made after finishing the drilling program but timescales usually dictate that an AFE is done sooner.)
- Issue the well design document for approval.
- Once the well design is known then the drilling program can be written to achieve the well design safely and cost effectively.

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Well Planning

- Write and Circulate Drilling Programmes for Each Hole Section
- BHA Programme
- Hole Size
- Bit Type and Size
- Deviation Requirements
- Type of BHA
- Required BHA Equipment and Sizes
- Directional Programme
- Sizing and Trajectory of the Hole
- Drilling Fluid Programme

- Optimum Mud Weight
- Type of Mud
- Properties of Mud
- Logging Programme
- Measurement While Drilling (MWD)
- Logging While Drilling (LWD)
- Open Hole Logging
- Cased Hole Logging

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Well Planning

- Write and Circulate Drilling Programmes for Each Hole Section
- Casing/Cementing Programme
- Formation/Fracture Pressure Knowledge
- To Determine Casing Setting Depth
- To Determine Maximum Allowable Mud Density
- Casing Setting Depths
- Casing Design
- Cementing Programs
- Bit Programme
- Size
- Type
- Nozzles sizes
- Well Completion Programme
- Perforation Programme
- Completion Type Selection
- Size of tubing, Xmas tree, types of connections, accessories to run, etc.
- These will depend on factors such as fluids produced, gas/oil ratio (GOR), production potential, tertiary recovery techniques planned, sand control requirements, etc.

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CONTINGENCY PLANS

- Taking Care of Events Occurring Outside Regular Mode of Operation
- Events Occurring Outside Regular Mode of Operation
- Materials
- Determine and Acquire All Materials to be Used
- Make Provision for Extra 10%
- Logistics Problems
- Community Hostilities
- Accidents and Injuries

- Natural Disasters
- Projections on Anticipated Problems
- Kick/Blow-out
- Sticking
- Logistics, etc
- Develop Appropriate Strategies, Preventive and Response Measures

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Federal University of Petroleum Resources Effurun

Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Lesson VIII: Well Control

Introduction to Natural Gas Processing

Sub-Topics

- Pressure Control
- Kicks and Kick Control
- Blow-out Prevention

Ogbon Advance Drilling Engineering Slide #2

Drilling Control Systems

Controlling the Drilling Process

Drilling process is controlled by the following ways

- Maintaining mud weight(density) to balance formation pressure as required.
- Monitoring the gas shows.
- Monitoring the rate of returns.
- Monitoring the mud pit volume.

□ In the event of any kick, the well is closed in through the annulus with the activation of Blow-Out Preventers (BOP).

□ The well is stabilized and higher mud weight(denser mud) is thereafter pumped in to Kill the well.

□ These processes are commonly referred to as Well Control Processes.

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Advance Drilling Engineering

Lesson VIII

Well Control

Ogbon Advance Drilling Engineering Slide #4

Introduction to Well Control

□ Introduction

Introduction to Natural Gas Processing

□ Drilling a well requires that formation pressure be control to enable the safe drilling of the well.

□ Wellbore hydrostatic pressure and bottom hole circulating pressure must be maintained at drilling programme pre-determined values or ranges

□ While theoretically, wells can be drilled overbalanced, balanced or underbalanced, the prevailing pressure control drilling technique has remained overbalanced drilling

□ Overbalanced drilling technique requires that the wellbore pressure be kept always above the formation pressure in both static and circulating conditions.

□ Underbalanced drilling which requires the wellbore pressure being kept minimally below the formation pressure is becoming common in view of its well production advantages of less or no skin, maximum productivity, etc.

□ Underbalanced drilling however requires high technological control to keep well pressure within bound to be able to keep unwanted “kicks” in check.

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Introduction to Well Control

□ Introduction

Introduction to Natural Gas Processing

During the drilling process, the formation pressure is controlled by the balance offered by drilling fluid pressure. $P_f = P_m$

Balance Drilling

If this balance is offset in favour of the formation pressure, (ie $P_f > P_m$), there will be formation fluid influx into the wellbore. This is what is referred to as Kick.

□ A Kick $P_f > P_m$

This is defined as any undesirable inflow of formation KICK

fluids into the wellbore from the reservoir that normally occurs as a result of a negative pressure differential across the formation

□ Kicks are inherent but major nuisance and danger of the drilling process.

□ Drilling process MUST prevent kicks and if they inevitably occur they must be safely controlled and circulated out of the wellbore

❑ Failure to achieve the above results into a “Blow-out” Kick not controlled which is a disaster. Blow-Out

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Causes of a Kick

❑ Kick Occurrence

Introduction to Natural Gas Processing

❑ Kick Occurs While Drilling

❑ Kick Occurs While Tripping

❑ Causes of Kick

Wells kick because the reservoir pressure of an exposed permeable formation is higher than the wellbore pressure at that depth this can be as a result of:

❑ Low Drilling Fluid Density than that of Formation Oil or Water

❑ Oil and/or salt water can also invade the wellbore from cuttings

Gas Cutting

❑ The presence of large volumes of gas in the returns can cause a drop in the average density and hydrostatic pressure of the drilling fluid.

❑ However, the appearance of gas cut mud at the surface should be access to be certain that it is not from cuttings before its concern results in over-weighting of the mud.

❑ The reduction of bottomhole pressure due to gas cutting at the surface is illustrated in the Table B.1.

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Causes of a Kick

❑ Causes of Kick Introduction to Natural Gas Processing

❑ Low Drilling Fluid Density than that of Formation Oil or Water

❑ Notice that the total reduction in hydrostatic pressure at 20,000 feet is only about 100 psi even though mud density is cut by 50 percent at the surface.

❑ This is because gas is very compressible and a very small volume of gas, which has an insignificant effect on mud density downhole, will approximately double in size each time the hydrostatic pressure is halved.

❑ Near the surface, this small volume of gas would have expanded many times resulting in a pronounced reduction of surface density

❑ Abnormal Reservoir Pressure

❑ Formation pressure due to a full column of salt water with average salinity for the area, is and referred to as normal.

❑ If the pressure is greater, it is known as abnormal.

❑ It is subnormal if the formation pressures below normal due to depleted zones or less than a full fluid column to the surface

□ Because salt water is a common formation fluid and averages about 67 lb/ft³ (8.944ppg) or 0.465 psi/ft, 0.465 psi/ft is considered the normal formation pressure gradient.

□ Normally pressured formations are usually drilled with about 70(9.344 ppg) to 75(10.01 ppg) pcf mud in the hole.

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Causes of a Kick

□ Causes of Kick

Introduction to Natural Gas Processing

□ Not Keeping the Hole Full on Trips

□ As drill pipe and drill collars are pulled from the hole during tripping operations, the fluid level in the hole drops and a volume of mud equal to the steel volume removed must be pumped into the annulus in order to maintain fluid level and mud hydrostatic pressure.

□ As a general rule, the hole should always be filled on trips before the reduction in hydrostatic pressure exceeds 75 psi.

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Causes of a Kick

□ Causes of a Kick

Introduction to Natural Gas Processing

□ Swabbing

□ Causes of Swabbing

□ Balled-up Bottom Hole Assembly – creates tighter hole for drillstring suction pull.

□ Pulling pipe too fast – creates greater instantaneous “vacuum”

□ Poor mud properties – high viscosity and gel mud cling to drillstring creating ball-up effect and thus swabbing

□ Heaving or swelling shale – this reduces hole clearance and produces the effect of balled-up drillstring.

□ Large OD Tools – Same as effect of balled-up BHA

□ Lost circulation to formation resulting in wellbore hydrostatic pressure reduction

□ Drilling into abnormally pressured permeable zones

□ Loss of Riser Drilling Fluid Column(offshore operations)

□ On floating drilling structures the loss of drilling fluid column in the riser may result in a reduction of hydrostatic pressure

in the wellbore and may cause the loss of primary well control. Loss could be due to

- Accidental disconnect
- Riser damage

Ogbon □ Displacement of Engineering

Advance Drilling riser with seawater Slide #10

Causes of a Kick

Introduction to Natural Gas Processing

- Causes of Kick
- Lost Circulation
- This involves the loss of whole mud to natural and/or induced fractures and to depleted reservoirs.

□ This results in a major drop in fluid level in the wellbore that can lower the mud hydrostatic pressure.

□ Common causes of lost circulation include:

□ High Mud Weight - Mud weigh > formation Fracture pressure

□ Running Drillstring Too Fast - Has Reverse Effect of Swabbing

□ Pressure due to Annular Circulation Friction

- Sloughing shale or Balled-up Tools
- Mud-Cap Drilling

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Causes of a Kick

- Causes of Kick Introduction to Natural Gas Processing
- Other Causes of Kick
- Annular flow after cement job
- Lost control during DST
- Drilling into adjacent wells
- Drilling through (shallow) gas zones at excessive rates
- Lost Circulation

□ Managed-Pressure Drilling(MPD)

□ The IADC Underbalanced Operations Committee defined MPD as is an adaptive drilling process used to precisely control the annular pressure profile throughout the wellbore.

□ The objectives are to ascertain the downhole pressure environment limits and to manage the annular hydraulic pressure profile

accordingly.

- Types of MPD include
- Constant Bottom Hole Pressure (CBHP)
- Dual Gradient Drilling (DGD)
- Return Flow Control (RFC) or HSE Method
- Pressurized Mud-Cap Drilling

Ogbon Advance Drilling Engineering Slide #12

Kick Indicators

- Kick Indicators
- During the drilling process, warning signals are always prevalent before a kick or a blowout can occur.
- The drilling crew must be at alert at all times to identify these warning signals and to react quickly by shutting in the well with only a small amount of formation fluids in the wellbore.
- These warning signals are referred to as “Kick Indicators”.
- Kick indicators are usually divided into three broad groups:
 - Positive Kick Indicators
 - Increase in Pit Volume
 - Increase in Flow Rate or Well Flows with pumps off
 - Secondary Kick Indicators
 - Decrease in Circulating Pressure
 - Gradual Increase in Drilling Rate
 - Drilling Breaks
 - Increase in Gas Cutting
 - Increase in Water Cutting or Chlorides

Ogbon Advance Drilling Engineering Slide #13

Kick Indicators

- Kick Indicators
- Abnormal Pressure Indicators
 - Decrease in Shale Density
 - Change in Cuttings Size and Shape
 - Increasing Fill on Bottom After a Trip or Improper Hole Fill
 - Increase in Flow Line Temperature
 - Increase in Rotary Torque
 - Increasing Tight Hole on Connections
- Any time the well experiences a positive indicator of a kick, immediate action must be taken to shut in the well.
- When a secondary abnormal pressure indicator of a kick is identified, confirmation steps should be taken to verify if the well is indeed kicking.

□ When Indicators of Abnormal Pressure are identified, it often means that the well is penetrating an abnormally pressured formation and remedial action may range from increasing the mud weight to setting casing.

□ Note that a flowing well must always be treated as a positive indication Of a kick, it should be shut in immediately.

Ogbon Advance Drilling Engineering Slide #14

Well Control Process

Introduction to Natural Gas Processing

□ Well Drilling Process Control

□ Pressure drilling Processes

□ Over Balanced Drilling

□ Balanced Drilling

□ Under Balanced Drilling

□ Drilling Engineering Association

(DEA) defined Underbalanced

(UB) drilling as deliberately

drilling into a formation where

the formation pressure or pore

pressure is greater than the Production Revenue Comparison: UBD, MPD, OBD

pressure exerted by the annular

fluid or gas column.

□ In this respect, “balanced”

pressure drilling is a

subcategory of underbalanced

drilling because the annular

pressure is expected to fall

below the formation pressure

during pipe movement

Ogbon Advance Drilling Engineering Slide #15

Well Control Process

Introduction to Natural Gas Processing

□ Well Drilling Process Control

□ Drilling Hazards

The main operational drilling hazards that result in Non Productive Time(NPT) and unwanted costs are

□ Lost Circulation

□ Stuck Pipe

□ Wellbore Instability

□ Kicks & Well Control Incidents

Differential Lost Wellbore Kicks &

Sticking Circulation Instability Well Control

□ Drilling Process Control or Well Control Process are all the processes or techniques employed by the drilling crew to prevent a kick and to control, circulate it out and kill the well when they do occur.

□ The successful detection and handling of threatened blowouts ('kicks') is a matter of maximum importance to our company.

Ogbon Advance Drilling Engineering Slide #16

Well Control Process

□ Loss Circulation Control

Introduction to Natural Gas Processing

□ Pressurized Mud-Cap Drilling

"Pressurized Mud-Cap Drilling" sometimes referred to as closed-hole circulation drilling (CHCD) involves drilling ahead blind (i.e., without returns to the surface) by pumping different fluid densities down the drill string and annulus simultaneously with a full annular fluid column maintained above a formation that is taking injected fluid and drilled cuttings.

□ The annulus fluid column is assisted by surface pressure (made possible with the use of a stationed Rotating Control Devices (RCD), maintained above a formation that is capable of accepting fluid and cuttings.

□ All fluid is lost to the thief zone.

□ The "mud cap" serves as an annular barrier, while the driller uses a lighter, less damaging and less expensive fluid to drill into the weak zone.

□ Mud-cap drilling permits continued drilling despite the presence of a pressured formation and a lost-circulation zone in the same interval of open hole.

□ MPD is fast becoming the technology of choice for deep water, carbonate-reservoirs and HPHT well control in this region.

□ Although mud-cap drilling has been employed in a limited manner in other oil producing regions of the world, Saudi Aramco is unique in the routine application of this methodology.

Ogbon Advance Drilling Engineering Slide #17

Well Control Process

□ Loss Circulation Control

Introduction to Natural Gas Processing

□ Pressurized Mud-Cap Drilling

Ogbon Advance Drilling Engineering Slide #18

Well Control Process

□ Loss Circulation Control

Introduction to Natural Gas Processing

□ Pressurized Mud-Cap Drilling

- The above figure illustrates the operations of PMCD method.
- The driller pumps the lightweight scavenger fluid down the drillpipe.
- After circulating around the bit, the fluid and cuttings are injected into a weak zone uphole below the last casing shoe.
- The heavy, viscous mud remains in the annulus as a mud cap above the weak zone exerting pressure and preventing the drill pipe fluid from returning to surface.
- The driller can apply optional backpressure if needed to maintain annular pressure control.
- The lighter drilling fluid improves PMCD method principle of operation ROP because of increased hydraulic horsepower and less chip holddown.

Ogbon Advance Drilling Engineering Slide #19

Well Control Process

- Loss Circulation Control

Introduction to Natural Gas Processing

- Pressurized Mud-Cap Drilling

Installation Illustration indicating equipment and operations of PMCD

Figure Illustrates How PMCD Works

Ogbon Advance Drilling Engineering Slide #20

Well Control Process

- Loss Circulation Control

Introduction to Natural Gas Processing

- Lost Circulation

- Pressurized Mud-Cap Drilling Benefits

□ It allows drilling to continue without incurring larger mud cost in thief zones.

□ It reduces well-control risk.

□ It results in faster penetration rates

Ogbon Advance Drilling Engineering Slide #21

Well Control Technique

The well must be shut in immediately when there is a positive indicator of a kick in the form of an increase in pit volume or flow rate.

If a secondary indicator of a kick is recognized then the well should be checked for flow before shutting in.

□ Shut-in Procedure while Drilling

1. SPACE OUT □ Pick up drill string until the spot tool joint clears the floor
2. SHUT DOWN □ Stop the mud pumps.
3. Check for Flow
4. If well is flowing, Open the HRC
5. SHUT IN □ Close the annular preventer or uppermost pipe ram preventer.
6. Close Choke
7. Confirm that the well is shut in and flow has stopped.
8. Open HCR valve.

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Well Control Technique

□ Post Shut-in Procedure while Drilling

1. Notify Supervisors, company man, etc,
2. Record Shut-in Casing Pressure – SICP
3. Record Shut-in Drill pipe Pressure – SIDP
4. Record Pit Gain
5. Record Time kick occurred.
6. Record Closing Pressure.
7. Prepare for Well Killing Implementation

Ogbon Advance Drilling Engineering Slide #23

Well Control Technique

□ Shut-in Procedure while Tripping

Once the well is detected to be flowing during a trip, it must be shut in immediately using the following "Three S" Shut-In Procedure

1. STAB VALVE □ Install Full-Open Safety Valve (open position) in drill string.

Close Safety Valve.

2. SPACE OUT □ Spot tool joint – set slips below top of tool joint
3. SHUT- IN □ Close the annular preventer or uppermost pipe ram and Choke

Confirm that the well is shut-in and flow has topped.

Open HCR valve.

4. Pick-up and stab the kelly or a pump-in line
5. Open Safety valve

6 Notify supervisor, etc

Ogbon Advance Drilling Engineering Slide #24

Well Control Technique

□ Post Shut-in Procedure While tripping

1. Notify Supervisors, company man, etc,
2. Record Shut-in Casing Pressure – SICP
3. Record Shut-in Drill pipe Pressure – SIDP
4. Record Pit Gain
5. Well TVD
6. Well TVD at Bit
7. Record Time & Date kick occurred.
8. Record Closing Pressure.
9. Prepare for Well Killing Implementation

□ Post Shut-in Procedure While Casing

1. Lower casing until a swage and valve can be stabbed.
2. Close the casing rams or annular preventer.
3. Stab the swage and valve.
4. Notify supervisor
5. Read and record pressures
6. Prepare to kill well

Ogbon Advance Drilling Engineering Slide #25

Well Control Techniques

Introduction to Natural Gas Processing

□ Primary Well Control Technique

□ This is the application of drilling fluid density to provide sufficient pressure to prevent formation fluid influx(overbalanced drilling) or maintain it at specified quantity(underbalance drilling)

Basic techniques involved;

□ Mud System Monitoring

Drilling Fluid Management

□ Ensure all valves to/from the active system are checked for integrity and adequate stock of reserve mud is held.

□ Ensure all pit monitoring systems are fully functioning and calibrated prior to entering possible hydrocarbon bearing zones.

□ Keep all mud treatments and transfers to the minimum possible.

Active Mud Volume

□ Monitor active system volume continuously and keep wellbore always full except while drilling surface hole and losses are expected or drilling blind as per programme

□ If returns are lost and mud tanks are depleted the wellbore should be

filled with water.

□ Observed pit level shall be closely and report any change to the Toolpusher/Drilling Supervisor.

Ogbon Advance Drilling Engineering Slide #26

Well Control Techniques

Introduction to Natural Gas Processing

□ Primary Well Control Technique

□ Mud System Monitoring

□ Drilling Fluid Management

□ Maintaining adequate drilling fluid density

□ Keeping wellbore full with adequate drilling density at all times

□ Monitor and effect changes in density, drilling fluid volumes and flow rates.

□ Minimize lost circulation and maintain suitable mud properties (lowest allowable density, reasonable PV, YP).

□ Monitoring During Tripping

□ During tripping the trip tank shall be used for accurate measurement of mud volume changes using electronic and manual floats. This is the responsibility of the Driller or Assistant Driller on the brake. It shall not be delegated to the Mud tester or other personnel.

□ When a drilling break occurs, drill no more than 2 ft into the break and then carry out a flow check.

Ogbon Advance Drilling Engineering Slide #27

Well Control Techniques

Introduction to Natural Gas Processing

□ Primary Well Control Technique

□ Flow Checks

□ Carry out Flow Checks in the following instances:

□ Drilling breaks

□ Prior to pulling out of hole with the bit in the shoe whilst POOH

□ Prior to pulling the bottom hole assembly through the rotary table at any time that standard kick indicators are observed.

□ Flow checks whilst drilling (in response to standard kick indications) and prior to and during tripping operations should be a minimum of 10 minutes duration

□ Note : Should flow be observed during a flow check then the well should be closed in immediately.

Ogbon Advance Drilling Engineering Slide #28

Well Control Techniques

Introduction to Natural Gas Processing

- Primary Well Control Technique
- Annular Pressure Monitoring
- A daily check shall be made on all casing annuli while operations are in progress.
- Any pressure observed shall be recorded and reported daily in the allocated section of the morning report.

□ Note:

A comprehensive check for annuli pressures shall be made prior to removing any mechanical barrier (eg. Xmas tree, Tubing head spool or casing head spool/housing).

Ogbon Advance Drilling Engineering Slide #29

Well Control Techniques

- Well Control Precautions
- Avoid kicks if possible
- Catch them early if they do occur
- Know how to control kicks if they do occur
- While Tripping
- Keep the hole full.
- Avoid excessive surge pressures.
- Avoid excessive swab pressures

Ogbon Advance Drilling Engineering Slide #30

Well Control Techniques

- Well Control Precautions
- While Tripping
- While Tripping Out of Hole
- A good drilling engineer will always stay on the rig floor during the first 20 stands of a trip.
 1. Make sure pipe is not pulled too fast.
 2. Make sure annulus is kept full.e.g. fill every 5 stands, and measure volume required to fill hole.
- If well is not taking enough fluid to replace volume of steel pulled from hole, fluid may have been swabbed into well.
- Measure fluid volume to fill hole:
 1. By counting pump strokes, or
 2. By direct measurement from trip tank.

Note: The second method is best.

Ogbon Advance Drilling Engineering Slide #31

Well Control Techniques

- Recommendations
- While Tripping
- Fill hole after each 5 stands of drill pipe when coming out of hole.
- Fill hole after each stand of drill collars when coming out of hole.
- When TIH after an extended period of time, break circulation gradually, at several different depths.
- Do not break circulation with bit and BHA just above the casing seat
- When breaking circulation,
- Start the pumps slowly,
- Rotate the drill string,
- Pick up on the drillstring at the same time
- This will reduce the initial pressure required to get the mud moving

Ogbon Advance Drilling Engineering Slide #32

Well Control Techniques

□ Secondary Well Control

Introduction to Natural Gas Processing

- This is the specific method used to control well in case of failure of primary and secondary well control.
- The well must be shut in immediately when there is a positive indicator of a kick in the form of an increase in pit volume or flow rate.
- If a secondary indicator of a kick is recognized then the well should be checked for flow before shutting in.
- Shut-in Procedure while Drilling
- 1. SPACE OUT Pick up drill string and spot tool joint.
- 2. SHUT DOWN Stop the mud pumps.
- 3. SHUT IN Close the annular preventer or uppermost pipe ram preventer.
- 4. Confirm that the well is shut in and flow has stopped.
- 5. Open HCR valve.

□ Post Shut-in Procedure while Drilling

1. Record Shut-in Casing Pressure – SICP
2. Record Shut-in Drill pipe Pressure – SIDP
3. Record Pit Gain
4. Record Time
5. Record Closing Pressure

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Well Control Techniques

□ Tertiary Well Control

Introduction to Natural Gas Processing

- This is the specific method used to control well in case of failure of primary

and secondary well control. These following examples are tertiary well control:

- Drill relief wells to hit adjacent well that is flowing and kill the well with heavy mud.
- They use this method to control the well that was firing on the platform.
- Dynamic kill by rapidly pumping of heavy mud to control well with Equivalent Circulating Density (ECD)
- Pump barite or gunk to plug wellbore to stop flowing
- Pump cement to plug wellbore

Ogbon Advance Drilling Engineering Slide #34

Causes of a Kick

- Causes of a Kick

Introduction to Natural Gas Processing

- Swabbing

□ Swabbing arises when the pulling of a drill string from the well produces a temporary bottomhole pressure reduction.

□ In many cases, the bottomhole pressure reduction may be large enough to cause the well to go underbalanced and allow formation fluids to enter the wellbore.

□ In most cases every time the well is swabbed in, a kick has been taken.

□ While the swab may not necessarily cause the well to flow or cause a pit gain increase, the well has produced formation fluids into the annulus, which have almost certainly lowered the hydrostatic pressure of the mud column.

□ Swab identification is mostly done by the hole filling for the volume of drillstring pulled out. If 10 bbl of steel is pulled and only 2 bbl of formation fluid had definitely swabbed into the well.

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Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Advance Drilling Engineering

Lesson VIII

Well Control

Ogbon Advance Drilling Engineering Slide #2

Well Killing Procedure Calculation

□ Driller's Method Calculation

□ Engineer's Method Calculations

Ogbon Advance Drilling Engineering Slide #3

Well Killing Procedure Calculation

Example

Well depth = 10,000'

Hole size = 10.5"

Drill pipe = 4.5", 16.60 #/ft

Drill Collars = 8" * 3.5" * 500 ft

Surface casing = 4,000', 13-3/8", 68 #/ft

Mud Weight = 10 ppg

Fracture gradient @ 4,000' = 0.7 psi/ft

Shut in Annulus Pressure, SICP = 400 psi

Shut in Drillpipe Pressure, SIDPP = 200 psi

Pit level increase = 20 barrels (kick size)

Drop the Z terms but consider Temperature

T at surface = 70 deg. F

Temperature gradient = 1.2 deg.F/100 ft

1. Determine the pressure at the casing seat at 4,000' when using the old mud (Driller's method) versus using the kill mud (Wait and Weight method) to circulate a gas kick out of the hole.

2. Determine the casing pressure at the surface when the top of the gas bubble has just reached the surface, for the same two mud weights

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used above. Advance Drilling Engineering Slide #4

Well Killing Procedure Calculation

200 psi

400 psi

13-3/8", 68 #/ft * 4,000 ft

4,000'

10 ppg old mud

4.5", 16.60 #/ft

9,500'

Kick 8" * 3.5" * 500 ft

10,000' Drill Bit

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Well Killing Procedure Calculation

□ Initial (Closed-In) Conditions

SICP = 400 psi

SIDPP = 200 psi

Pit gain = 20 barrels

1. Bottom Hole Pressure

psi

$P_{10,000} = 200 + (10,000 \text{ ft}) * (0.520) = 5,400 \text{ psi}$

ft

2. Drill Collars Annular Vol/ft

$(d_h^2 - d_c^2) (10.52) (8.2)$

2 2

VDC_Ann OR = 0.04495 bbl / ft

1029 1029

3. Drill Pipe Annular Vol/ft in Open Hole

$(d_h^2 - d_p^2) (10.52) (4.52)$

2 2

Vdp 2 ann = 0.0875 bbl / ft

1029 1029

4. Drill Pipe Annular Vol/ft in Cased Hole

$(d_{2ca}^2 - d_p^2) (13.3752) (4.52)$

2

Vdp1 ann = 0.154 bbl / ft

1029 1029

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Well Killing Procedure Calculation

vdp1.csg = 0.154 bbl/ft

4,000'

vdp2,hole = 0.0875 bbl/ft

9,500'

vdc,hole = 0.04495 bbl/ft

10,000'

Ogbon Advance Drilling Engineering 7 Slide #7

Well Killing Procedure Calculation

5. Height of Kick

Pit gain 20 bbl

h 10,000 = 445 ft

DC Annular Capacity 0.04495 bbl/ft

6. Hydrostatics in the Annulus

$PH = SICP + PH_{old \text{ mud}} + PKICK = 5,400 + 400 + 0.0520 * 10 * (10,000 - 445) = P_{kick_10,000}$

7. Hydrostatic Pressure across Kick Fluid

$$\rho_{\text{kick}} = 10,000 - 5,400 - 400 = 0.0520 \times 10 \times 9,555 = 31.4 \text{ psi}$$

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Well Killing Procedure Calculation

Shut-in Well Parameters

Ogbon Advance Drilling Engineering Slide #9

Well Killing Procedure Calculation

8. Calculate Kick Density

Weight of Kick Fluid

$$W_{\text{kick}} = \rho_{\text{kick}} \times \text{Area}$$

lb

$$= 31.4 \times (10.5^2)$$

$$= 8^2$$

) in 2

$$= 1,141 \text{ lb}$$

in 2 4

Volume of Kick

$$= (10.52^2 \times 8^2)$$

$$A_{\text{kick}} \times h_{\text{kick}} =$$

$$2^3$$

$$V_{\text{kick}} = \text{in} \times 445 = 112.25 \text{ ft}$$

$$4 \times 144 \text{ in}^2 / \text{ft}^2$$

$$\rho_{\text{kick}} = \text{Weight} / \text{Volume} = 1141 / 112.25 = 10.165 \text{ lb/ft}^3$$

Volume 112.25

Kick is gaseous

Ogbon Advance Drilling Engineering Slide #10

Well Killing Procedure Calculation

Driller's Method – Kick at Casing shoe

What is the pressure at 4,000 ft when the top of the kick fluid first reaches that point?

1. Volume of kick at 4000 ft

As the kick fluid moves up the annulus, it expands. If the expansion follows the gas law, then

$$P_1 V_1 = P_2 V_2$$

$$0.0 \times B_1$$

$$Z_1 n R T_1 = Z_2 n R T_2$$

$$0.0 \times B_1 = 0.0 \times B_2$$

[surface] [bottom]

Now, since R is constant and ignoring changes in compressibility factor, we

have

P V P V

0 0 B B

T T

0 B

[surface] [bottom]

P T V

V 0 B 0

B P T

B 0

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Well Killing Procedure Calculation

Driller's Method – Kick at Casing shoe

What is the pressure at 4,000 ft when the top of the kick fluid first reaches that point?

1. Volume of kick at 4000 ft Recall: $T_{surf} = 70 \text{ deg. F}$; Temp. gradient = $1.20 \text{ deg.F/100 ft}$

P T

10,000 4,000 $T_{4000} = 70 + 1.2 \times 4000 = 70 + 48 = 118 \text{ OF}$

V V * T

4,000 10,000 P

100

4,000 10,000

$T_{10000} = 70 + 1.2 \times 10000 = 70 + 120 = 190 \text{ OF}$

100

2. At the Casing shoe, Kick is the hole section with drill pipe and open hole

$V_{4000} = V_{dp, hole} \times \text{Height of Kick @ 4000 ft}$

5,400 118 460

*

$= 0.08743 \times h_{4,000}$

20 * P 190 460

4,000

1,098,444

h 4,000

P_{4,000}

P₄₀₀₀,

3. Recall that h_{4000}

BHP P_{4,000} $\Delta P_{K_4,000}$ ΔP_{MA}

weight 1,141lbs

$\Delta P_{K_4,000}$ 16.1psi

π

area 10.52 4.52 in

4

Ogbon Advance Drilling Engineering Slide #12

Well Killing Procedure Calculation

□ Driller's Method – Kick at Casing shoe

What is the pressure at 4,000 ft when the top of the kick fluid first reaches that point?

3. Recall that

BHP □ P_{4,000} □ ΔP_{K_4,000} □ ΔP_{MA}□ 5,400 □ P_{4,000} □ 16 □ 0.52 * (6,000 □ h 4,000)

OR

5,384 □ P_{4,000} □ 3,120 □ 0.52 * h 4,000

□ 1,098,444 □

2,264 □ P_{4,000} □ (0.52) * □ □ □□ P_{4,000} □

This gives rise to the quadratic equation

□ P_{4,000}

2

□ 2,264 P_{4,000} □ 571,191 □ 0

2,264 □ □ 2,2642 □ 4 * 571,191 □

0.5

□ P_{4,000} □

2

The solutions are

P_{4,000} □ 2,493 psi □ □ 0.6233 psi/ft □ 0.7□ P_{4,000} □ 2,493 psi

Ogbon Advance Drilling Engineering Slide #13

Well Killing Procedure Calculation

□ Driller's Method – Kick at the Surface

What is the pressure when the top of the kick fluid first reaches the surface?

This will require two equations

1. When the bubble rises, it expands and the volume of the bubble at the surface is given by:

□ P □ □ T □

V₀ □ V_{10,000} □ □ 10,000 □ □ □ 0 □ □□ P₀ □ □ T_{10,000} □

□ 5400 □ □ 70 □ 460 □

□ (0.13006) h₀ □ 20 □ □ □ □ □□ P₀ □ □ 70 □ 120 □ 460 □

677,084

□ h₀ □ □ eqn. 1

Po

weight 1,141 lb

$\Delta PK, 0 \approx 10.85 \approx 11 \text{ psi}$

π

area $\approx 12.4152 \approx 4.52 \text{ in}^2$

4

Now

From BHP $\approx P_0 \approx \Delta PK_0 \approx \Delta PMA$

$5,400 \approx P_0 \approx 11 \approx 0.52 * (10,000 \approx h_0)$

Recall that 677,084

$h_0 \approx$

P_0

Ogbon Advance Drilling Engineering 14 Slide #14

Well Killing Procedure Calculation

$\approx$ Driller's Method – Kick at the Surface

What is the pressure when the top of the kick fluid first reaches the surface?

So $\approx 677,084 \approx$

$\approx 5,400 \approx P_0 \approx 11 \approx 0.52 \approx 10,000 \approx$

$\approx P_0 \approx$

$\approx 5,400 \approx 5,200 \approx 11 \approx P_0 \approx P_0 \approx 352,084$

2

$\approx P_0 \approx 189 \approx P_0 \approx 352,084 \approx 0$

2

And the is also a quadratic equation

$189 \approx 189^2 \approx (4)(352,084) \approx 2$

1

$\approx P_0 \approx$

2

$\approx P_0 \approx 695.34 \text{ psi} \approx 695 \text{ psi}$

677,084

$\approx h_0 \approx 973.74 \approx 974 \text{ ft}$

695.34

$\approx P_4,000 \approx P_0 \approx 0.52 * (4,000 \approx 974) \approx \Delta PK_0$

$\approx 695 \approx 1,574 \approx 11 \approx 2,280 \text{ psi}$

Alternatively

$P_4,000 \approx P_{10,000} \approx (0.52) * (10,000 \approx 4,000)$

$\approx 5,400 - 3,120 \approx 2,280 \text{ psi}$

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Well Killing Procedure Calculation

$\approx$ Engineer's Method – Kick at Casing Shoe

Calculate the pressure at 4,000 ft when the top of the bubble reaches this point.

Recall that $P \propto T$

$V_{4,000} \propto V_{10,000} \propto 10,000 \propto 4,000$

$P_{4,000} \propto T_{10,000} \propto M$

$5,400 \propto 578$

$(0.08743) h_{4,000} \propto 20 \propto$

650

$P_{4,000}$

1,098,444

$h_{4,000}$

$P_{4,000} \propto M$

Also

M_1

$BHP \propto P_{4,000} \propto \Delta P_{K_4,000} \propto \Delta P_M \propto \Delta P_{M1}$

Drill String Inside Capacity = $DP_{cap.} + DC_{cap}$

$bbl \propto bbl$

$0.01422 * 9,500 \text{ ft} \propto 0.0119 * 500 \text{ ft} \propto \text{Old Mud}$

$\text{ft} \propto \text{ft}$

$141 \text{ bbl} \propto \text{Quantity of } 10.0 \text{ ppg mud below the bubble.}$

Old Mud

Since this is the original mud that filled the drill string

Kill Mud

Ogbon Advance Drilling Engineering Slide #16

Well Killing Procedure Calculation

$\propto$ Engineer's Method – Kick at Casing Shoe

Calculate the pressure at 4,000 ft when the top of the bubble reaches this point.

Recall from previous calculation,

$\Delta P \propto 16 \text{ psi}$

$K_{4,000}$

M

141bbl

$P \propto 0.052 * 10 *$

$M \propto 0.08743 \text{ bbl/ft}$

$\Delta P \propto 839 \text{ psi}$

M

Now $BHP \propto P \propto \Delta P \propto \Delta P \propto \Delta P$

$4,000 \propto K_{4,000} \propto M \propto M_1$

M

SIDPP

Density of Kill Mud(MW) ρ_{MW}

KILL OLD $0.052 * 10,000$

M1

200

$\rho_{10} - \rho_{10} = 0.385 - 10.385$ ppg

0.052×10000

ρ_{141}

$\Delta P = 0.052 * 10.38 * 6,000$ h

M1 $4,000$ 0.08743

BHP $P + \Delta P + \Delta P + \Delta P$

Old Mud

$4,000$ K $4,000$ M M1

$5,400$ $P + 16 = 839 + 2,368 + (0.5398) h$

Old Mud

$4,000$ $4,000$

Kill Mud

Ogbon Advance Drilling Engineering Slide #17

Well Killing Procedure Calculation

Engineer's Method – Kick at Casing Shoe

Calculate the pressure at $4,000$ ft when the top of the bubble reaches this point.

Since

$5,400$ $P + 16 = 839 + 2,368 + (0.5398) h$

$4,000$ $4,000$

ρ_{10} M

$1,098,444$

$2,177$ $P + (0.5398) h$

$4,000$ $P_{4,000}$

ρ_{10}

And multiplying both side of equation by P and rearrange

$P^2 + 2,177 P + 592,940 = 0$ M

$4,000$ $4,000$

M1

Solving the quadratic equation, we have

0.5

$2,177 \pm \sqrt{2,177^2 - 4 * 592,940}$

$P =$

$4,000$ 2

$P = 2,422$ psi 0.61 psi/ft Old Mud

4,000

Old Mud

Kill Mud

Ogbon Advance Drilling Engineering 18 Slide #18

Well Killing Procedure Calculation

□ Engineer's Method – Kick at the Surface

Calculate the pressure at 4,000 ft when the top of the bubble reaches this point.

Volume of Gas Bubble at Surface

□P □T □

□ 10,000 □ 0 □

V □V □ □

0 10,000 □P □T

□ 0 □ 10,000 □

□ □

□ 5,400 □ 530 □

□ 0.13006 * h □ 20 □ □ 650 □

0 □ P

□ 0 □

677,084

□h □

0 P

Recall that 0

P □ P □ ΔP □ ΔP □ ΔP

10,000 0 K0 M M1

Consequently

□P □ 11 psi

K,0

Assume all 10 lb/gal mud is inside the 13 3/8" csg.

Then the height of 10 lb mud:

141.0 bbl

h □ 1,084 ft

M 0.13006 bbl/ft

Ogbon Advance Drilling Engineering Slide #19

Well Killing Procedure Calculation

□ Engineer's Method – Kick at the Surface

Calculate the pressure when the bubble reaches top.

The Hydrostatic Column across the mud columns can therefore be given as

1. Old Mud(M)

$$\Delta P = 0.052 * 10 * 1,084 = 564 \text{ psi}$$

M

2. Kill Mud(M1)

$$\Delta P = 0.052 * 10.38 * (10,000 - h = 1,084)$$

M1 0

$$P = P + \Delta P = \Delta P = \Delta P$$

$$10,000 - 0 - K0 - M - M1$$

$$5,400 - P = 11 - 564 = 0.5398 * (8,916 - h)$$

0 0

Solving

$$12.14 - P = (0.5398) h$$

0 0

Substituting for $h_0 = 677,084$ in the above equation, we get

$$P_0 =$$

$$= 677,084$$

$$12 - P = (0.5398) \text{ and } P_2 = 12P = 365,490 - 0$$

$$0 - P = 0 - 0$$

$$0 - 0$$

Solving 1

$$12 - 122 = 4 * 365,490 = 2$$

$$P = 610.59 = 611 \text{ psi}$$

0 2

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Well Killing Procedure Calculation

Engineer's Method - Kick at the Surface

Calculate the pressure when the bubble reaches top.

The Hydrostatic Column across the mud columns can therefore be given as

3. Height of bubble at Surface

$$677,084 - 677,084$$

$$h = 1,109 \text{ ft}$$

$$P = 610.59$$

0

Check Pressure at 4000 ft - Surface

$$P = P + \Delta P = \Delta P = \Delta P$$

$$4,000 - 0 - K0 - M - M1$$

$$611 - 11 - 569 = 0.052 * 10.38 * (4,000 - 1,109 - 1,093)$$

$$= 2,161 \text{ psi} = 0.54 \text{ psi/ft}$$

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Well Killing Procedure Calculation

□ Engineer's Method – Kick at the Surface

Summary and Comparison of the Engineer's Method Calculations

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Well Killing Procedure Calculation

Summary and Comparison of the Two Methods

Bubble At 10,000 ft At Casing Shoe At Surface

Driller's Engineer's Driller's Engineer's Driller's Engineer's

Method Method Method Method Method Method

P4,000 2,480 2,480 2,493 2,422 2,280 2,161

P0 400 400 413 342 695 611

From the above although the Engineer's Method results in lower values, the difference is quite acceptable.

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Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Advance Drilling Engineering

Lesson II

Drilling Service Contracts

Ogbon Advance Drilling Engineering Slide #2

Lesson II: Drilling Service Contract

Introduction to Natural Gas Processing

Sub-Topics

□ Introduction

□ Types of Contracts in the Oil & Gas Service

Industry

□ Essentials of Drilling Services Contract

Ogbon Advance Drilling Engineering Slide #3

Introduction to Drilling Service Contracts

Introduction to Natural Gas Processing

□ The Drilling Service Contract is an agreement between the operator and the service company to drill and complete a well OR to render certain specialized services within the drilling and completion processes.

□ The service companies are expected by this contract to render certain specialized services based on expert knowledge, skill and technology, equipment and materials.

- Major areas of service include:
 - Logistics
 - Provision of drilling rig
 - Drilling the well
 - Provision and engineering of mud
 - Provision of equipment and materials
 - Casing
 - Mud logging
 - Well logging
 - Directional drilling
 - Directional survey
 - MWD and LWD
 - Completions
 - This agreement allocates responsibilities to both the Operator and contractor
- Ogbon Advance Drilling Engineering Slide #4

Introduction to Drilling Service Contracts

Introduction to Natural Gas Processing

- Operators Responsibilities majorly include payment for services rendered and supervision of the operation to guarantee successful delivery of the well.
- Contractor must render services in accordance to contract specifications
- Features of Drilling Service Contracts
- Good, workable, and cooperative arrangements laced with goodwill between the operator and drilling contractor
- Driving the hardest bargain possible to the point of picking at every contract clause, every possible charge-back item, strongest possible indemnities, mobilization items, etc., will usually result in hard feelings and a less productive operation.
- If the relationship is very adversarial and convoluted, it will result in a difficult working relationship
- When an operator decides to coordinate the drilling contractor's equipment and personnel down through the driller's position rather than communicate through the tool pusher results in problems.

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Introduction to Drilling Service Contracts

Introduction to Natural Gas Processing

- Who Formulates the Contract
- The Operator
- The International Association of Drilling Contractors ("IADC")
- The IADC is a world-wide drilling contractor trade organization

that drafts and distributes drilling contract forms which favour drilling contractors as opposed to operators

□ When it is a buyers' market for rigs, the operator will have a strong position to use his formulated contract with few negotiated changes; however, in a sellers' market, the drilling contractor will try to use his formulated contract, which of course favours the contractor's position.

□ Types of Contract

There are four main types of contracts commonly used in the oil and gas service Industry; namely:

- Drilling contracts
- Well services contracts
- Seismic acquisition agreements
- Master Service Contract

Ogbon Advance Drilling Engineering Slide #6

Types of Drilling Service Contracts

Introduction to Natural Gas Processing

□ Types of Contract

□ Drilling Contracts

□ This is the contract agreement between the operator and the service company to drill and complete a well.

□ The nature, major parts and operation are dealt with in subsequent sections.

□ Seismic Acquisition Agreements

□ Seismic Data are usually provided by Seismic companies who have licences/agreement from government or land owner to carry out geophysical and seismic surveys.

□ The process of seismic investigation, data acquisition, analysis and interpretation for 3-D seismic data is cost prohibitive to the geophysicists

□ Operators are expected to purchase or procure a licence from the seismic companies for either outright ownership of data or usage of data

□ Seismic Acquisition/Licensing Agreement spells the extent of data ownership or usage.

□ 3-D seismic data are needed by independent operators to raise money from banks for projects.

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Types of Drilling Service Contracts

□ Types of Contract

Introduction to Natural Gas Processing

□ Seismic Acquisition Agreements

□ Generally as with drilling contracts, the big companies have leverage and directly purchase exclusive rights to the data, and may even sell or license the data itself after it has used it while smaller operators, may only acquire seismic data rights by licensing the data directly from a geophysical company or indirectly from a brokerage company.

□ Usually, license agreement are quite restrictive with respect to disclosure of seismic data to third parties, the extent of liability of contracted geophysicist to the seismic company, etc.

□ Operator will normally insist on broad possible rights to use and show the data acquired while seismic companies will insist on limited disclosure.

□ Agreements also relate to who processes the seismic data

□ All issues need to be clearly stated in Seismic Licensing Agreements

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Types of Drilling Service Contracts

□ Types of Contract

Introduction to Natural Gas Processing

□ Well services contracts

□ Costs for well services contracts may be set by day rates, or based upon a set price for the service provided.

□ These costs include elements of both services and equipment.

□ Logistics are normally handled by operator

□ Master Service Contract

□ These are “Master Service Agreements” formulated by “big” oil and gas producing companies when they expect an ongoing relationship with a reliable service company.

□ It is generally in favour of the producer.

□ On the other hand, service companies can also author “Master Service Agreement” for small national oil companies that conduct their own operations

Ogbon Advance Drilling Engineering Slide #9

Types of Drilling Services Contract

Introduction

□ Payment Terms of Drilling Services Contract to Natural Gas Processing

Payment for services rendered is normally in form of footage, day or combination

□ Footage Contract:

□ Operator pays the drilling company a certain stipulated amount for each foot drilled from the surface to a certain depth or to some specified objective

- The contract provides that the drilling contractor furnishes the drilling crew, drilling equipment, and certain specified services.
- Riskier for the contractor because he assumes more of well-related risk due to any unforeseen down time the rig may have.
- Footage contract attracts higher cost to the operator.
- Under all forms, the operator will generally reserve the right to order that drilling cease at any point in time
- Day-work compensation rates may apply when the drilling is suspended or delayed by the operator.

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Types of Drilling Services Contract

Introduction to Natural Gas Processing

□ Payment Terms of Drilling Services Contract

□ Day-work:

□ Payment is based on the number of days the rig or equipment or men are on site.

□ When the rig or equipment or men are on “stand-by”, the rate is usually lower.

□ In modern time, a substantial shift has been towards the Daywork contract from footage and turnkey; and contractors are more comfortable with its provision guaranteeing them a fixed amount of revenue per operating day regardless of what was done or how efficiently the hole is drilled.

□ Combination Agreements:

□ A combination of Daywork and Footage contracts that stipulate when each type is used for certain operations.

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Types of Drilling Services Contract

Introduction to Natural Gas Processing

□ Payment Terms of Drilling Services Contract

□ Turnkey:

□ In the Turnkey contract, the drilling contractor is obligated to drill a complete well for a lump sum fixed fee which the operator agrees to pay contractor upon completion of the well.

□ Drilling contractor assumes all the cost and responsibility of the well site, contracts with third parties for equipment and services and has no operator supervision onsite.

□ Contractor takes some or all the risk associated with delay and mechanical failure in drilling a well to a certain depth

- Typically awarded to companies that the operator trusted and has worked with closely over time.
- A pure turnkey contract provides the highest risk and highest reward for the contractor.
- If there are difficulties during the operation and the turnkey depth is not reached, the contractor is not paid

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Types of Drilling Services Contract

Introduction to Natural Gas Processing

- Payment Terms of Drilling Services Contract
- Turnkey:
 - The risk born by the contractor are quite substantial compared to daywork operations
 - Main features of turnkey contracts:
 - Obligation to drill a well to a certain depth and/or formation.
 - A fixed contract price(sum) which the contractor earns entirely if the contractor performs and completes the well to the specified depth and/or formation with whatever other obligations that are specified in the “turnkey obligation” section of the contract.
 - The operation and the drilling methods are controlled by the contractor because he takes all the risk of losing the well.
 - Turnkey contract fees are usually higher due to obvious risk taken by the contractor.
 - This risk transfer accounts for the higher cost to the operator of turnkey operations.

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Types of Drilling Services Contract

Introduction to Natural Gas Processing

- Payment Terms of Drilling Services Contract
- Turnkey:
 - Expert, skilled and knowledgeable crew are basic requirement of the contractor if he must succeed with Turnkey contract.
 - However, it should be noted that the operator is still responsible for the cost and obligation of completing and equipping the well.
 - A turnkey contract, however, may be tailor-made to particular drilling conditions, placing specified risks on the operator and providing additional compensation to the drilling contractor.

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Types of Drilling Service Contracts

Introduction to Natural Gas Processing

- Payment Terms of Drilling Services Contract
- Alliancing contracts
- In this type of contract, a lead contractor is employed to subcontract and manage all the services needed to drill a well, but the operator still stays closely involved.
- Effectively, the lead contractor provides most of the resources of a drilling department, plus areas of specialist expertise.
- Partners align around a common goal of delivering the well safely in shortest duration at minimum cost to specification.
- Benefits of these strategies include.
- If a true team spirit emerges where people work cooperatively together for achieving the same goals, costs can possibly be cut on long-term (development) projects.
- However, one guiding principle should be that the operator retains the technical ability to plan and supervise the wells.
- This means keeping competent drilling people in place.

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Types of Drilling Service Contracts

Introduction to Natural Gas Processing

- Payment Terms of Drilling Services Contract
- Alliancing contracts
- Main 3C-factors needed for an alliance:
 - Commitment,
 - Communication,
 - Competence.
- These take time to get in place.
- An alliance will not swing smoothly into action from the start therefore management also needs the commitment to see it through the initial hiccups.

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

- Components of Drilling Service Contracts
- Contract Duration
- The length of the contract is commonly determined by the number of wells the operator wants to drill.
- Long term
- Usually driven by market conditions and rig availability in market.

- New rig owners' financial institutions may require a reasonable payback on the loan before the contractor can sign a contract.
- In long term contract, operator should do all it can to mould the contractor into its culture and routine. This usually results in
 - Higher efficiency,
 - Safer operation and
 - Less trouble time
 - Less downtime for the operator and the drilling contractor
 - A more team-oriented effort between the parties
 - A more cost-effective, trouble-free operation—a truly “win/win” situation
- Short term
- Contractors will insist on short term contract during rig scarcity and high oil prices while operators will do same when rigs are surplus oil prices are low

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

□ Components of Drilling Service Contracts

□ The Work Scope/Specifications:

This is one of the most important parts of the contract and it consist of:

- Wellbore diameter of each section of the well
- Depth of each section of the well
- The drilling muds to be used in each section
- The deviational and directional requirement of each section
- The equipment, technology and services to be employed in each section.

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

□ Components of Drilling Service Contracts

□ Payment Rates/Fees Terms

- The rates to be paid for service rendered by the service contractor and the manner of payment determines the type of service contract
- Types of service contracts by rates and mode of payment include
 - Footage rate
 - Day work rate
 - Turnkey rate
 - Combination and

- Alliance
- Other Fees Commonly negotiated in service contracts include
- Standby rates
- Mobilization rates
- Commissioning rates
- Decommissioning rates

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

- Components of Drilling Service Contracts
- Liability or Risk Allocation Clauses
- Drilling is a high risky process associated with huge cost.
- Problem arising from drilling are associated with substantial damage to property and in personal injuries.
- Problem may also occur at thousands of feet down hole where direct observation is difficult, the nature and cause may be difficult to ascertained and appropriated.
- Consequently, operators endeavour to eliminate uncertainty with regard to responsibility for consequences relating to problems that arise by ensuring that responsibilities for damages are clearly stated.
- The goal is to ensure that when various scenarios come to pass the partners responsible for damages are well enunciated, and to make sure that there are neither gaps in insurance coverage, nor double coverage.

- The above ensure that protracted litigation is avoided

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

- Components of Drilling Service Contracts
- Liability or Risk Allocation Clauses
- Risk allocation scheme include:
- Indemnification Clause
- Insurance Requirement
- Other Contractual Protection Clauses
- These mitigate the effects of casualty risks, foster certainty in the case of an accident, and reduce litigation costs.
- The understanding of the Basic Risk Allocation Principles common to IADC drilling contracts form is essential to negotiating acceptable contracts and protecting clients'

interests.

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Essentials of Drilling Services Contract

Introduction

- Components of Drilling Service Contracts to Natural Gas Processing
- Liability or Risk Allocation Clauses
- Indemnification Clause
- Indemnity clauses exempt(indemnify) parties from penalties and liabilities.
- Types of Indemnity Clauses prevalent in oil & gas industry drilling service contracts include:
 - “Knock for Knock” indemnification provision □ “Bury your own Dead”
 - Both operator and contractor separately assumes all risk associated with its own equipment and personal, regardless of fault, and agrees to indemnify the other party for claims arising out of injuries suffered to their respective employees and properties.
 - When each party “buries his own dead”, he is in best position to reduce risk of failure by controlling their respective personnel and equipment.
 - It also takes out the element of fault which further reduces likelihood of protracted litigation.
 - This however does not apply to gross negligent or breach of safety acts.
 - “Knock for Knock” indemnity does not also apply to third parties and their properties
 - Certain states do have anti-indemnity laws forbidding, prohibiting or limiting agreements that require indemnity for a party’s own negligence

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Essentials of Drilling Services Contract

Introduction

- Components of Drilling Service Contracts to Natural Gas Processing
- Liability or Risk Allocation Clauses
- Indemnification Clause

Example of Simple Indemnity Clause

Contractor shall release, defend, indemnify and hold the Company Group harmless from and against any and all claims, damages, liabilities, and expenses (including attorneys’ fees and all costs of defense) for bodily injury to, illness or death, or any damage to or loss of property, of any member of Contractor Group to the extent such bodily injury, illness, death, damage or loss arises out of or is incident to the performance of the Services, including loading, unloading, ingress, and egress of personnel or cargo, regardless of the cause, even though caused in whole or in part by a pre-existing condition, or the negligence (sole or concurrent), strict liability, or the unseaworthiness, unairworthiness or defective

condition of vessels, craft or premises owned, supplied, hired, chartered or borrowed under other agreements or otherwise of or by Company Group, excluding in each case to the extent that such injury, illness, death, damage, or loss was caused or contributed to by the gross negligence or willful misconduct of the party seeking defense, indemnity or release.

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

☐ Components of Drilling Service Contracts

☐ Liability or Risk Allocation Clauses

☐ Indemnification Clause

☐ Operator's Indemnity

Operator shall indemnify Contractor from and against any claim concerning:

☐ Personal injury to or loss of life of any personnel of Company group.

☐ Loss of or damage to any property of Company group, arising out of or in connection with the Work.

☐ Pollution from reservoir and property of Company

☐ The loss of or damage to property, materials or equipment of Contractor Group, which occurs whilst in-hole below the rotary table, unless due to fair wear and tear or caused by Contractor's negligence. To the extent that loss or damage shall be limited to the replacement value of Contractor Group's equipment.

☐ The loss of or damage to any property of Contractor Group when being used by Company

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

☐ Components of Drilling Service Contracts

☐ Liability or Risk Allocation Clauses

☐ Indemnification Clause

☐ Operator's Indemnity

☐ The loss of or damage to any well or hole

☐ The blowout, fire, explosion, cratering or any uncontrolled well condition (including the cost to control a wild well and the removal of debris)

☐ The damage to any reservoir, geological formation or underground strata or the loss of oil or gas therefrom

□ The use of radioactive tools in relation to the Work or any contamination resulting therefrom (including retrieval and/or containment and clean up) - This applies regardless of any form of liability, whether strict or by negligence, in whatever form, on the part of Contractor group.

□ The loss or damage suffered by a Third Party in connection with the Work, to the extent that any such loss or damage is not caused by the negligence or breach of duty (whether statutory or otherwise) of Ogbon Company. Advance Drilling Engineering Slide #25

Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

□ Components of Drilling Service Contracts

□ Liability or Risk Allocation Clauses

□ Indemnification Clause

□ Operator's Indemnity

Operator shall indemnify Contractor from and against any claim concerning:

□ Personal injury to or loss of life of any personnel of Company group.

□ Loss of or damage to any property of Company group, arising out of or in connection with the Work.

□ Pollution from reservoir and property of Company

□ The loss of or damage to property, materials or equipment of Contractor Group, which occurs whilst in-hole below the rotary table, unless due to fair wear and tear or caused by Contractor's negligence. To the extent that loss or damage shall be limited to the replacement value of Contractor Group's equipment.

□ The loss of or damage to any property of Contractor Group when being used by Company

Ogbon Advance Drilling Engineering Slide #26

Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

□ Components of Drilling Service Contracts

□ Liability or Risk Allocation Clauses

□ Indemnification Clause

□ Operator's Indemnity

□ The loss of or damage to any well or hole

□ The blowout, fire, explosion, cratering or any uncontrolled well condition (including the cost to control a wild well and the removal of debris)

- The damage to any reservoir, geological formation or underground strata or the loss of oil or gas therefrom
- The use of radioactive tools in relation to the Work or any contamination resulting therefrom (including retrieval and/or containment and clean up) - This applies regardless of any form of liability, whether strict or by negligence, in whatever form, on the part of Contractor group.
- The loss or damage suffered by a Third Party in connection with the Work, to the extent that any such loss or damage is not caused by the negligence or breach of duty (whether statutory or otherwise) of Ogbon Company. Advance Drilling Engineering Slide #27

Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

- Components of Drilling Service Contracts
- Liability or Risk Allocation Clauses
- Indemnification Clause
- Contractor's Indemnity

Contractor shall indemnify Operator from and against any claim concerning:

- The infringement of patents or other industrial property rights arising out of or in connection with the Work or Company's use of the Work, except where such infringement necessarily arises from the use of Company Provided Items or Company's instructions
- The loss or damage suffered by a Third Party in connection with the Work, to the extent that any such loss or damage is caused by the negligence or breach of duty (whether statutory or otherwise) of Contractor Group.
- The pollution occurring on the premises of Contractor Group or originating from the property and equipment of Contractor Group located above the rotary table, arising from or relating to the performance of the Work.

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

- Components of Drilling Service Contracts
- Liability or Risk Allocation Clauses
- Indemnification Clause
- "Pass Through" Indemnification provision.
- This is a provision in an oilfield contract that allows the indemnitee to pass the indemnity that he has received under the contract through to

other persons as a matter of contract, e.g. its contractors, subcontractors, affiliates, etc.

- This is commonly associated with contractor's contracts with third parties or sub-contractor contract
- It requires subcontractors to take off risk on damages resulting from actions of the operator and that are quite disproportionate to the value of their contract
- A fall back position is for the subcontractor to take on huge insurance coverage that eats deep into his profit thereby making business quite suffocating in an ordinarily very lucrative business of the oil and gas industry.

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

- Components of Drilling Service Contracts
- Liability or Risk Allocation Clauses
- Indemnification Clause
- "Pass Through" Indemnification provision.
- Essential contract for Operator generally contains a "Broad Reciprocal Indemnity" meaning both contractors and sub-contractors

□ Drilling contractor will want indemnity for Operator's people and property and people and property of Operator's other contractors

□ With broad reciprocal in drilling contract, Operator will owe indemnity to drilling contractor every time there is an accident

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

- Components of Drilling Service Contracts
- Waiver or Consequential Damages Clause
- A waiver absolves a party from the consequential damages arising from actions of the other(s) that may affect him or the other(s).
- For example contractor will insist on a waiver clause absolving him of any and all consequential damages arising from a blow out.
- On the other hand, the operator would like to have full and complete compensation if the process of drilling damages his reservoir.
- Nevertheless, no waiver will be allowed by law for damages arising from acts of "gross negligence" from any party

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

□ Components of Drilling Service Contracts

□ Liability or Risk Allocation Clauses

□ Risk allocation scheme include:

□ Insurance Requirement

□ Company shall provide and maintain the insurances according to applicable laws and regulations upon commencement of the Work and shall remain in force throughout the contract duration to cover its liabilities under the Contract

□ Contractor shall at its own expense procure and maintain insurance upon commencement of the Work and shall remain in force throughout the contract duration to cover its liabilities under the Contract

□ Contractor's insurance policies shall be taken out with first class insurers and according to the best insurance terms available on the market.

□ Company shall be included in Contractor's insurances as protected additional insured to the extent of the liability assumed by Contractor under this Contract.

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

□ Components of Drilling Service Contracts

□ Force Majeure Terms

□ "an unexpected event that crucially affects somebody's ability to do something and can be used in law as an excuse for not having carried out the terms of an agreement (formal)"

The English dictionary - Microsoft® Encarta® 2007.

□ Effects of Force Majeure Term in Contract

□ Neither of the parties shall be considered in breach of an obligation under the Contract to the extent the party can establish that fulfilment of the obligation has been prevented by Force Majeure.

□ Each party shall cover its own costs resulting from Force Majeure.

□ Notification of Force Majeure.

□ The party invoking Force Majeure shall as soon as possible notify the other party of the Force Majeure situation. Such notice shall also include the cause of the delay and the presumed duration thereof.

□ Cancellation due to Force Majeure.

□ Each party is entitled to cancel the Contract if the Force Majeure situation

lasts without interruption, or it is evident that it will do so, for more than 60 - sixty - Days.

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Essentials of Drilling Services Contract

Introduction to Natural Gas Processing

□ Components of Drilling Service Contracts

□ Other Components Include:

□ Termination provisions

□ Assignment of Contract

□ Amendments Clause

□ Notices and Claims Clauses

□ Confidentiality clause

□ Numerous exhibits on:

□ Customs

□ Confidentiality

□ Items required by law (e.g., equal opportunity)

□ Policies of both companies, etc.

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The Contract Document

Introduction to Natural Gas Processing

□ The document consisting of:

□ The Conditions of Contract

□ The Purchase Orders

□ A Separate Form of Agreement (if required)

□ The flowing Appendices:

□ Appendix A: Scope of Work

□ Appendix B: Compensation

□ Appendix C: Contract Schedule

□ Appendix D: Administration requirements

□ Appendix E: Company Provided Items

□ Appendix F: Health, environment and Safety

□ Appendix G: Company's requirements to Contractors Equipment

□ Appendix H: Contractors specifications.

□ Appendix I: Contractors proprietary information

□ Appendix J: Subcontractors

□ Appendix K: Guarantees

□ This document is also referred to as the Contract

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Course:

PNG 521

Advance Drilling Engineering

Ogbon Advance Drilling Engineering Slide #1

Advance Drilling Engineering

Lesson III

Well Planning - 4

Ogbon Advance Drilling Engineering Slide #2

Well Completion Design

Well Completion Design

Ogbon Advance Drilling Engineering Slide #3

Casing

Well Design

Completion Design

□ Introduction to Completion

□ Factors Affecting Completion Design

□ Completion Design

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Well Completion Design

Introduction to Well Completion

□ Once a natural gas or oil well is drilled, and it has been verified that commercially viable quantities of natural gas are present for extraction, the well must be 'completed' to allow for the flow of petroleum or natural gas out of the formation and up to the surface.

□ Well Completion is the preparation of the drilled well for the production process.

□ It involves all the processes and equipment installation required to:

□ Let the flow of the reservoir fluid into the wellbore

□ Let the lifting of the fluid from the well bottom to the surface

□ Pump the reservoir fluid out of the wellbore bottom

□ Pump fluid into the Reservoir (Injection Well).

□ The well is the only communication with the reservoir

□ The effectiveness of that communication is a large factor in reservoir drainage, recovery, as well as overall project economics

□ The individual well completion must be designed to yield maximum overall profitability on a field basis

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Well Completion Design

Introduction to Well Completion

□ Completed Well

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Well Completion Design

Introduction to Well Completion

- Objective of Completion Design
- To optimize productivity/injectivity performance,
- To make sure the well produces reliably and safely,
- To optimize the implementation of artificial lift,
- To optimize equipment lifetime and make it easy to change all or some of the equipment in the future without much difficulty.
- To produce hydrocarbons in an efficient and safe manner.
- To satisfy the overall life of field / well economics and economics of the entire project.
- Functions of Well Completion
- Well completion creates a dependable pathway to the surface for the hydrocarbons.
- Provide means for the safe and efficient production of oil and gas from the reservoir.

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WellCasing Design

Completion Design

Introduction to Well Completion

- Data Sources for Well Completion Design

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Well Completion Design

Factors for Consideration in Well Completion Design

- Simple and Low Cost Completion - The ideal completion is the lowest cost completion (initial and operating costs)
- Estimation of the production characteristics throughout the life of the well in both the installation equipment mechanical considerations and the reservoir considerations.
- Integrity and Deliverability throughout the anticipated well lifetime
- INCLUSION of Allowance of changing future operating and maintenance expense conditions (such as workovers and interventions) during the initial designs and over its design life such as scale and wax precipitation, corrosion, rising water or gas cut, etc.

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Well Completion Design

Factors for Consideration in Well Completion Design

- Purpose of the well -

- Exploration Wells: To obtain preliminary data on a reservoir, nature and characteristics of the fluids on the reservoir, pay zone characteristics (pressure, temperature, permeability and productivity)
- Appraisal or Confirmation Well: To obtain refined or complete the data from exploration and determine the off-wellbore characteristics (permeability, reservoir boundaries, existence of faults/discontinuity)
- Developmental Well: To determine the total number of wells bring the field on the production stream, maximum production and field recovery.

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Well Completion Design

Factors for Consideration in Well Completion Design

- Parameters related to the Environment

Constrains related to country/site/location of well i.e land or offshore.

Environmental restrictions involve:

- Difficulty in obtaining supplies
- Available space
- Available utilities
- Safety rules and regulations
- Certain operations that may or may not be possible
- Meteorological and oceanographic conditions must also be taken into consideration where applicable.

- Parameters related to the Drilling

- Types of rigs (drilling/completions
- Well profile- horizontal/directional
- Drilling and Casing program
- Drilling in the pay zone(s) and drilling fluids-
- Formation damage and fluid contamination.
- Cementing the production casing
- Wellbore diameter

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Well Completion Design

Factors for Consideration in Well Completion Design

- Parameters related to the Reservoir
- Reservoir pressure and its changes- need for injection wells?
- Interface between fluids and their changes- water encroachment
- Number of levels to be produced
- Rock characteristics and fluid types
- Production profile and the number of wells required.
- Parameters related to the Production
- Safety- compliance with various regulations

- Flowing well or artificial lift-
- Operating conditions- space availability, operating fluids-erosion/corrosion
- Anticipated measurement, maintenance or work over operations-
- Other factors
- Completion techniques:
- General configuration,
- Pay zone to bore hole connection,
- Treatment of the formation,
- Equipment installed,
- Artificial lift

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Well Completion Design

Completion Design

- Basic Steps of Design of Completion String.
- The Conceptual Design
- Specification of the bottom hole completion technique
- Selection of production conduit
- Assessment of completion string facilities
- Evaluation of well performance.
- Selection of the Well Architecture
- Detailed Design
- Specification of All components
- Assessment of the ultimate well and completion performance
- Assessment of the sensitivity to reservoir data used for the design.
- Evaluation of the Detailed Design :
- Optimizing completion performance
- Optimizing well performance

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Well Completion Design

Completion Design

- Principal Decision Areas

Several aspects of the well need to be taken into consideration in well completion design.

- Casing string set and cemented before drilling the reservoir section
9.625" (9 5/8"); 8.5"; or Smaller bits of 6".
- Typical liner size
7" inside a 8" hole, or 5" inside a 6" hole.
- Sand Control
- Artificial Lift Systems

□ Note that wells can be either Producers (oil or gas) or injectors (water or gas)

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Well Completion Design

Completion Design

□ Principal Decision Areas

□ Angle of Penetration through the reservoir section (except truly horizontal wells).

□ Reservoir Management through field life is a critical part of hydrocarbon asset management.

□ Completion Cost:

An appropriate risk analysis must be conducted of low initial completion cost versus long term effective reservoir management.

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Well Completion Design

Completion Design

□ Areas of Study that Need to be Involved in the Full Completion Design

□ Reservoir Inflow Performance Modeling □ to determine productivity of the reservoir

□ Vertical Lift Performance □ to determine tubing size / artificial lift requirements

□ Well Completion Type Selection □ to maximize productivity and optimize cost / recovery

□ Material Selection □ tubing metallurgy

□ Fluid Production Chemistry Issues □ Waxing and scaling potential

□ Completion Component Selection □ packers, safety valves, etc.

□ Tubing Stress Analysis □ to Determine integrity, burst and collapse rating, Ogbon

etc. Advance Drilling Engineering Slide #16

Well Completion Design

Completion Design

□ Areas of Study that Need to be Involved in the Full Completion Design

□ Equipment & Services Procurement / Contracting Writing Procedures, etc.

□ Performing the Operations □ Reliability and Expertise of Service Company

□ Writing a post-completion report and analyzing lessons learned

□ Other company business objectives that need to be considered include;

□ Safety, Efficiency and Risk analysis

□ These Constitute an integral part of the whole process.

□ They will help to deliver an optimized completion.

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Well Completion Design

Completion Design

□ Well Inflow Performance Relationship (IPR) Modelling

□ Relationship between the well flow rate(q) and the associated bottom hole flowing pressure (P_{wf}).

□ To determine productivity of the reservoir

□ Normally affected by any damage to the wellbore

Ono

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Well Completion Design

Completion Design

□ Tubing Performance Curve or Lift Curve

Pressure Drops in Producing Well

Katz Producing Gas Well Flow Equation

Ogbon Advance Drilling Engineering Slide #19

Well Completion Design

Completion Design

□ Tubing Performance Curve or Lift Curve

□ Relationship Between the Well Flow Rate and the Tubing Pressure Drop

□ Normally Determined by the Wellhead Pressure, Tubing Size and Flow Rate

Ono

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Well Completion Design

Completion Design

□ Optimum Production Flow Rate

□ The intersection between the two curves determines the well optimum production flow rate

□ Any other desired flow rate has to be augmented by providing the pressure differential between the two curves with an artificial lift.

Ono

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Well Completion Design

Completion Design

INCREASE of PWF INCREASE of PWF

Due to Fluid Hold-Up Due to increasing friction

Ono

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Well Completion Design

Completion Design

Designing for Optimum Tubing Size for Depleting Reservoir

□ Effect of Reservoir Pressure Depletion on the IPR curve.

Ono

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Well Completion Design

Completion Design

Designing for Optimum Tubing Size for Depleting Reservoir

□ Effect of Reservoir Pressure Depletion on the IPR curve.

Ono

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Well Completion Design

Completion Design

Designing for Optimum Tubing Size for Depleting Reservoir

□ Determined by tubing size that has optimum production with all possible reservoir IPRs throughout its production life.

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Well Completion Design

Completion Design

□ Well Performance

Inflow performance Relationship (IPR)/Vertical Flow Performance (VFP □Tubing Performance) Curves.

Fig. 3.3

Ono

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Well Completion Design

Completion Design

Designing for Optimum Tubing Size for Depleting Reservoir

It has been believed that the tubing size selected should still ensure producing under the maximum lifting efficiency at the end of the flowing period. The formula of selecting tubing diameter is shown in Equation

where

d - internal diameter of tubing, mm;

□1 - relative density of liquid;

Q1 - production rate at the end of the flowing period, t/d;

L - tubing length, m;

pwf - flowing bottomhole pressure at the end of the flowing period, 105 Pa;

pwh - flowing tubing pressure at the end of the flowing period, 105 Pa.

The production rate at the end of the flowing period can be determined by the intersections of the future inflow performance IPR curve and tubing TPR curve.

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Well Completion Design

Completion Design

Designing for Optimum Tubing Size for Depleting Reservoir

Example

A certain well has predictive $Q_1 = 75 \times 10^3$ kg/d, minimum wellhead tubing pressure $P_{wh} = 2 \times 10^5$ Pa, flowing bottomhole pressure $P_{wf} = 23 \times 10^5$ Pa, well depth $L = 1000$ m, and relative density of produced fluid $\rho_1 = 0.9$. Make selection of the tubing size with the longest flowing period.

Solving Process.

Tubing with an inside diameter closer to this value, such as 2 7/8-in. (inside diameter 62.0 mm) tubing, can be used.

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Paper 5 of 5 — PEE 521 Advanced Drilling engineering by Engr Gift

Level 500 • Second Semester • Handout

ADVANCED DRILLING ENGINEERING

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DEPARTMENT OF PETROLEUM ENGINEERING
COLLEGE OF ENGINEERING AND TECHNOLOGY
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
EFFURUN

ADVANCED DRILLING ENGINEERING COURSE CONTENT

- Advanced Drilling Methods and Technologies.
- Drilling hydraulics.
- Rheology of Newtonian and non-Newton fluids.
- Physical/Chemical Properties of Drilling Fluids
- Rotary drilling hydraulics.

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Advanced Drilling Methods and Technologies

- The drilling industry is changing rapidly in the areas of technology, safety, environment, etc.
- The driving forces are largely economic. High returns on any oil & gas field project are determined by:
 - Production (Revenue)
 - Reduction in capital and operating costs
- The main area of E & P operations expenditure is drilling and reduction in drilling cost is possible by the methods and technologies involved.
- The appropriate choice and application of drilling technology has

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Advanced Drilling Methods and Technologies

Advanced drilling methods use a variety of technologies to improve safety, efficiency and the time of drilling an oil well.

Modern and advanced well drilling methods may be grouped into these categories in terms of their mechanism of drilling:

- Percussion.
- Abrasion.

☐Hydraulic.

☐High Power Vibration

☐High Power Laser

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Advanced Drilling Practices

Oil and gas exploration use several drilling methods and platforms based on the formations, geographic location, soil type and other factors. Here are five of the most common drilling methods in used:

☐ Percussion or Cable Drilling

☐ Rotary Drilling

☐ Dual-Wall Reverse-Circulation Drilling (see animated video)

☐ Electro-Drilling

☐ Directional Drilling

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Modern Drilling Methods

☐ Percussion or Cable Drilling

Percussion drilling, also known as cable drilling, is a drilling method that uses a heavy bit or hammer, suspended by a cable, to break up the ground through repeated impacts. This method is particularly useful for site investigations and geotechnical work, especially in situations where casing is needed to prevent borehole collapse

☐ Rotary Drilling

Rotary drilling is a method of creating boreholes or wells by rotating a drill bit against the earth's surface, using a rotary drilling rig to apply downward force and rotational energy. This technique is widely used in various industries like oil and gas, mining, and construction for creating large diameter holes in different ground conditions.

DRILLING AND WELL

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Modern Drilling Methods

☐ Directional Drilling

Directional drilling is the practice of drilling non-vertical bores to reach specific underground targets or to install utilities in a non-linear path. It involves intentionally deviating a wellbore from its natural path using specialized tools and techniques. This method is used in both oil and gas and utility installation applications .

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Drilling Platforms:

Drilling platforms can be grouped into two types based on the environment of operations

□ Onshore Drilling Rigs: These are rigs that are use on land. They provide stable platforms for drilling. There are 4 types of onshore or land rigs:

- Conventional drilling rigs
- Directional drilling rigs
- Hydraulic fracturing drilling rigs
- Coiled tubing drilling rig

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Drilling Platforms:

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Drilling Platforms:

□ Offshore Drilling Rigs:

Offshore drilling rigs are large, often massive, structures used for drilling oil and natural gas wells in marine environments. They can be either fixed platforms anchored to the ocean floor or floating structures like drillships and semi-submersibles. These rigs are self-contained, equipped with drilling equipment, living quarters for workers, and facilities for processing and storing extracted resources

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Offshore Drilling Rigs (Bottom supported rigs)

□ Barges:

They are designed to work in shallow water (less than 20 ft deep).

The rig is floated to the drill site, and the lower hull is sunk to rest on the sea bottom.

The large surface area of the lower hull keeps the rig from sinking into the soft mud and provides a stable drilling platform.

□ Submersibles:

They are barge that are designed to work in deeper water (up to 50 ft deep);

They have extensions that allow it to raise its upper hull above the water

level.

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Offshore Drilling Rigs (Bottom support)

□ Platforms

□ Platforms use a jacket (a steel tubular framework anchored to the ocean bottom) to support the surface production equipment, living quarters, and drilling rig.

□ Multiple directional wells are drilled from the platform by using a rig with a movable substructure.

□ The rig is positioned over preset wellheads by jacking across on skid beams.

□ After all the wells are drilled, the rig and quarters are removed from the platform.

□ Smaller platforms use a jackup rig to drill the wells.

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Offshore Drilling Rigs (Bottom support)

□ Jack-ups

Jack-ups are similar to platforms except that the support legs are not permanently attached to the seafloor – The weight of the rig is sufficient to keep it on location

The rig's legs can be jacked down to drill and jacked up to move to a new location – when under tow, a floating hull buoys the jack-up.

The derrick is cantilevered over the rear to fit over preset risers if necessary.

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Offshore Drilling Rigs (Floating Drilling Rigs)

□ They are not attached or resting on the ocean bottom and have special features to facilitate this:

□ They are held on location by anchors or dynamic positioning

□ The drill string and riser are isolated from wave motion by motion compensators.

There are two categories of floating drilling rigs:

□ Semi-submersible drilling rig

□ Drill ships

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Offshore Drilling Rigs (Floating Drilling Rigs)

Semi-submersible rigs:

□ They consist of a large triangular, rectangular or pentangular platform,

connected with submerged hulls by means of large columns which vary in number from 3 to 8, according to the shape of the vessel

- They are kept vertical over the site by means of mooring or dynamic positioning systems

- Semis are subjected to wave motion, their drilling apparatus is located in the center where wave motion is minimal.

- Semis are flooded to a drilling draft where the lower pontoons are below the active wave base, thereby stabilizing the motion.

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Offshore Drilling Rigs (Floating Drilling Rigs)

Drill ships:

- Drilling ships are used for operating in deep waters, often under extreme environmental conditions, such as drilling in arctic areas

- Drilling ships represents an optimal solution of drilling exploratory wells in remote areas, far removed from supply points, as it can carry all the material necessary for drilling even a particularly difficult well.

- Drilling ships are kept in a vertical position over the well by means of mooring or dynamic positioning systems

- Drilling ships when moored can be used for drilling in depths of up to about 1,000 m, while for greater depths dynamic positioning systems must be used, and with these the ship is capable of operating in 3,000 m of water

- The drilling apparatus on a drill ship is mounted in the center of the ship over a moon pool, which is a reinforced hole in the bottom of the ship through which the drill string is raised and lowered

- The ship can be turned into the oncoming wind or currents for better stability

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Drilling from floating offshore rigs

The techniques for drilling offshore wells from floating rigs are basically the same as those used for onshore wells.

In general, the factors involved in drilling wells from floating rigs are due to the following:

- the wellhead is located on the seabed;

- the submarine BOPs are located on the subsea wellhead and are controlled hydraulically or electrically from the surface;

- the BOPs are connected to the rig by means of a marine riser (enabling the drilling fluid to circulate upwards);

- the marine riser, connected at the top of the BOP stack, has a ball joint on its base and a slip joint above sea level to offset the horizontal and vertical movements of the rig;

- the lines for preventing blowouts (kill lines and choke lines) run from the surface manifold on the rig to the subsea wellhead, as independent lines fixed to the outside of

the marine riser.

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Offshore drilling operations

Drilling starts after the rig has been positioned vertically over the well, by means of mooring or dynamic positioning systems.

□ The first operation consists in placing on the seabed (by means of a pipe string) of a temporary guide base (a strong steel framework with a central hole which has a tapered inlet at the top, provided with four guidelines and a number of steel pins that become embedded in the seabed and prevent displacement).

□ The pipe string is then disconnected, leaving only the temporary guide base on the seabed, with the four guidelines that connect up with the rig.

□ Now, the drilling phase can begin, so as to place the first casing (the foundation pile) and it is carried out using the circulation of seawater, the cuttings do not rise to the surface but are scattered over the seabed.

□ The foundation pile is taken down to a depth of a few tens of metres .

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Offshore drilling operations

At this point the foundation pile is lowered into the hole again using a light frame and the four guidelines.

□ The foundation pile ends with a permanent guide structure, characterized by four tubular columns placed at the apexes, through which run the guidelines.

□ The four columns serve in the subsequent phases to guide the submarine BOPs to the wellhead with precision.

□ The permanent guide structure contains the housing for the wellhead, to which the successive casings will be anchored.

□ The subsea wellhead has a different system of flanging and anchoring the casings and it is shaped in such a way as to enable the lock-up of the hydraulic connector to which the BOPs are coupled.

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Offshore drilling operations

Next, the foundation pile is fully cemented by means of a drill string.

□ Once the cement has set, drilling continues, boring the second section of the hole inside the foundation pile, in which the second casing is inserted.

□ Once the conductor pipe has also been fully cemented, the well has a stable structure and it is possible to install the submarine BOPs.

□ After this drilling continues with the sequence of operations typical of onshore wells.

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Offshore drilling operations

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Offshore drilling operations

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Offshore drilling operations

Subsea BOPs

□ In offshore drilling BOPs have the same function as those used in onshore wells, but are connected in a single complex (the BOP stack) before being mounted on the wellhead, so as to reduce assembly times at the sea bottom.

□ They are lodged in a square-section cage structure with female columns at the apexes, into which the male tubular columns of the permanent guide structure fit.

□ The BOP stack is lowered and fastened to the wellhead by means of a hydraulically controlled connection, ensuring hydraulic sealing.

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Offshore drilling operations

Subsea BOPs

□ In the event of temporarily abandoning the well because of adverse weather and sea conditions, it

is possible to suspend the pipes on the shaped rams of the lower BOP, to unscrew the pipes and to close the well with the upper blind rams – at this point it is possible to disconnect also the marine riser and possibly to abandon the site.

□ The marine riser can be reconnected when the sea and weather conditions improve.

□ The hydraulic lines controlling the various functions of the BOP stack converge in a connector block, to which is connected the bundle of flexible pipes for their control from the surface.

□ The BOPs can be operated in a similar way to that used in onshore wells, known as the direct system.